MATHEMATICS FOR TECHNOLOGY

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Mathematics for Technology

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CHAPTER OVERVIEW

1: Module 1- Measurement and Geometry

1.1: Measurement

Measurement

There are two major systems of measurement in use today. They are the *United States system* and the *metric system*.

The **metric system of measurement** is the decimal system of measure developed in the 1790s in France. This system which is also referred to as SI is the official unit used by every country except the United States. These units use a standard set of prefixes such as milli, centi, kilo, deci

The **US Customary Units** is a system of measurement primarily used in the United States which was validated in 1832. This system of measurement came from the English units, the units of measure used in the British Empire before the formulation of the Imperial system on January 1, 1826, and before American Independence. The customary units are widely used in the areas of commerce and industry. Consumer products and industrial manufacturing materials are weighed or measured using the customary units as evident in their labels.

The common customary units of measure used are:

- inch, foot, yard, and mile for lengths
- acre for area
- cubic inch, cubic foot, and cubic yard for volume and capacity
- ounce, pint, quart, and gallon for fluid volume
- pint and quart for dry volume

This section begins by defining measurement and its common uses.

1.1.1: Measurement and the United States System

6.1.1 Learning Objectives

- Define measurement.
- Perform arithmetic calculations on units of length, weight, and capacity.
- Convert from one unit of measure in the United States system to another unit of measure.
- Solve application problems using each form of measurement.

Measurement

Before we work the *United States measurement system* and the *metric system*, let's gain a clear understanding of the concept of measurement.

Definition: Measurement

Measurement is comparison to some standard.

Definition: Standard Unit of Measure

The concept of measurement is based on the idea of direct comparison. This means that measurement is the result of the comparison of two quantities. The quantity that is used for comparison is called the **standard unit of measure**.

Over the years, standards have changed. Quite some time in the past, the standard unit of measure was determined by a king. For example,

1 inch was the distance between the tip of the thumb and the knuckle of the king.
1 inch was also the length of 16 barley grains placed end to end.

Today, standard units of measure rarely change. Standard units of measure are the responsibility of the Bureau of Standards in Washington D.C.

Some desirable properties of a standard are the following:

1. *Accessibility*. We should have access to the standard so we can make comparisons.
2. *Invariance*. We should be confident that the standard is not subject to change.
3. *Reproducibility*. We should be able to reproduce the standard so that measurements are convenient and accessible to many people.

Definition: Length, Weight, Capacity

Length is a linear measurement to describe the distance between two objects.

Weight is used to describe how heavy or light an object is.

Capacity is used to describe the amount of liquid (or other pourable substance) that will fill an object.

The United States System of Measurement

Some of the common units (along with their abbreviations) for the United States system of measurement are listed in the following table.

Unit Conversion Table	
Length	1 foot (ft) = 12 inches (in) 1 yard (yd) = 3 feet (ft) 1 mile (mi) = 5,280 feet
Weight	1 pound (lb) =16 ounces (oz) 1 ton (T) = 2,000 pounds
Liquid Capacity	1 tablespoon (tbsp) = 3 teaspoons (tsp) 1 fluid ounce (fl oz) = 2 tablespoons 1 cup (c) = 8 fluid ounces 1 pint (pt) = 2 cups 1 quart (qt) = 2 pints 1 gallon (gal) = 4 quarts

Conversions in the United States System

It is often convenient or necessary to convert from one unit of measure to another. For example, it may be convenient to convert a measurement of length that is given in feet to one that is given in inches. Such conversions can be made using *unit fractions*.

Definition: Unit Fraction

A **unit fraction** is a fraction with a value of 1 in the numerator or the denominator.

Unit fractions are formed by using two equal measurements. One measurement is placed in the numerator of the fraction, and the other in the denominator. **Placement depends on the desired conversion**.

Placement of Units
Place the unit being converted *to* in the **numerator**.
Place the unit being converted *from* in the **denominator**.

For example,

Equal Measurements	Unit Fraction
1 ft = 12 in	$\frac{1 \text{ ft}}{12 \text{ in}}$ or $\frac{12 \text{ in}}{1 \text{ ft}}$
1 pt = 16 fl oz	$\frac{1 \text{ pt}}{16 \text{ fl oz}}$ or $\frac{16 \text{ fl oz}}{1 \text{ pt}}$
1 wk = 7 da	$\frac{7 \text{ da}}{1 \text{ wk}}$ or $\frac{1 \text{ wk}}{7 \text{ da}}$

Example 1

Make the following conversions. If a fraction occurs, convert it to a decimal rounded to two decimal places.

Convert 11 yards to feet.

Solution

Looking in the unit conversion table under *length*, we see that $1 \text{ yd} = 3 \text{ ft}$. There are two corresponding unit fractions, $\frac{1 \text{ yd}}{3 \text{ ft}}$ and $\frac{3 \text{ ft}}{1 \text{ yd}}$. Which one should we use? Look to see which unit we wish to convert to. Choose the unit fraction with this unit in the *numerator*. We will choose $\frac{3 \text{ ft}}{1 \text{ yd}}$ since this unit fraction has feet in the numerator. Now, multiply 11 yd by the unit fraction. Notice that since the unit fraction has the value of 1, multiplying by it does not change the value of 11 yd.

$$\begin{aligned} 11 \text{ yd} &= \frac{11 \text{ yd}}{1} \cdot \frac{3 \text{ ft}}{1 \text{ yd}} && \text{Divide out common units.} \\ &= \frac{11 \cancel{\text{ yd}}}{1} \cdot \frac{3 \text{ ft}}{1 \cancel{\text{ yd}}} && \text{(Units can be added, subtracted, multiplied, and divided, just as numbers can.)} \\ &= \frac{11 \cdot 3 \text{ ft}}{1} \\ &= 33 \text{ ft} \end{aligned}$$

Thus, $11 \text{ yd} = 33 \text{ ft}$.

Example 2

Convert 36 fl oz to pints.

Solution

Looking in the unit conversion table under *liquid volume*, we see that $1 \text{ pt} = 16 \text{ fl oz}$. Since we are to convert to pints, we will construct a unit fraction with pints in the numerator.

$$\begin{aligned} 36 \text{ fl oz} &= \frac{36 \text{ fl oz}}{1} \cdot \frac{1 \text{ pt}}{16 \text{ fl oz}} && \text{Divide out common units.} \\ &= \frac{36 \cancel{\text{ fl oz}}}{1} \cdot \frac{1 \text{ pt}}{16 \cancel{\text{ fl oz}}} \\ &= \frac{36 \cdot 1 \text{ pt}}{16} \\ &= \frac{36 \text{ pt}}{16} && \text{Reduce} \\ &= \frac{9}{4} \text{ pt} && \text{Convert to decimals: } \frac{9}{4} = 2.25 \end{aligned}$$

Thus, $36 \text{ fl oz} = 2.25 \text{ pt}$.

Example 5

Your want to lay out a red carpet walkway for your grandma's 75th birthday. The distance from the dropoff area to the entrance of the party venue is 18 feet. The fabric you want is only sold in yards. How many yards of fabric do you need?

Solution

The unit fraction needed for converting from feet to yards is $\frac{1 \text{ yd}}{3 \text{ ft}}$.

$$\begin{aligned} 18 \text{ ft} &= \frac{18 \text{ ft}}{1} \cdot \frac{1 \text{ yd}}{3 \text{ ft}} && \text{Divide out common units.} \\ &= \frac{18 \cancel{\text{ ft}}}{1} \cdot \frac{1 \text{ yd}}{3 \cancel{\text{ ft}}} \\ &= \frac{18 \text{ yd}}{1 \cdot 3} \\ &= 6 \text{ yd} \end{aligned}$$

Thus, $18 \text{ ft} = 6 \text{ yd}$. You would need to purchase 6 yards of fabric for the red carpet.

Try It Now 1

Make the following conversions. If a fraction occurs, convert it to a decimal rounded to two decimal places.

1. Convert 2 mi to feet.
2. Convert 26 ft to yards.
3. Convert 9 qt to pints.

Answer

1. 10,560 ft
2. 8.67 yd.
3. 18 pt

1.1.1.1: Exercises

1. In the U.S. system, abbreviations can be difficult to associate with the correct unit of measurement. What unit do these abbreviations belong to?
    a. lb
    b. oz
    c. ft
    d. mi
    e. yd
    f. pt
    g. qt
2. Convert the following:
    a. 52 inches to feet
    b. 257 miles to inches
    c. 11 feet to yards
    d. 47 ounces to pounds
    e. 2.3 tons to ounces
    f. 7 gallons to quarts
    g. 10 cups to quarts
    h. 3.6 quarts to cups
    i. 29 cups to gallons
    j. 3 pints to fluid ounces
    k. 417 ounces to ton
    l. $8\frac{1}{2}$ gallons to cups
    m. 2 mi to inches
    n. 3 cups to tablespoons
    o. 16 tsp to cups
3. A recipe calls for 3 teaspoons of soy sauce. You have a 5 pt container. How many teaspoons could you get from your container of soy sauce?
4. The coach of a local sports team brought 6 gallons of water that needs to be put into bottles that only hold one pint each. How many bottles will they be able to fill?
5. An 8 ft length of 4 inch wide crown molding costs $14. How much will it cost to buy 40 ft of crown molding?
6. A chain weighs 10 pounds per foot. How many ounces will 4 inches weigh?
7. Sugar contains 15 calories per teaspoon. How many calories are in 1 cup of sugar?
8. The store is selling lemons at 2 for $1. Each yields about 2 tablespoons of juice. How much will it cost to buy enough lemons to make a 9-inch lemon pie requiring $\frac{1}{2}$ cup of lemon juice?

1.1.2: The Metric System of Measurement

6.1.2 Learning Objectives

- Describe some of the advantages of the base ten number system
- Define prefixes of the metric measures
- Convert from one unit of measure in the metric system to another unit of measure

The Advantages of the Base Ten Number System

The metric system of measurement takes advantage of our base ten number system. The advantage of the metric system over the United States system is that in the metric system it is possible to convert from one unit of measure to another simply by multiplying or dividing the given number by a power of 10. This means we can make a conversion simply by moving the decimal point to the right or the left.

Prefixes

Common units of measure in the metric system are the meter (for length), the liter (for capacity), and the gram (for mass). The units are abbreviated m for meter, L for liter, and g for gram. To each of the units a prefix can be attached. The **metric prefixes** along with their meaning are listed below.

Metric Prefixes and Abbreviations

giga (g): billion
mega (m): million
kilo (k): thousand
hecto (h): hundred
deka (dk): ten
deci (d): tenth
centi (c): hundredth
milli (m): thousandth

For example, if length is being measured,

1 kilometer is equivalent to 1000 meters. 1 km = 1000 m
1 centimeter is equivalent to one hundredth of a meter. 1 cm = 0.01 m
1 millimeter is equivalent to one thousandth of a meter. 1 mm = 0.001 m

Conversion from One Unit to Another Unit

Let's note three characteristics of the metric system that occur in the metric table of measurements.

1. In each category, the prefixes are the same.
2. We can move from a *larger to a smaller* unit of measure by moving the decimal point to the *right*. (or multiplying by 10 for each step)
3. We can move from a *smaller to a larger unit* of measure by moving the decimal point to the *left*. (or dividing by 10 for each step)

The following table provides a summary of the relationship between the basic unit of measure (meter, gram, liter) and each prefix, and how many places the decimal point is moved and in what direction.

kilo hecto deka unit deci centi milli

Basic Unit to Prefix		
unit to deka	1 to 10	Divide by 10
unit to hecto	1 to 100	Divide by 10^2
unit to kilo	1 to 1,000	Divide by 10^3
unit to deci	1 to 0.1	Multiply by 10
unit to centi	1 to 0.01	Multiply by 10^2
unit to milli	1 to 0.001	Multiply by 10^3

Conversion Table

Listed below, in the unit conversion table, are some of the common metric units of measure.

Unit Conversion Table		
Length	$1 \text{ kilometer (km)} = 1\text{,000 meters } (m)$	$1,000 \times 1\text{m}$
	$1 \text{ hectometer (hm)} = 100 \text{ meters}$	$100 \times 1\text{m}$
	$1 \text{ dekameter (dam)} = 10 \text{ meters}$	$10 \times 1\text{m}$
	1 meter (m)	$1 \times 1\text{m}$
	$1 \text{ decimeter (dm)} = \frac{1}{10} \text{ meter}$	$.1 \times 1\text{m}$
	$1 \text{ centimeter (cm)} = \frac{1}{100} \text{ meter}$	$.01 \times 1\text{m}$
	$1 \text{ millimeter (mm)} = \frac{1}{1,000} \text{ meter}$	$.001 \times 1\text{m}$
Mass	$1 \text{ kilogram (kg)} = 1\text{,000 grams } (g)$	$1,000 \times 1\text{g}$
	$1 \text{ hectogram (hg)} = 100 \text{ grams}$	$100 \times 1\text{g}$
	$1 \text{ dekagram (dag)} = 10 \text{ grams}$	$10 \times 1\text{g}$
	1 gram (g)	$1 \times 1\text{g}$
	$1 \text{ decigram (dg)} = \frac{1}{10} \text{ gram}$	$.1 \times 1\text{g}$
	$1 \text{ centigram (cg)} = \frac{1}{100} \text{ gram}$	$.01 \times 1\text{g}$
	$1 \text{ milligram (mg)} = \frac{1}{1,000} \text{ gram}$	$.001 \times 1\text{g}$
Volume	$1 \text{ kiloliter (kL)} = 1\text{,000 liters } (L)$	$1,000 \times 1\text{L}$
	$1 \text{ hectoliter (hL)} = 100 \text{ liters}$	$100 \times 1\text{L}$
	$1 \text{ dekaliter (daL)} = 10 \text{ liters}$	$10 \times 1\text{L}$
	1 liter (L)	$1 \times 1\text{L}$
	$1 \text{ deciliter (dL)} = \frac{1}{10} \text{ liter}$	$.1 \times 1\text{L}$
	$1 \text{ centiliter (cL)} = \frac{1}{100} \text{ liter}$	$.01 \times 1\text{L}$
	$1 \text{ milliliter (mL)} = \frac{1}{1,000} \text{ liter}$	$.001 \times 1\text{L}$

Distinction Between Mass and Weight

There is a distinction between mass and weight. The **weight** of a body is related to gravity whereas the mass of a body is not. For example, your weight on the earth is different than it is on the moon, but your mass is the same in both places. **Mass** is a measure of a body's resistance to motion. The more massive a body, the more resistant it is to motion. Also, more massive bodies weigh more than less massive bodies.

Converting Metric Units

To convert from one metric unit to another metric unit:

1. Determine the location of the original number on the metric scale (pictured in each of the following examples).
2. Multiply or divide the original number by 10 raised to the same number of places as is necessary to move to the metric unit you wish to go to. (If changing from a larger to a smaller unit, multiply by 10. If changing from a smaller to a larger unit, divide by 10)

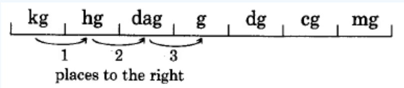
We can also convert from one metric unit to another using unit fractions. Both methods are shown in the following example.

Example 1

Convert 3 kilograms to grams.

Solution

a. 3 kg can be written as 3.0 kg. Then,



$3.0\text{ kg} = 3\,\underbrace{000}_{1\,2\,3}.\text{ g}$

Since we are moving from a larger unit to a smaller unit. Multiply the original number by 10 three times.

$3 \times 10^3 = 3000$ grams

Thus, $3\text{ kg} = 3{,}000\text{ g}$.

b. We can also use unit fractions to make this conversion.

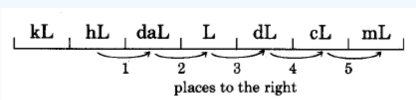
Since we are converting to grams, and $1{,}000\text{ g} = 1\text{ kg}$. we choose the unit fraction $\dfrac{1{,}000\text{ g}}{1\text{ kg}}$ since grams is in the numerator.

$$\begin{aligned} 3\text{ kg} &= 3\text{ kg} \cdot \frac{1{,}000\text{ g}}{1\text{ kg}} \\ &= 3\,\cancel{\text{kg}} \cdot \frac{1{,}000\text{ g}}{1\,\cancel{\text{kg}}} \\ &= 3 \cdot 1{,}000\text{ g} \\ &= 3{,}000\text{ g} \end{aligned}$$

Example 2

Convert 67.2 hectoliters to milliliters.

Solution



$67.2\text{ hL} = 67\,\underbrace{20000}_{1\,2\,3\,4\,5}.\text{ mL}$

Since we are moving from a larger unit to a smaller unit. Multiply the original number by 10 five times.

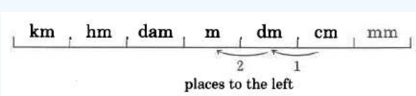
$67.2 \times 10^5 = 6{,}720{,}000$ milliliters

Thus, $67.2\text{ hL} = 6{,}720{,}000\text{ mL}$.

Example 3

Convert 100.07 centimeters to meters.

Solution



$100.07\text{ cm} = 1.\underbrace{00}_{2\,1}07\text{ m}$

Since we are moving from a smaller unit to a larger unit. Divide the original number by 10 two times.

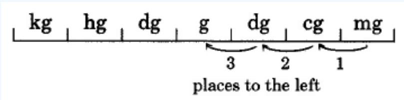
$100.07 \div 10^2 = 1.0007$ meters

Thus, $100.07\text{ cm} = 1.0007\text{m}$.

Example 4

Convert 0.16 milligrams to grams.

Solution



$0.16 \text{ mg} = 0.\underbrace{000}_{3\ 2\ 1}16 \text{ g}$

Since we are moving from a smaller unit to a larger unit. Divide the original number by 10 three times.

$0.16 \div 10^3 = 0.00016$ grams

Thus, $0.16 \text{ mg} = 0.00016$.

Example 5

A medication is sold in 5 milligram size capsules. The pharmacist receives the medication that that is used to fill the capsules in 3.5 kilogram bags. How many capsules can be produced?

Solution

First we need to convert from kilograms to milligrams. Since we are moving from a larger to a smaller unit. Multiply the original number by 10 six times.

$3.5 \times 10^6 = 3,500,000$ milligrams

Each capuse holds 5 milligrams, so we divide the number we just found by 5 to determine the number of capsules that can be filled.

$3,500,000 \div 5 = 700,000$ capsules

Thus, 3.5 kilograms of the medication produce 700,000 capsules.

Try It Now 1

Make the following conversions. If a fraction occurs, convert it to a decimal rounded to two decimal places.

1. 411 kilograms to grams
2. 5.626 liters to centiliters
3. 80 milliliters to kiloliters
4. 150 milligrams to centigrams
5. 2.5 centimeters to meters

Answer

1. 411,000 g
2. 562.6 cL
3. 0.00008 kL
4. 15 cg
5. 0.025 m

1.1.2.1: Exercises

1. In the metric system, the prefix tells what power of ten the basic unit is being multiplied by. What power of ten does the prefix mean?
   a. milli
   b. kilo
   c. hecto
   d. centi
   e. giga
   f. mega
   g. deci
2. State the symbol or abbreviation for each prefix listed in number 1.
3. Convert the following:
   a. 9 meters to centimeters
   b. 520 milliliters to liters
   c. 48.66 liters to deciliters
   d. 2 millimeters to hectometers
   e. 211 centimeters to meters
   f. 6325 milliliters to liters
   g. 1278 grams to kilograms
   h. 3.1 kilograms to grams
   i. 67 millimeters to centimeters
   j. 1.24 kilograms to milligrams
   k. 6 kilograms to grams.
   l. 99.04 dekaliters to centiliters
4. A wire costs $2 per meter. How much will 3 kilometers of wire cost?

1.1.3: Conversion Between the Metric and US Customary Systems of Measurement

6.1.3 Learning Objectives

- To estimate equivalence between the Metric Units and US Customary Units
- To convert Metric Units to US Customary Units
- To convert US Customary Units to Metric Units

Estimate equivalence between the Metric Units and US Customary Units

Before converting metric units to US customary units and vice versa, you need to familiarize the approximate metric equivalents of common customary units. The table below summarizes the commonly used metric and US customary units and their approximate equivalence. The notations enclosed are the abbreviations of each unit.

Metric Units	**US Customary Units**
Lengths	
1 meter (m)	3.28 feet (ft. or ’)
1 kilometer (km)	0.62 mile (m)
1 centimeter (cm)	0.39 inch (in. or ”)
0.9144 meter (m)	1 yard (yd.)
1.852 kilometers (km)	1 nautical mile (nmi. or NM)
Weight	
1 kilogram (kg)	2.2 pounds (lb. or #)
28.3 grams (g)	1 ounce (oz.)
1 milligram (mg)	0.002 teaspoon (tsp.)
Fluid Capacity	
29.57 milliliters (mL)	1 fluid ounce (fl. Oz.)
1 liter (L)	4.23 cups (c)
3.79 liters (L)	1 gallon (gal.)
473.176 milliliters (mL)	1 pint (pt.)
0.95 liters (L)	1 quart (qt.)
158.987 liters (L)	1 barrel (bbl.)

Additional Common Conversions

Metric Units	US Customary Units
Area	
6.45 square centimeters (cm^2)	1 square inch (in^2)
1 square meter (m^2)	1.196 square yard (yd^2)
4046.873 square meters (m^2)	1 acre (ac.)
Volume and Capacity	
16.39 milliliter (mL)	1 cubic inch (in^3)
28.32 liters (L)	1 cubic foot (ft^3)
764.55 liters (L)	1 cubic yard (yd^3)
1233.482 cubic meters (m^3)	1 cubic foot (ft^3)

Convert Metric Units to US Customary Units

You have familiarized the commonly used equivalence between metric and US customary units. At this point, you will learn to convert metric units to US customary units.

To convert metric units to US customary units, use their estimate equivalence. Write the equivalence as a ratio, and multiply it to the given measurement (in metric) to convert. Be sure that the units of measure to convert cancels out. For instance, to convert liters to gallons, be sure to cancel out liters and retain gallons as the final unit.

Now, let us consider some examples.

Example 1

The official regulators of a shipment company classify a parcel weighing more than 51 pounds (lbs.) as a heavy shipment. Andrea packed 24 terracotta plant containers to be shipped to her customers. If each plant container weighs 1.3 kilograms (kg), will her package be classified as a heavy shipment?

Solution

To determine whether the parcel is a heavy shipment or not, you need to convert the total weight of the plant containers to pounds.

Compute for the total weight.

$$\begin{aligned} \text{Total Weight } &= 24 \times 1.3 \text{ kg} \\ &= 31.2 \text{ kg} \end{aligned}$$

Convert the total weight to pounds. Use the metric to US customary units' equivalence: 1 kg = 2.2 lbs.

$$31.2 \cancel{\text{kg}} \times \frac{2.2 \text{ lbs}}{1 \cancel{\text{kg}}} = 68.64 \text{ lbs}$$

Since the total weight is more than 51 lbs., the shipment is considered heavy.

Try it Now 1

Angelica who lives in Las Vegas, Nevada, visited her friend in Canada. She knows that Canada uses the metric system, so their speed limits are posted in kilometers per hour. On her way, she sees a road sign that says, “max 60 km/h”. She knows the sign means “a maximum speed of 60 kilometers per hour” but she is unsure if this is over the speed limit in Las Vegas, Nevada. How can Angelica convert 60 km/h to miles per hour? Use the metric to US customary units’ equivalence: 1 kilometer (km) = 0.62 mile.

Answer

In Las Vegas, the limit is 65 miles per hour (mph). Since 37.2 mph is below this limit, 60km/h is not over the speed limit.

Example 2

Half of a meter of ribbon is required for wrapping presents enclosed in a rectangular box. How many presents using the same box size can be wrapped with 10 yards of ribbon?

Solution

Convert half a meter (0.5 m) to yards. Use the metric to US customary units’ equivalence: 1 yard = 0.9144 meter.

$$0.5 \cancel{\text{ m}} \times \frac{1 \text{ yd}}{0.9144 \cancel{\text{ m}}} = 0.5468 \text{ yd}$$

The result means that one present requires 0.5468 yards of ribbon. Calculate the number of presents by dividing 10 yards by 0.5468 yards.

$$10 \text{ yds } \div \frac{0.5468 \text{ yd}}{1 \text{ present}} = 10 \cancel{\text{ yd}} \times \frac{1 \text{ present}}{0.5468 \cancel{\text{ yd}}} = 18.29 \text{ presents}$$

Approximately, 18 presents can be wrapped with 10 yards of ribbons.

Try it Now 2

Crystal and her friends are planning for a road trip, which they have determined to be 2,000 kilometers (km) long. They want to rent a van that can travel 100 km using 9 liters of gas. Suppose the gas price is $2.74 a gallon (gal.), how many gallons of gas they will need? How much money will the group of friends need to save for gas? Use the metric to US customary units’ equivalence: 4.405 liters (L)= 1 gallon (gal.).

Answer

The team needs 40.86 gallons of gas for the trip, which will cost them $111.96.

Try it Now 3

Christine plans to put up a short-course swimming pool at her house. A short-course swimming pool usually measures 22.86 meters in length, at least 18.29 meters in width, and 2.5 meters in depth. Convert the dimensions of the pool to inches. Use the metric to US customary units’ equivalence: 1 centimeter (cm) = 0.39 inch (in.) and 1 centimeter (cm) = 0.01 meter (m).

Answer

Approximately, the pool measures 891.54 in by 713.31 in by 97.5 in.

Try It Now 4

Drug dosage for children is based on body weights, milligrams per pound (mg/lb). A 5-year-old child with Meningitis who weighs 20 kilograms was prescribed Ceftriaxone. The dosage requirement is 220 milligrams per pound per day (mg/lb/day) once daily. The drug is prediluted in 40 mg/mL concentration. Calculate the dose in teaspoons. Use the metric to US customary units' equivalence: 1 kilogram (kg) = 2.2 pounds (lb.) and 1 mg = 0.0002 tsp.

Answer

The dose is approximately 2 teaspoons.

Convert US Customary Units to Metric Units

You have learned how to convert metric units to US customary units. Now, you will learn to convert US customary units to metric units.

Example 3

Daniel drove 40 miles in two hours. What distance (in meters) did he cover in one minute? Use the US customary to metric units' equivalence: 0.62 mile (mi.) = 1 kilometer (km), 1 km is equal to 1,000 meters (m), and 1 hour = 60 minutes.

Solution

Convert 40 miles in two hours to kilometers per hour.

$$\frac{40\ \cancel{\text{mi}}}{2\ \text{hrs}} \times \frac{1\ \text{km}}{0.62\ \cancel{\text{mi}}} = 32.2581\ \text{km/hr}$$

Convert 32.2581 kilometers per hour to meters per hour.

$$\frac{32.2581\ \cancel{\text{km}}}{1\ \text{hr}} \times \frac{1000\ \text{m}}{1\ \cancel{\text{km}}} = 32,258.1\ \text{m/hr}$$

Convert 32,258.1 meters per hour to meters per minute.

$$\frac{32,258.1\ \text{m}}{1\ \cancel{\text{hr}}} \times \frac{1\ \cancel{\text{hr}}}{60\ \text{min}} = 537.635\ \text{m/min}$$

Daniel has covered 537.635 meters in one minute.

Try It Now 5

A supermarket has a car parking spot that is 9 feet wide and 18 feet long. Its ground-level parking has 50 parallel parking spots (side by side) of the same sizes. What is the area of the parking lot in square centimeters? Use the US customary units' equivalence: 1 foot (ft.) = 12 inches (in.) and 1 square inch (in^2) = 6.45 square centimeters (cm^2). (Hint: You will need to use the formula to find the area of a rectangle)

Answer

The area of the parking lot is 7,523,280 square centimeters.

Example 4

Allison wants to bake a chocolate cake. The recipe requires 1¾ cups of all-purpose flour. A supermarket sells all-purpose flour in grams instead of cups. All-purpose flour comes in the following sizes: 80 grams, 400 grams, and 800 grams. If Allison plans to buy just enough flour for 1 recipe, which size will she buy? Use the US customary units equivalence: 1 cup (c.) = 8 ounces (oz.) and 1 ounce (oz.) = 28.3 grams (g).

Solution

Converting 1¾ to a fraction or decimal will help when converting cups to ounces. In this example 1¾ = 1.75 will be used.

$$1.75 \cancel{\text{ cups}} \times \frac{8 \text{ oz}}{1 \cancel{\text{ cup}}} = 14 \text{ oz}$$

Convert ounces to grams.

$$14 \cancel{\text{ oz}} \times \frac{28.3 \text{ g}}{1 \cancel{\text{ oz}}} = 396.2 \text{ g}$$

Since 80 grams is less than the required recipe amount and 800 grams is much higher than 396.2 grams, Allison must buy the 400 gram package.

Try It Now 6

Melody's house garage is 45 feet long while David's garage is 10 yards and 10 feet long. Which house has a longer garage? How much is the difference (in meters)? Use the US customary to metric equivalence: 3.28 feet (ft.) = 1 meter (m) and 1 foot (ft.) = 0.3333 yard (yd.)

Answer

Melody's garage is longer by 1.52 meters.

Try it Now 7

How many liters of water are needed to fill an Olympic swimming pool with a total volume of 3,300 cubic yards? Assume that the water level is as deep as the pool. Use the US customary to metric units' equivalence: 1 cubic yard (yd^3) = 746.56 liters (L).

Answer

At most 2,463,648 liters of water are needed to fill the swimming pool.

Try It Now 8

Robert's new car requires at most 57 liters of fuel. If the current average price for regular gas is $2.864 per gallon, how much will it cost him to fill his car's tank? Use the US customary to metric units' equivalence: 1 gallon = 3.79 liters

Answer

It will cost Robert approximately $43.32 to fill his car's tank.

1.1.3.1: Exercises

Section 6.1.3 Exercises

1. Convert the following between U.S. and Metric measurement systems.
    a. 52 inches to centimeters
    b. 2 miles to meters
    c. 11 feet to decimeters
    d. 47 ounces to grams
    e. 4580 milliliters to cups
    f. 22 millimeters to inches
    g. 2.3 tons to kilograms
    h. 2,610 grams to pounds
    i. 7 gallons to liters
    j. 10 cups to quarts
    k. 3.6 quarts to cups
    l. 29 cups to gallons
    m. 3 pints to milliliters
    n. 8 meters to yards
2. If you drove 100 km/hour on the freeway, were you exceeding the speed limit?
3. Would you expect to pay more for 10 gallons (40 quarts) of gasoline or for 40 liters of gasoline?
4. Which is the better deal: apples priced at $1.69 per pound or the same apples priced $3.72 per kilogram?
5. If you ran 100 yards in the same time as I ran 100 meters, who ran faster?

1.2: Geometry

Geometry

Geometry is used in our every day when we decorate, design, and arrange objects in a space.

The most commonly used aspects of geometry are:

- Perimeter and circumference, when dealing with the length of the edges of an object.
- Area, when dealing with the amount of surface an object covers.
- Volume, when dealing with how much an object can hold.

Each of these will be explored in the following sections.

1.2.1: Perimeter and Area of Geometric Figures

6.2.1 Learning Objectives

- Define polygon
- Find the perimeter of a polygon
- Find the areas of common polygons

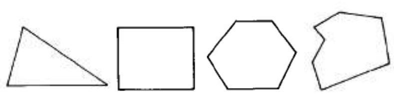
Polygons

We can make use of conversion skills with denominate numbers to make measurements of geometric figures such as rectangles, triangles, and circles. To make these measurements we need to be familiar with several definitions.

Definition: Polygon

A **polygon** is a closed plane (flat) figure whose sides are line segments (portions of straight lines).

Polygons



Not polygons

Perimeter

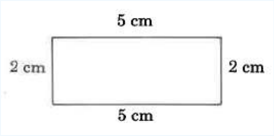
Definition: Perimeter

The **perimeter** of a polygon is the distance around the polygon.

To find the perimeter of a polygon, we simply add up the lengths of all the sides.

Example 1

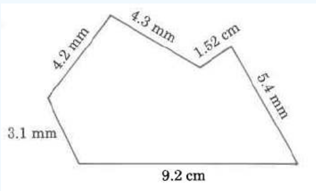
Find the perimeter of the rectangle.



Solution

$$\begin{aligned}\text{Perimeter} &= 2\text{ cm} + 5\text{ cm} + 2\text{ cm} + 5\text{ cm}\\ &= 14\text{ cm}\end{aligned}$$

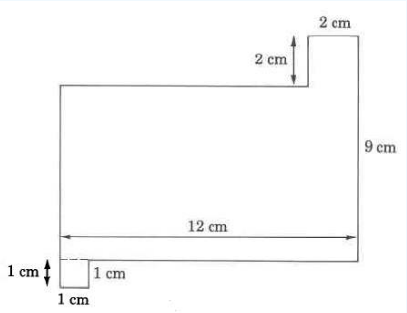
Example 2



Solution

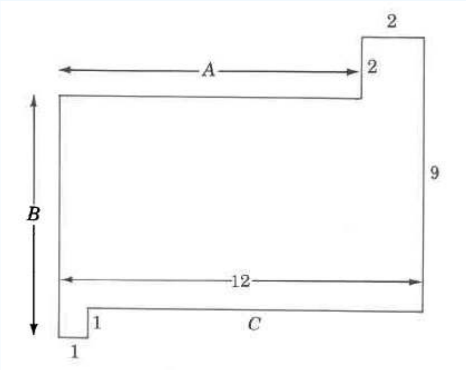
$$\begin{aligned} \text{Perimeter} \quad = \quad & 3.1 \text{ mm} \\ & 4.2 \text{ mm} \\ & 4.3 \text{ mm} \\ & 1.52 \text{ mm} \\ & 5.4 \text{ mm} \\ & \underline{+\, 9.2 \text{ mm}} \\ & 27.72 \text{ mm} \end{aligned}$$

Example 3



Solution

Our first observation is that three of the dimensions are missing. However, we can determine the missing measurements using the following process. Let A, B, and C represent the missing measurements. Visualize



A = 12m - 2m = 10m

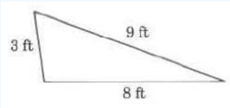
B = 9m + 1m - 2m = 8m

C = 12m - 1m = 11m

$$\begin{aligned} \text{Perimeter} \quad = \quad & 8 \text{ m} \\ & 10 \text{ m} \\ & 2 \text{ m} \\ & 2 \text{ m} \\ & 9 \text{ m} \\ & 11 \text{ m} \\ & 1 \text{ m} \\ & \underline{+\, 1 \text{ m}} \\ & 44 \text{ m} \end{aligned}$$

Try It Now 1

Find the perimeter of this triangle.

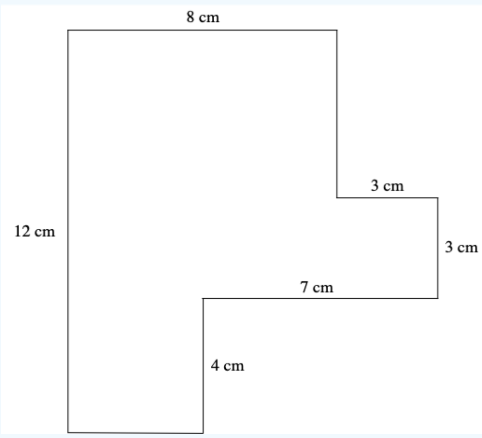


Answer

20 ft

Try It Now 2

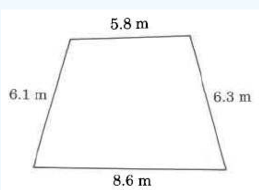
Find the perimeter of this polygon.



Answer

46 cm

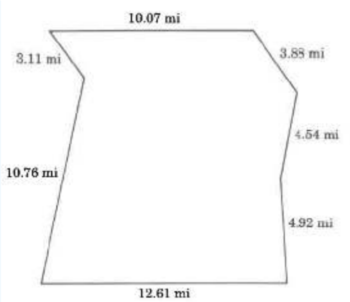
Try It Now 3



Answer

26.8 m

Try It Now 4



Answer

49.89 mi

Quite often it is necessary to multiply one denominate number by another. To do so, we multiply the number parts together and the unit parts together. For example,

$$\begin{aligned} 8 \text{ in} \cdot 8 \text{ in} &= 8 \cdot 8 \cdot \text{in} \cdot \text{in} \\ &= 64 \text{ in}^2 \end{aligned}$$

$$\begin{aligned} 4 \text{ mm} \cdot 4 \text{ mm} \cdot 4 \text{ mm} &= 4 \cdot 4 \cdot 4 \cdot \text{mm} \cdot \text{mm} \cdot \text{mm} \\ &= 64 \text{ mm}^3 \end{aligned}$$

Sometimes the product of units has a physical meaning. In this section, we will examine the meaning of the products $(\text{length unit})^2$ and $(\text{length unit})^3$

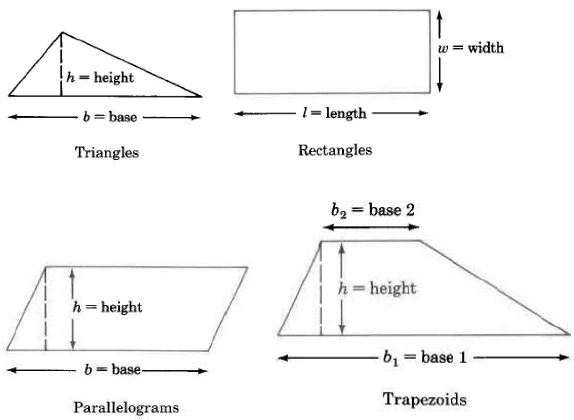
The Meaning and Notation for Area

The product $(\text{length unit}) \cdot (\text{length unit}) = (\text{length unit})^2$, or, square length unit (sq length unit), can be interpreted physically as the *area* of a surface.

Area
The **area** of a surface is the number of square length units contained in the surface.

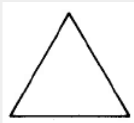
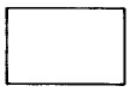
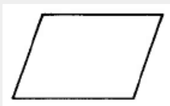
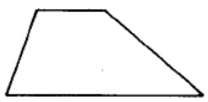
For example, 3 sq in means that 3 squares, 1 inch on each side, can be placed precisely on some surface. (The squares may have to be cut and rearranged so they match the shape of the surface.)

We will examine the area of the following geometric figures.



Area Formulas

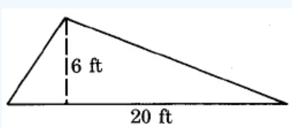
We can determine the areas of these geometric figures using the following formulas.

	Figure	Area Formula	Statement
	Triangle	$A_T = \frac{1}{2} \cdot b \cdot h$	Area of a triangle is one half the base times the height.
	Rectangle	$A_R = l \cdot w$	Area of a rectangle is the length times the width.
	Parallelogram	$A_P = b \cdot h$	Area of a parallelogram is base times the height.
	Trapezoid	$A_{Trap} = \frac{1}{2} \cdot (b_1 + b_2) \cdot h$	Area of a trapezoid is one half the sum of the two bases times the height.

Finding Areas of Some Common Geometric Figures

Example 4

Find the area of the triangle.



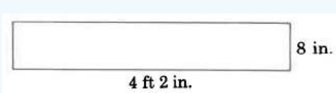
Solution

$$\begin{aligned} A_T &= \frac{1}{2} \cdot b \cdot h \\ &= \frac{1}{2} \cdot 20 \cdot 5 \text{ sq ft} \\ &= 10 \cdot 6 \text{ sq ft} \\ &= 60 \text{ sq ft} \\ &= 60 \text{ ft}^2 \end{aligned}$$

The area of this triangle is 60 sq ft, which is often written as 60 ft^2.

Example 5

Find the area of the rectangle.



Solution

Let's first convert 4 ft 2 in to inches. Since we wish to convert to inches, we'll use the unit fraction $\frac{12 \text{ in}}{1 \text{ ft}}$ since it has inches in the numerator. Then,

$$\begin{aligned} 4 \text{ ft} &= \frac{4 \text{ ft}}{1} \cdot \frac{12 \text{ in}}{1 \text{ ft}} \\ &= \frac{4 \cancel{\text{ft}}}{1} \cdot \frac{12 \text{ in}}{1 \cancel{\text{ft}}} \\ &= 48 \text{ in} \end{aligned}$$

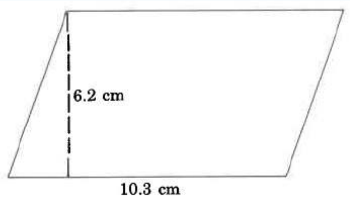
Thus, $4 \text{ ft } 2 \text{ in} = 48 \text{ in} + 2 \text{ in} = 50 \text{ in}$

$$\begin{aligned} A_R &= l \cdot w \\ &= 50 \text{ in} \cdot 8 \text{ in} \\ &= 400 \text{ sq in} \end{aligned}$$

The area of this rectangle is 400 sq in.

Example 6

Find the area of the parallelogram.



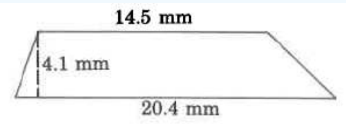
Solution

$$\begin{aligned} A_P &= b \cdot h \\ &= 10.3 \text{ cm} \cdot 6.2 \text{ cm} \\ &= 63.86 \text{ sq cm} \end{aligned}$$

The area of this parallelogram is 63.86 sq cm.

Example 7

Find the area of the trapezoid.



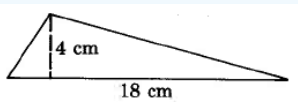
Solution

$$\begin{aligned} A_{Trap} &= \frac{1}{2} \cdot (b_1 + b_2) \cdot h \\ &= \frac{1}{2} \cdot (14.5 \text{ mm} + 20.4 \text{ mm}) \cdot (4.1 \text{ mm}) \\ &= \frac{1}{2} \cdot (34.9 \text{ mm}) \cdot (4.1 \text{ mm}) \\ &= \frac{1}{2} \cdot (143.09 \text{ sq mm}) \\ &= 71.545 \text{ sq mm} \end{aligned}$$

The area of this trapezoid is 71.545 sq mm.

Try It Now 5

Find the area of each of the following geometric figures.

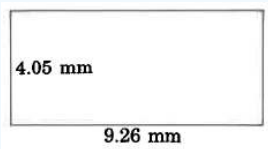


Answer

36 sq cm

Try It Now 6

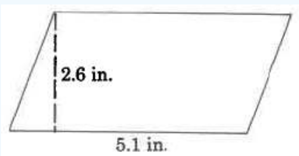
Find the area of the rectangle



Answer

37.503 sq mm

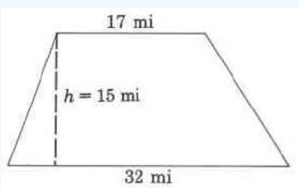
Try It Now 7



Answer

13.26 sq in.

Try It Now 8



Answer

367.5 sq mi

Composite Figures

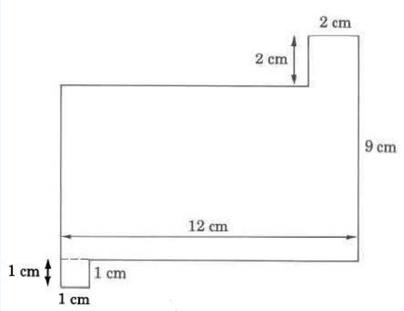
Definition: Composite Figures

A **composite figure** is a figure made up of two or more geometric figures.

When determining the area of a composite figure it is best to determine what shapes the composite figure is made of first. Once you have determined the shapes, find the area for those individual shapes and add them together.

Example 8

Find the area of the composite figure.



Solution

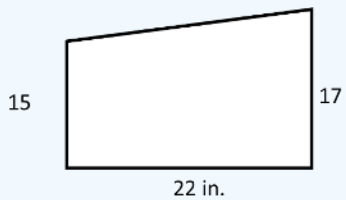
This figure is made up of two squares and one rectangle. There is a square with 1 cm sides and one with 2 cm sides. The rectangle has a 12 cm side, but to figure out the other side we need to subtract 9 cm–2 cm =7 cm.

$$\begin{aligned} A_{figure} &= l \cdot w + l \cdot w + l \cdot w \\ &= 1 \text{ cm} \cdot 1 \text{ cm} + 2 \text{ cm} \cdot 2 \text{ cm} + 12 \text{ cm} \cdot 7 \text{ cm} \\ &= 1 \text{ cm}^2 + 2 \text{ cm}^2 + 84 \text{ cm}^2 \\ &= 87 \text{sq cm} \end{aligned}$$

The area of the composite figure is 87 sq. cm.

Try It Now 9

Find the area of the composite figure.



Answer

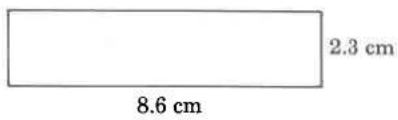
This shape is composed of a rectangle and a triangle.

The area of the composite figure is 352 sq in.

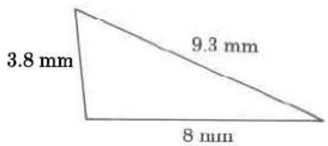
1.2.1.1: Exercises

1. Find each perimeter.

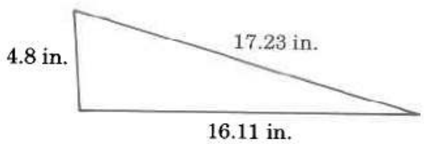
a.



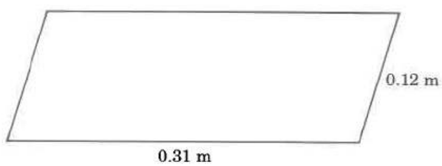
b.



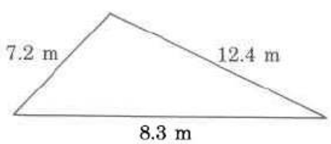
c.



d.

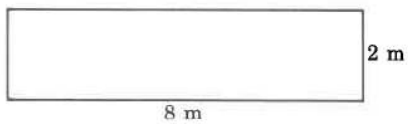


e.

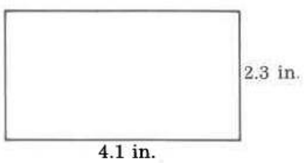


2. Find the area for the following:

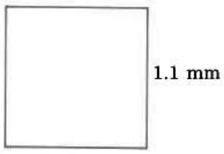
a.



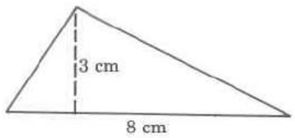
b.



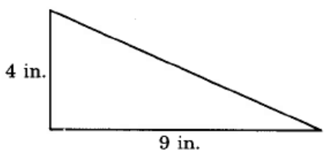
c.



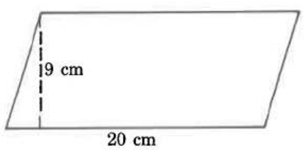
d.



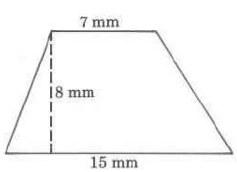
e.



f.

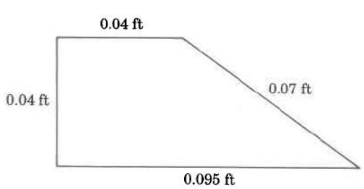


g.

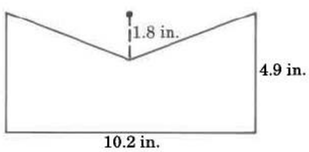


3. Find the area for the following figures below:

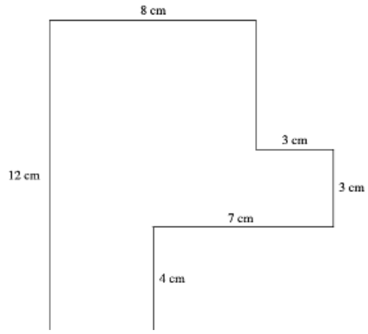
a.



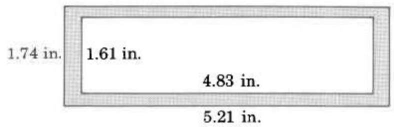
b.



c.



4. Find the area of the shaded region.



1.2.2: Circles

6.2.2 Learning Objectives

- Identify the radius and diameter of a circle.
- Use π or its approximate value in formulas.
- Find the circumference of a circle.
- Find the area of a circle.
- Find the perimeter or area of composite figures involving circles/

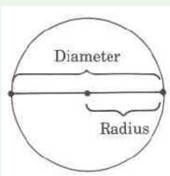
Circumference/Diameter/Radius

Definition: Part of a Circle

The **circumference** of a circle is the distance around the circle.

A **diameter** of a circle is any line segment that passes through the center of the circle and has its endpoints on the circle.

A **radius** of a circle is any line segment having as its endpoints the center of the circle and a point on the circle.
The radius is one half the diameter.



The Number π

The symbol π, read "pi," represents the nonterminating, nonrepeating decimal number 3.14159 … . This number has been computed to millions of decimal places without the appearance of a repeating block of digits.

For computational purposes, π is often approximated as 3.14. We will write $\pi \approx 3.14$ to denote that π is approximately equal to 3.14. The symbol "≈" means "approximately equal to." Use the π key on your calculator when evaluating for the best approximation.

Formulas

To find the circumference of a circle, we need only know its diameter or radius. We then use a formula for computing the circumference of the circle.

Definition: Formula

A **formula** is a rule or method for performing a task. In mathematics, a formula is a rule that directs us in computations.

Formulas are usually composed of letters that represent important, but possibly unknown, quantities.

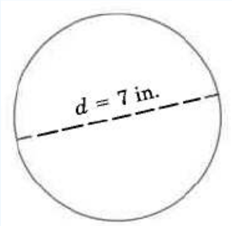
If C, d, and r represent, respectively, the circumference, diameter, and radius of a circle, then the following two formulas give us directions for computing the circumference of the circle.

Circumference Formulas

1. $C = \pi d$
2. $C = 2\pi r$

Example 1

Find the exact circumference of the circle.



Solution

Use the formula $C = \pi d$.

$C = \pi \cdot 7 \text{ in}$

By commutativity of multiplication,

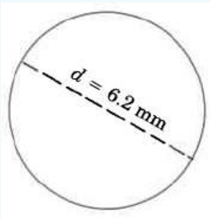
$C = 7 \text{ in} \cdot \pi$

$C = 7\pi \text{ in}$, exactly

This result is exact since π has not been approximated.

Example 2

Find the approximate circumference of the circle.



Solution

Use the formula $C = \pi d$.

$C \approx (3.14)(6.2)$

$C \approx 19.648 \text{ mm}$

This result is approximate since π has been approximated by 3.14.

Example 3

Find the approximate circumference of a circle with radius 18 inches.

Solution

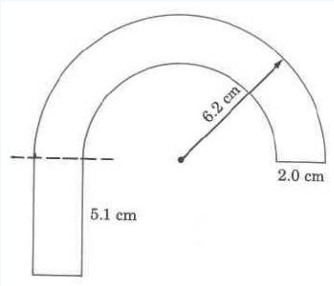
Since we're given that the radius, r, is 18 in, we'll use the formula $C = 2\pi r$.

$C \approx (2)(3.14)(18 \text{ in})$

$C \approx 113.04 \text{ in}$

Example 4

Find the approximate area of the figure.



Solution

We notice that we have two semicircles (half circles).

The larger radius is 6.2 cm.

The smaller radius is $6.2 \text{ cm - } 2.0 \text{ cm} = 4.2 \text{ cm}$.

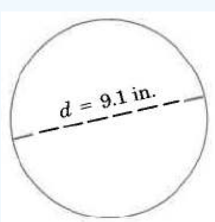
The width of the bottom part of the rectangle is 2.0 cm.

$$\begin{array}{lrll}
\text{Perimeter} & = & 2.0 \text{ cm} & \\
 & & 5.1 \text{ cm} & \\
 & & 2.0 \text{ cm} & \\
 & & 5.1 \text{ cm} & \\
 & & (0.5)\cdot(2)\cdot(3.14)\cdot(6.2 \text{ cm}) & \text{Circumference of outer semicircle.} \\
 & + & \underline{(0.5)\cdot(2)\cdot(3.14)\cdot(4.2 \text{ cm})} & \text{Circumference of inner semicircle.} \\
 & & & 6.2 \text{ cm - } 2.0 \text{ cm} = 4.2 \text{ cm} \\
 & & & \text{The 0.5 appears because we want the} \\
 & & & \text{perimeter of only half a circle.}
\end{array}$$

$$\begin{array}{llr}
\text{Perimeter} & \approx & 2.0 \text{ cm} \\
 & & 5.1 \text{ cm} \\
 & & 2.0 \text{ cm} \\
 & & 5.1 \text{ cm} \\
 & & 19.468 \text{ cm} \\
 & & \underline{+13.188 \text{ cm}} \\
 & & 48.856 \text{ cm}
\end{array}$$

Try It Now 1

Find the exact circumference of the circle.

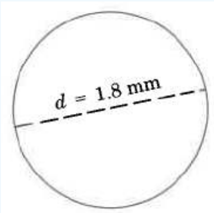


Answer

9.1π in

Try It Now 2

Find the approximate circumference of the circle.



Answer

5.652 mm

Try It Now 3

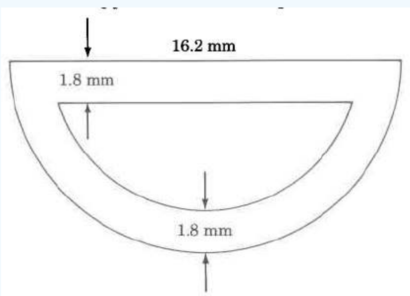
Find the approximate circumference of the circle with radius 20.1 m.

Answer

126.228 m

Try It Now 4

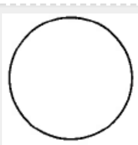
Find the approximate outside perimeter of



Answer

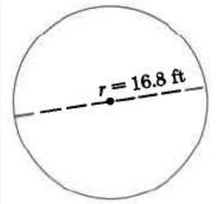
41.634 mm

Area of a Circle

	Figure	Area Formula	Statement
	Circle	$A_c = \pi r^2$	Area of a circle is π times the square of the radius.

Example 5

Find the approximate area of the circle.



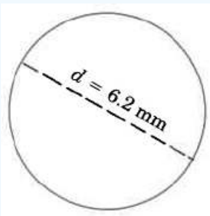
Solution

$$\begin{aligned} A_c &= \pi \cdot r^2 \\ &\approx (3.14) \cdot (16.8 \text{ ft})^2 \\ &\approx (3.14) \cdot (282.24 \text{ sq ft}) \\ &\approx 888.23 \text{ sq ft} \end{aligned}$$

The area of this circle is approximately 886.23 sq ft.

Example 6

Find the approximate area of the circle.



Solution

In this case we are given the diameter instead of the radius. We need to find the radius before we can calculate the area.

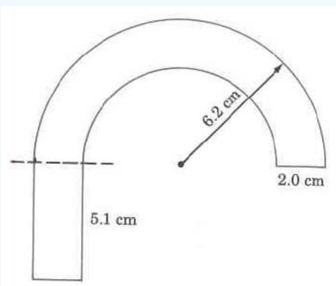
$r = \frac{d}{2} = \frac{6.2}{2} = 3.1\text{mm}$

$$\begin{aligned} A_c &= \pi \cdot r^2 \\ &\approx (3.14) \cdot (6.2 \text{ ft})^2 \\ &\approx (3.14) \cdot (38.44 \text{ sq mm}) \\ &\approx 120.7 \text{ sq mm} \end{aligned}$$

The area of this circle is approximately 120.7 sq mm.

Example 4

Find the approximate area of the figure.



Solution

We notice that we have two semicircles (half circles) and a small rectangle

The larger radius is 6.2 cm.

The smaller radius is $6.2 \text{ cm - } 2.0 \text{ cm} = 4.2 \text{ cm}$.

The width of the bottom part of the rectangle is 2.0 cm.

The area of this shape will be $A = \text{area of the rectangle} + \text{area of the outer semicircle} - \text{area of the inner semicircle}$

Area of the rectangle: $A = 5.1 \cdot 2 = 10.2 \text{ cm}$

Area of the outer semicircle $A = \dfrac{\pi(6.2)^2}{2} = 9.7 \text{ cm}$

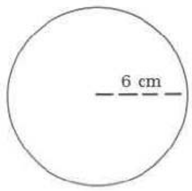
Area of the inner semicircle $A = \dfrac{\pi(4.2)^2}{2} = 6.6 \text{ cm}$

$A = \text{area of the rectangle} + \text{area of the outer semicircle} - \text{area of the inner semicircle} = 10.2\text{cm} + 9.7\text{cm} - 6.6\text{cm} = 13.3\text{cm}$

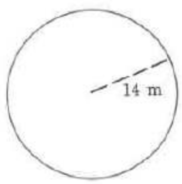
1.2.2.1: Exercises

1. Find the area and circumference of the following figures.

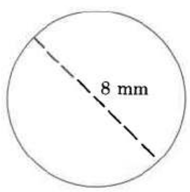
    a.

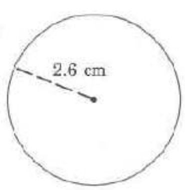
    b.

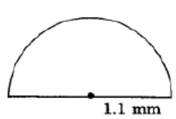
    c.

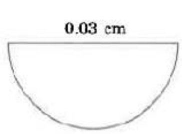
    d.

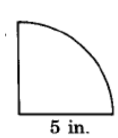
    e.

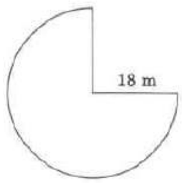
    f.

    g.

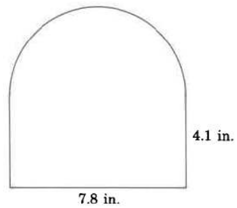
    

    h.

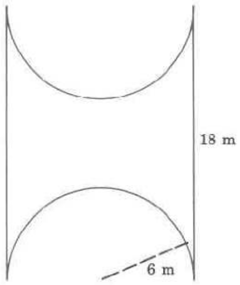


2. Find the area and perimeter of the following composite figures

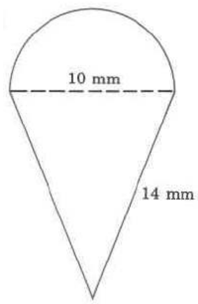
a.



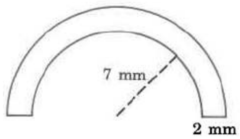
b.



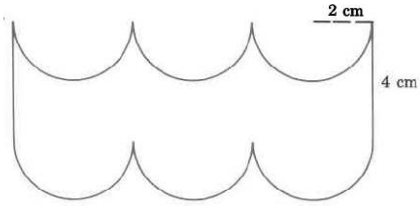
c.



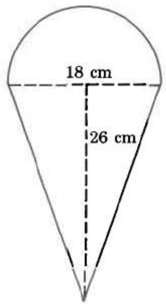
d.



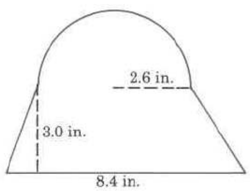
e.



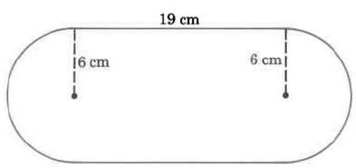
f.



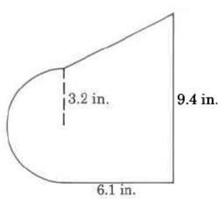
g.



h.



i.



1.2.3: Solid Geometric Figures and Objects

6.2.3 Learning Objectives

- Find the volume of some common geometric objects

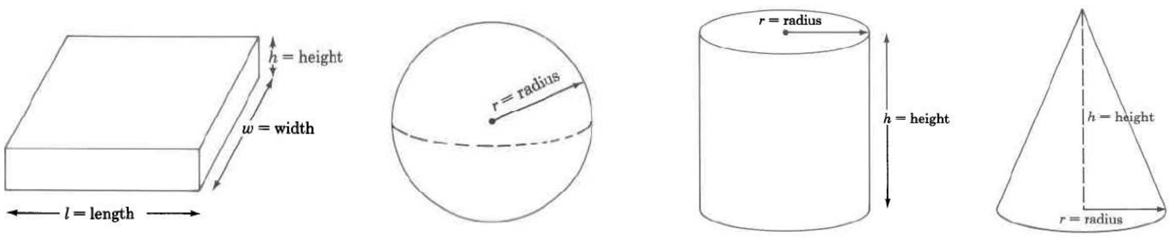
The Meaning and Notation for Volume

The product $(\text{length unit})(\text{length unit})(\text{length unit}) = (\text{length unit})^3$, or cubic length unit (cu length unit), can be interpreted physically as the *volume* of a three-dimensional object.

Definition: Volume

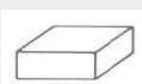
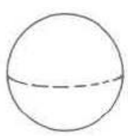
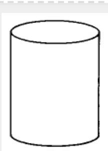
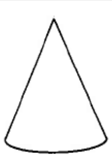
The **volume** of an object is the amount of cubic length units contained in the object.

For example, 4 cu mm means that 4 cubes, 1 mm on each side, would precisely fill some three-dimensional object. (The cubes may have to be cut and rearranged so they match the shape of the object.)



Rectangular solid Sphere Cylinder Cone

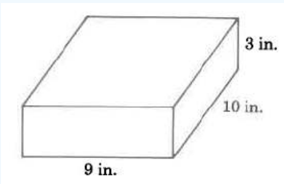
Volume Formulas

	Figure	Volume Formula	Statement
	Rectangular solid	$\begin{aligned} V_R &= l \cdot w \cdot h \\ &= (\text{area of base}) \cdot (\text{height}) \end{aligned}$	The volume of a rectangular solid is the length times the width times the height.
	Sphere	$V_s = \frac{4}{3} \cdot \pi \cdot r^3$	The volume of a sphere is $\frac{4}{3}$ times π times the cube of the radius.
	Cylinder	$\begin{aligned} V_{Cyl} &= \pi \cdot r^2 \cdot h \\ &= (\text{area of base}) \cdot (\text{height}) \end{aligned}$	The volume of a cylinder is π times the square of the radius times the height.
	Cone	$\begin{aligned} V_c &= \frac{1}{3} \cdot \pi \cdot r^2 \cdot h \\ &= (\text{area of base}) \cdot (\text{height}) \end{aligned}$	The volume of a cone is $\frac{1}{3}$ times π times the square of the radius times the height.

Finding Volumes of Some Common Geometric Objects

Example 1

Find the volume of the rectangular solid.



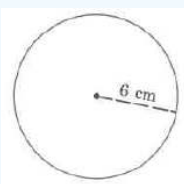
Solution

$$\begin{aligned} V_R &= l \cdot w \cdot h \\ &= 9 \text{ in} \cdot 10 \text{ in.} \cdot 3 \text{ in} \\ &= 270 \text{ cu in} \\ &= 270 \text{ in}^3 \end{aligned}$$

The volume of this rectangular solid is 270 cu in.

Example 2

Find the approximate volume of the sphere.



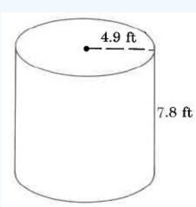
Solution

$$\begin{aligned} V_S &= \frac{4}{3} \cdot \pi \cdot r^3 \\ &\approx \left(\frac{4}{3}\right) \cdot (3.14) \cdot (6 \text{ cm})^3 \\ &\approx \left(\frac{4}{3}\right) \cdot (3.14) \cdot (216 \text{ cu cm}) \\ &\approx 904.32 \text{ cu cm} \end{aligned}$$

The approximate volume of this sphere is 904.32 cu cm, which is often written as 904.32 cm^3.

Example 3

Find the approximate volume of the cylinder.



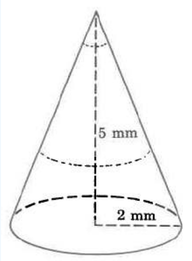
Solution

$$\begin{aligned} V_{Cyl} &= \pi \cdot r^2 \cdot h \\ &\approx (3.14) \cdot (4.9 \text{ ft})^2 \cdot (7.8 \text{ ft}) \\ &\approx (3.14) \cdot (24.01 \text{ sq ft}) \cdot (7.8 \text{ ft}) \\ &\approx (3.14) \cdot (187.278 \text{ cu ft}) \\ &\approx 588.05292 \text{ cu ft} \end{aligned}$$

The volume of this cylinder is approximately 588.05292 cu ft. The volume is approximate because we approximated π.

Example 4

Find the approximate volume of the cone. Round to two decimal places.



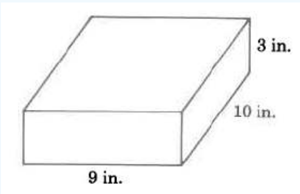
Solution

$$\begin{aligned} V_c &= \frac{1}{3} \cdot \pi \cdot r^2 \cdot h \\ &\approx (\frac{1}{3}) \cdot (3.14) \cdot (2 \text{ mm})^2 \cdot (5 \text{ mm}) \\ &\approx (\frac{1}{3}) \cdot (3.14) \cdot (4 \text{ sq mm}) \cdot (5 \text{ mm}) \\ &\approx (\frac{1}{3}) \cdot (3.14) \cdot (20 \text{ cu mm}) \\ &\approx 20.9\overline{3} \text{ cu mm} \\ &\approx 20.93 \text{ cu mm} \end{aligned}$$

The volume of this cone is approximately 20.93 cu mm. The volume is approximate because we approximated π.

Try It Now 1

Find the volume of the rectangular solid.

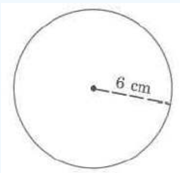


Answer

21 cu in

Try It Now 2

Find the volume of the sphere. Use the π key on your calculator to find the approximate volume.

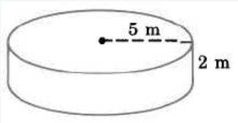


Answer

904.32 cu ft

Try It Now 3

Find the volume of the cylinder. Use the π key on your calculator to find the approximate volume.

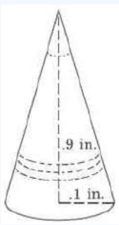


Answer

157 cu m

Try It Now 4

Find the volume of the cone. Use the π key on your calculator to find the approximate volume.



Answer

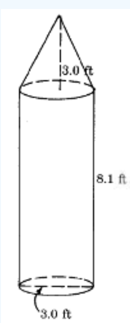
0.00942 cu in

Finding the Volume of Composite Figures

Similar to finding the area of composite figures, we will first determine the separate shapes that the figure is made of. Then use the formulas for the volumes of the individual shapes and add them together.

Example 5

Find the approximate volume of the composite figure. Use the π key on your calculator to find the approximate volume if necessary. Round to two decimal places.



Solution

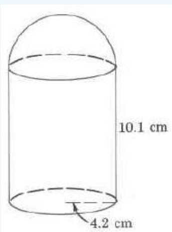
This shape is made up of a cone and a cylinder. We can tell from the base of the cylinder, that the diameter of the cone is 3.0 ft.

$$\begin{aligned}
V &= V_{cone} + V_{cylinder} \\
&= \frac{1}{3} \cdot \pi \cdot r^2 \cdot h + \pi \cdot r^2 \cdot h \\
&\approx (\frac{1}{3}) \cdot \pi \cdot (1.5 \text{ ft})^2 \cdot (3.0 \text{ ft}) + \pi \cdot 1.5 \text{ ft}^2 \cdot 8.1 \text{ ft.} \\
&\approx (\frac{1}{3}) \cdot \pi \cdot (2.25 \text{ ft}^2) \cdot (3.0 \text{ ft}) + \pi \cdot 2.25 \text{ ft}^2 \cdot 8.1 \text{ ft.} \\
&\approx (\frac{1}{3}) \cdot \pi \cdot (6.75 \text{ cu ft}) + \pi \cdot 18.225 \text{ cu ft} \\
&\approx 7.06858 \text{ cu ft} + 57.25553 \text{ cu ft} \\
&\approx 64.32 \text{ cu ft}
\end{aligned}$$

The volume of this figure is approximately 64.32 cu ft. The volume is approximate because we approximated π.

Try It Now 6

Find the volume of the composite figure.



Answer

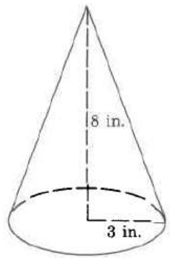
This shape is composed of a hemisphere and a cylinder.

$V \approx 870$ cu cm

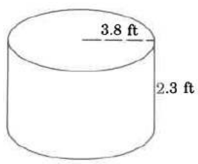
1.2.3.1: Exercises

1. Find the approximate volume

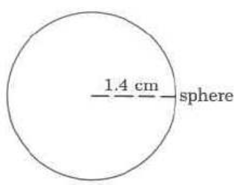
   a.



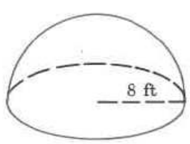
   b.



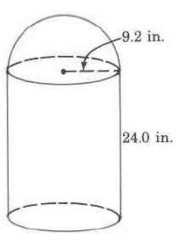
   c.



   d.



   e.



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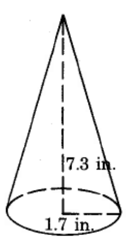
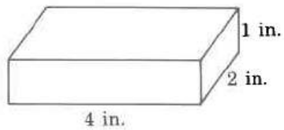


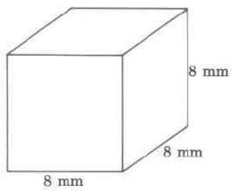
2. Find the exact volume

2. Find the exact volume.

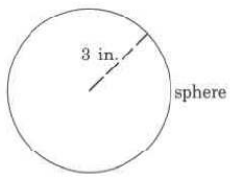
a.



b.



c.



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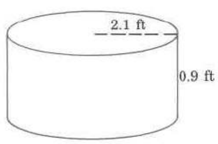

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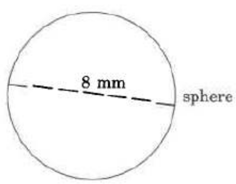

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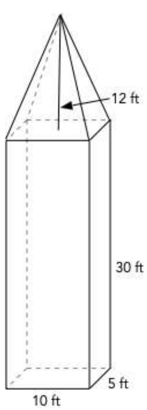
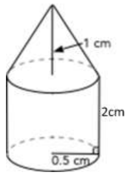
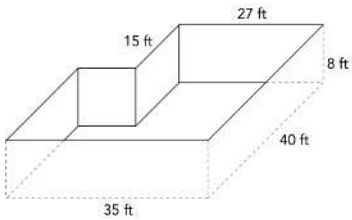


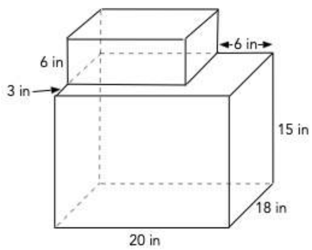
g.



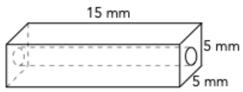
3. Sketch a rectangular prism.
4. How much soil must be removed from a lot to create a basement for a new house?



5. Two boxes in the shape of rectangular prisms are stacked one on top of the other. What is the combined volume of the boxes?



6. A glass bead used to make jewelry is a rectangular prism with square ends. The stringing hole is 1.5 mm in diameter. What is the volume of the glass bead?



1.3: Percents

1.5 Learning Objectives

- Find what percent one number is of another
- Change fractions to percents
- Rewrite decimals as percents
- Find a percent of a number
- Solve applications involving percents

In the 2004 vice-presidential debates, Edwards's claimed that US forces have suffered "90% of the coalition casualties" in Iraq. Cheney disputed this, saying that in fact Iraqi security forces and coalition allies "have taken almost 50 percent" of the casualties[1]. Who is correct? How can we make sense of these numbers?

Percent literally means “per 100,” or “parts per hundred.” When we write 40%, this is equivalent to the fraction $\frac{40}{100}$ or the decimal 0.40. Notice that 80 out of 200 and 10 out of 25 are also 40%, since $\frac{80}{200}=\frac{10}{25}=\frac{40}{100}$.

Example 1

243 people out of 400 state that they like dogs. What percent is this?

Solution

$\frac{243}{400}=0.6075=\frac{60.75}{100}$. This is 60.75%.

Notice that the percent can be found from the equivalent decimal by moving the decimal point two places to the right.

Example 2

Write each as a percent: a) $\frac{1}{4}$ b) 0.02 c) 2.35

Solution

a) $\frac{1}{4}=0.25=25\%$ b) $0.02=2\%$ c) $2.35=235\%$

Percents

If we have a *part* that is some *percent* of a *whole*, then

$\text{percent }=\frac{\text{part}}{\text{whole}}$, or equivalently, $\text{part }=\text{percent}\cdot\text{whole}$

To do the calculations, we write the percent as a decimal.

Notice that the equation translated to English would be "part is percent **of** the whole." So direct multiplication is used.

In some cases the question will ask for the percent **off** the whole, which is different.

If we have a *part* that is some *percent* off a *whole*, then

$\text{part }=(100\%-\text{percent})\cdot\text{whole}$

Since the statement is the percent **off**, to find the correct value for the part, the percent must be subtracted from 100% first.

Example 3

The sales tax in a town is 9.4%. How much tax will you pay on a $140 purchase?

Solution

Here, $140 is the whole, and we want to find 9.4% *of* $140. We start by writing the percent as a decimal by moving the decimal point two places to the left (which is equivalent to dividing by 100). We can then compute:

$\text{tax} = 0.094(140) = \$13.16$ in tax.

Example 4

In the news, you hear "tuition is expected to increase by 7% next year." If tuition this year was $1200 per quarter, what will it be next year?

Solution

The tuition next year will be the current tuition plus an additional 7%, so it will be 107% of this year's tuition:

$\$1200(1.07) = \$1284.$

Alternatively, we could have first calculated 7% of $1200: $\$1200(0.07) = \$84.$

Notice this is *not* the expected tuition for next year (we could only wish). Instead, this is the expected *increase*, so to calculate the expected tuition, we'll need to add this change to the previous year's tuition:

$\$1200 + \$84 = \$1284.$

Try it Now 1

A TV originally priced at $799 is on sale for 30% off. There is then a 9.2% sales tax. Find the price after including the discount and sales tax.

Answer

The sale price is $\$799(1-0.30) = \$799(0.70) = \$559.30$. After tax, the price is $\$559.30(1.092) = \$610.76$.

Example 5

The value of a car dropped from $7400 to $6800 over the last year. What percent decrease is this?

Solution

To compute the percent change, we first need to find the dollar value change: $\$6800 - \$7400 = -\$600$ Often we will take the absolute value of this amount, which is called the **absolute change**: $|-600| = 600$.

Since we are computing the decrease relative to the starting value, we compute this percent out of $\$7400$:

$\frac{600}{7400} = 0.081 = 8.1\%$ decrease. This is called a **relative change**.

Absolute and Relative Change

Given two quantities,

Absolute change = $|\text{ending quantity} - \text{starting quantity}|$

Relative change: $\frac{\text{absolute change}}{\text{starting quantity}}$

Absolute change has the same units as the original quantity.

Relative change gives a percent change.

The starting quantity is called the **base** of the percent change.

The base of a percent is very important. For example, while Nixon was president, it was argued that marijuana was a “gateway” drug, claiming that 80% of marijuana smokers went on to use harder drugs like cocaine. The problem is, this isn’t true. The true claim is that 80% of harder drug users first smoked marijuana. The difference is one of base: 80% of marijuana smokers using hard drugs, vs. 80% of hard drug users having smoked marijuana. These numbers are not equivalent. As it turns out, only one in 2,400 marijuana users actually go on to use harder drugs[2].

Example 6

There are about 75 QFC supermarkets in the U.S. Albertsons has about 215 stores. Compare the size of the two companies.

Solution

When we make comparisons, we must ask first whether an absolute or relative comparison. The absolute difference is $215–75 = 140$. From this, we could say “Albertsons has 140 more stores than QFC.” However, if you wrote this in an article or paper, that number does not mean much. The relative difference may be more meaningful. There are two different relative changes we could calculate, depending on which store we use as the base:

Using QFC as the base, $\frac{140}{75} = 1.867$.

This tells us Albertsons is 186.7% larger than QFC.

Using Albertsons as the base, $\frac{140}{215} = 0.651$.

This tells us QFC is 65.1% smaller than Albertsons.

Notice both of these are showing percent *differences*.

We could also calculate the size of Albertsons relative to QFC: 215/75, which means Albertsons is 2.867 or 286.7% times the size of QFC. Likewise, we could calculate the size of QFC relative to Albertsons: 75/215, which tells us that QFC is 34.9% of the size of Albertsons.

Example 7

Suppose a stock drops in value by 60% one week, then increases in value the next week by 75%. Is the value higher or lower than where it started?

Solution

To answer this question, suppose the value started at \$100. After one week, the value dropped by 60%:

$\$100 - \$100(0.60) = \$100 - \$60 = \$40.$

In the next week, notice that base of the percent has changed to the new value, \$40. Computing the 75% increase:

$\$40 + \$40(0.75) = \$40 + \$30 = \$70.$

In the end, the stock is still \$30 lower, or $\frac{\$30}{\$100} = 30\%$ lower, valued than it started.

Try it Now 2

The U.S. federal debt at the end of 2001 was \$5.77 trillion, and grew to \$6.20 trillion by the end of 2002. At the end of 2005 it was \$7.91 trillion, and grew to \$8.45 trillion by the end of 2006[3]. Calculate the absolute and relative increase for 2001-2002 and 2005-2006. Which year saw a larger increase in federal debt?

Answer

2001-2002: Absolute change: \$0.43 trillion. Relative change: 7.45%

2005-2006: Absolute change: \$0.54 trillion. Relative change: 6.83%

2005-2006 saw a larger absolute increase, but a smaller relative increase.

Example 8

A Seattle Times article on high school graduation rates reported "The number of schools graduating 60 percent or fewer students in four years – sometimes referred to as "dropout factories" – decreased by 17 during that time period. The number of kids attending schools with such low graduation rates was cut in half."

a) Is the "decrease by 17" number a useful comparison?

b) Considering the last sentence, can we conclude that the number of "dropout factories" was originally 34?

Solution

a) This number is hard to evaluate, since we have no basis for judging whether this is a large or small change relative to the number of such schools. If the number of "dropout factories" dropped from 20 to 3, that would be a very significant change, but if the number dropped from 217 to 200, that would be less of an improvement.

b) The last sentence provides relative change which helps put the first sentence in perspective. We can estimate that the number of "dropout factories" was probably previously around 34. However, it's possible that students simply moved schools rather than the school improving, so that estimate might not be fully accurate.

Example 9

In the 2004 vice-presidential debates, Edwards's claimed that US forces have suffered "90% of the coalition casualties" in Iraq. Cheney disputed this, saying that in fact Iraqi security forces and coalition allies "have taken almost 50 percent" of the casualties. Who is correct?

Solution

Without more information, it is hard for us to judge who is correct, but we can easily conclude that these two percents are talking about different things, so one does not necessarily contradict the other. Edward's claim was a percent with coalition forces as the base of the percent, while Cheney's claim was a percent with both coalition and Iraqi security forces as the base of the percent. It turns out both statistics are in fact fairly accurate.

Try it Now 3

In the 2012 presidential elections, one candidate argued that "the president's plan will cut $716 billion from Medicare, leading to fewer services for seniors," while the other candidate rebuts that "our plan does not cut current spending and actually expands benefits for seniors, while implementing cost saving measures." Are these claims in conflict, in agreement, or not comparable because they're talking about different things?

Answer

Without more information, it is hard to judge these arguments. This is compounded by the complexity of Medicare. As it turns out, the $716 billion is not a cut in current spending, but a cut in future increases in spending, largely reducing future growth in health care payments. In this case, at least the numerical claims in both statements could be considered at least partially true. Here is one source of more information if you're interested: http://factcheck.org/2012/08/a-campaign-full-of-mediscare/

We'll wrap up our review of percents with a few cautions.

First, when talking about a change in quantities that are already measured in percents, we have to be careful in how we describe the change.

Example 10

A politician's support increases from 40% of voters to 50% of voters. Describe the change.

Solution

We could describe this using an absolute change: $|50\% - 40\%| = 10\%$. Notice that since the original quantities were percents, this change also has the units of percent. In this case, it is best to describe this as an increase of 10 **percentage points**.

In contrast, we could compute the percent change: $\frac{10\%}{40\%} = 0.25 = 25\%$ increase. This is the relative change, and we'd say the politician's support has increased by 25%.

Second, pay attention to shifting base values.

Example 11

An employer asks his employee to take a temporary 10% pay cut for 6 months because the company has hit some hard times. At the end of the 6 months the employer will give the employee a 10% raise. Will the employee make more, less, or the same amount as before the pay cut? Is that fair?

Solution

Suppose the employee makes $4500 per month. The employer is proposing to take 10% off the current salary.

$\text{part} = \$4500\,(1-0.10) = \$4500\,(0.90) = \$4050$

After 6 months pass the employer promises to give a 10% raise on the salary. The employee will retain 100% of what they have plus 10%.

$\text{part} = \$4050\,(1+0.10) = \$4050\,(1.10) = \$4455$

Since the value that the percent is based on has changed the employee would not be restored to their original salary in this scenario.

Last, beware of averaging percents.

Example 12

A basketball player scores on 40% of 2-point field goal attempts, and on 30% of 3-point of field goal attempts. Find the player's overall field goal percentage.

Solution

It is very tempting to average these values, and claim the overall average is 35%, but this is likely not correct, since most players make many more 2-point attempts than 3-point attempts. We don't actually have enough information to answer the question. Suppose the player attempted 200 2-point field goals and 100 3-point field goals. Then they made 200(0.40) = 80 2-point shots and $100(0.30) = 30$ 3-point shots. Overall, they made 110 shots out of 300, for a $\frac{110}{300} = 0.367 = 36.7\%$ overall field goal percentage.

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[1] www.factcheck.org/cheney_edwards_mangle_facts.html

[2] http://tvtropes.org/pmwiki/pmwiki.php/Main/LiesDamnedLiesAndStatistics

[3] www.whitehouse.gov/sites/defa...s/hist07z1.xls

1.3.1: Exercises

Section 1.5.1

1. Write each percent as a fraction.
   a. 87%
   b. 40%
   c. 110%
2. Write as a percent.
   a. 5858
   b. 4545
3. Write each percent as a decimal.
   a. 27%
   b. 8.328%
   c. 125%
   d. 0.5%
4. Change each decimal to a percent.
   a. 0.34
   b. 2.5
   c. 0.0045
5. 50 is 40% of what number?
6. A shirt that usually sells for $25 is on sale for $21. What is the percent decrease in the price of the shirt?
7. A car salesman has a commission rate of 12% on the cars sold. If the commission on one sale was $1836, what was the purchase price of the car?
8. You purchase an item originally priced at $240 that is discounted 30%. What is the sale price of the item?
9. You pay sales tax of 9.5% on your purchase. What is the total cost for your purchase with the discount and the sales tax?
10. The sales clerk accidentally added the sales tax based on the original price of your purchase before giving you a 30% discount. Would this error cause the item to cost you more or less or the same as what you figured in part b)?
11. A store advertises a 40% off clearance sale. When you arrived at the store, you received a coupon for an additional 10% off. The coupon is applied after the 40% discount.
    a. What is the total discount if the item is normally priced at $30?
    b. What is the total discount percent?
    c. Why do you think the store uses two discount percents instead of one? Explain.
12. Consider the following.
    a. A parenting blog states that 12-year-old girls typically weigh 200% of what 6-year-old girls weigh? Does that mean a girl's weight doubles between ages 6 and 12?
    b. Could it be true that an 8-year-old girl weighs 30% more than a 6-year- old girl?
13. Out of 230 racers who started the marathon, 212 completed the race, 14 gave up, and 4 were disqualified. What percentage did not complete the marathon?
14. Patrick left an $8 tip on a $50 restaurant bill. What percent tip is that?
15. Ireland has a 23% VAT (value-added tax, similar to a sales tax). How much will the VAT be on a purchase of a €250 item?
16. Employees in 2012 paid 4.2% of their gross wages towards social security (FICA tax), while employers paid another 6.2%.
    a. How much will someone earning $45,000 a year pay towards social security out of their gross wages?
    b. What is the total amount contributed towards the worker's social security?
17. A project on Kickstarter.com was aiming to raise $15,000 for a precision coffee press. They ended up with 714 supporters, raising 557% of their goal. How much did they raise?

18. Another project on Kickstarter for an iPad stylus raised 1,253% of their goal, raising a total of $313,490 from 7,511 supporters. What was their original goal?
19. The population of a town increased from 3,250 in 2008 to 4,300 in 2010. Find the absolute and relative (percent) increase.
20. The number of CDs sold in 2010 was 114 million, down from 147 million the previous year[1]. Find the absolute and relative (percent) decrease.
21. A company wants to decrease their energy use by at least 15%. If their next bill is $1,700 a month, were they successful? Why or why not?
22. A store is hoping an advertising campaign will increase their number of customers by at least 30%. They currently have about 80 customers a day.
    a. How many customers will they have if their campaign is successful?
    b. If they increase to 120 customers a day, were they successful? Why or why not?
23. An article reports “attendance dropped 6% this year, to 300.” What was the attendance before the drop?
24. An article reports “sales have grown by 30% this year, to $200 million.” What were sales before the growth?
25. The Walden University had 47,456 students in 2010, while Kaplan University had 77,966 students. Complete the following statements:
    a. Kaplan’s enrollment was ___% larger than Walden’s.
    b. Walden’s enrollment was ___% smaller than Kaplan’s.
    c. Walden’s enrollment was ___% of Kaplan’s.
26. In the 2012 Olympics, Usain Bolt ran the 100m dash in 9.63 seconds. Jim Hines won the 1968 Olympic gold with a time of 9.95 seconds.
    a. Bolt’s time was ___% faster than Hines’.
    b. Hines’ time was ___% slower than Bolt’s.
    c. Hines’ time was ___% of Bolt’s.
27. A store has clearance items that have been marked down by 60%. They are having a sale, advertising an additional 30% off clearance items. What percent of the original price do you end up paying?
28. Which is better: having a stock that goes up 30% on Monday than drops 30% on Tuesday, or a stock that drops 30% on Monday and goes up 30% on Tuesday? In each case, what is the net percent gain or loss?
29. Are these two claims equivalent, in conflict, or not comparable because they’re talking about different things?
    a. “16.3% of Americans are without health insurance”[2]
    b. “only 55.9% of adults receive employer provided health insurance”[3]
30. Are these two claims equivalent, in conflict, or not comparable because they’re talking about different things?
    a. “We mark up the wholesale price by 33% to come up with the retail price”
    b. “The store has a 25% profit margin”
31. Are these two claims equivalent, in conflict, or not comparable because they’re talking about different things?
    a. “Every year since 1950, the number of American children gunned down has doubled.”
    b. “The number of child gunshot deaths has doubled from 1950 to 1994.”
32. Are these two claims equivalent, in conflict, or not comparable because they’re talking about different things?[4]
    a. “75 percent of the federal health care law’s taxes would be paid by those earning less than $120,000 a year”
    b. “75 percent of those who would pay the penalty [health care law’s taxes] for not having insurance in 2016 would earn under $120,000”
33. Are these two claims equivalent, in conflict, or not comparable because they’re talking about different things?
    a. “The school levy is only a 0.1% increase of the property tax rate.”
    b. “This new levy is a 12% tax hike, raising our total rate to $9.33 per $1000 of value.”
34. A high school currently has a 30% dropout rate. They’ve been tasked to decrease that rate by 20%. Find the equivalent percentage point drop.
35. A politician’s support grew from 42% by 3 percentage points to 45%. What percent (relative) change is this?

36. Marcy has a 70% average in her class going into the final exam. She says "I need to get a 100% on this final so I can raise my score to 85%." Is she correct?
37. In a survey of households in the United States 9.4% of the households had 2 dogs. If there are about 117 million households in the U.S., about how many households have 2 dogs?
38. A typical 180-lb man is 111 lbs of water and 27 pounds of fat. What percent of the weight is water? What percent of the weight is fat?
39. Between the last two censuses, a town's population grew from 3200 to 3650. What is the percent increase in the population?
40. In 1901 life expectancy at birth was 49 years. At the end of the century life expectancy at birth was 77 years. What is the increase in life expectancy during the 20^{th} century? Complete this sentence: Life expectancy in 2000 was ______ percent of life expectancy in 1901.
41. If you bought a stock for $250 and later sold it for $1250, by what percent did your stock increase in value? What percent of the purchase price was the sales price?
42. At work, you get a 2% cost-of-living adjustment (COLA). *True of False*: Over a 5-year period, your pay will increase by 10%. If this is false, by what percent will your pay increase over the 5-year period?
43. All merchandise at a store is on sale at 30% off. You have a coupon that gives you 10% off the purchase price. What will you pay for a jacket originally priced at $80? What percent off of the original price is your final price?
44. Your first three exam scores in a class are 74%, 89%, and 83%. What percent do you need to receive on the last exam to have an 85% overall in the class?
45. What is the markup percent for an item that has a wholesale price of $1.13 and a retail price of $1.69?
46. If you buy a refrigerator priced at $1325 and the sales tax rate is 6.5%, how much sales tax would you pay on the purchase? What would the total price of the refrigerator be?
47. Find the sales tax rate in a region where you paid $62.65 tax on a big screen TV priced at $895.
48. Why is this conclusion incorrect? "In a survey of a class, 70% want pizza for their class party and 20% want cake. Therefore, 90% want pizza or cake for their party."
49. Why is this reasoning incorrect? "You received a score of 80% on five tests in a class and 90% on one test. Therefore, your average test score is 85% for the six tests."
50. Suppose you have one quart of water/juice mix that is 50% juice, and you add 2 quarts of juice. What percent juice is the final mix?
51. During the 2004-2005 NBA, Kobe Bryant attempted 387 3-point shots and made 131 of them. Express his performance as a fraction, a decimal and a percent.
52. The following table describes the students enrolled at a small two-year college.

class	full-time	part-time
freshman	262	116
sophomore	214	239

a. What percent of the freshmen are full-time students?
b. What percent of the students are enrolled part-time?
c. What percent of the part-time students are sophomores?

1.3.2: Proportions and Rates

1.5.1 Learning Objectives

- Compare 2 quantities as a rate or ratio
- Solve a proportion
- Solve applications of proportions
- Solve applications using dimensional analysis

If you wanted to power the city of Seattle using wind power, how many windmills would you need to install? Questions like these can be answered using rates and proportions.

Rates

A **ratio** compares two quantities with the same units by division.

A **rate** is the ratio (fraction) of two quantities with different units.

A **unit rate** is a rate with a numerator or a denominator of one.

Example 1

Your car can travel 300 miles on a tank of 15 gallons. Express this as a rate and as a unit rate.

Solution

Expressed as a rate, $\frac{300 \text{ miles}}{15 \text{ gallons}}$. We can divide to find a unit rate: $\frac{20 \text{ miles}}{1 \text{ gallon}}$, which we could also write as $20\frac{\text{miles}}{\text{gallon}}$, or just 20 miles per gallon.

Proportion Equation

A **proportion** equation is an equation showing the equivalence of two rates or ratios.

Example 2

Solve the proportion $\frac{5}{3}=\frac{x}{6}$ for the unknown value x.

Solution

This proportion is asking us to find a fraction with denominator 6 that is equivalent to the fraction $\frac{5}{3}$. We can solve this by multiplying both sides of the equation by 6, giving $x=\frac{5}{3} \cdot 6=10$.

You could also note that the denominator of the right-hand ratio is twice as large as the denominator of the left-hand ratio, so the numerator on the right must also be twice as large as the numerator on the left.

Example 3

A map scale indicates that $\frac{1}{2}$ inch on the map corresponds with 3 real miles. How many miles apart are two cities that are $2\frac{1}{4}$ inches apart on the map?

Solution

We can set up a proportion by setting equal two $\frac{\text{map inches}}{\text{real miles}}$ rates, and introducing a variable, x, to represent the unknown quantity – the mile distance between the cities.

$$\frac{\frac{1}{2}\text{ map inch}}{3\text{ miles}} = \frac{2\frac{1}{4}\text{ map inches}}{x\text{ miles}} \quad \text{Multiply both sides by } x \text{ and rewriting the mixed number}$$

$$\frac{\frac{1}{2}}{3} \cdot x = \frac{9}{4} \quad \text{Multiply both sides by 3}$$

$$\frac{1}{2}x = \frac{27}{4} \quad \text{Multiply both sides by 2 (or divide by } \frac{1}{2}\text{)}$$

$$x = \frac{27}{2} = 13\frac{1}{2}\text{ miles}$$

Many proportion problems can also be solved using **dimensional analysis**, the process of multiplying a quantity by rates to change the units.

Example 4

Your car can travel 300 miles on a tank of 15 gallons. How far can it travel on 40 gallons?

Solution

We could certainly answer this question using a proportion: $\frac{300\text{ miles}}{15\text{ gallons}} = \frac{x\text{ miles}}{40\text{ gallons}}$.

However, we earlier found that 300 miles on 15 gallons gives a rate of 20 miles per gallon. If we multiply the given 40 gallon quantity by this rate, the *gallons* unit "cancels" and we're left with a number of miles:

$$40\text{gallons} \cdot \frac{20\text{ miles}}{\text{gallon}} = \frac{40\ \cancel{\text{gallons}}}{1} \cdot \frac{20\text{ miles}}{\cancel{\text{gallon}}} = 800\text{ miles}$$

Notice if instead we were asked "how many gallons are needed to drive 50 miles?" we could answer this question by inverting the 20 mile per gallon rate so that the *miles* unit cancels and we're left with gallons:

$$50\text{miles} \cdot \frac{1\text{ gallon}}{20\text{ miles}} = \frac{50\ \cancel{\text{miles}}}{1} \cdot \frac{1\text{ gallon}}{20\ \cancel{\text{miles}}} = \frac{50\text{ gallons}}{20} = 2.5\text{ gallons}$$

Dimensional analysis can also be used to do unit conversions. Here are some unit conversions for reference.

Unit Conversions

Length

$$\begin{array}{ll} 1\text{ foot (ft)} = 12\text{ inches (in)} & 1\text{ yard (yd)} = 3\text{ feet (ft)} \\ 1\text{ mile} = 5,280\text{ feet} & \\ 1000\text{ millimeters } mm = 1\text{ meter (m)} & 100\text{ centimeters (cm)} = 1\text{ meter} \\ 1000\text{ meters (m)} = 1\text{ kilometer (km)} & 2.54\text{ centimeters (cm)} = 1\text{ inch} \end{array}$$

Weight and Mass

$$\begin{array}{ll} 1\text{ pound (lb)} = 16\text{ ounces (oz)} & 1\text{ ton} = 2000\text{ pounds} \\ 1000\text{ milligrams (mg)} = 1\text{ gram (g)} & 1000\text{ grams} = 1\text{ kilogram (kg)} \\ 1\text{ kilogram} = 2.2\text{ pounds (on earth)} & \end{array}$$

Capacity

$$\begin{array}{ll} 1\text{ cup} = 8\text{ fluid ounces (fl oz)}^* & 1\text{ pint} = 2\text{ cups} \\ 1\text{ quart} = 2\text{ pints} = 4\text{ cups} & 1\text{ gallon} = 4\text{ quarts} = 16\text{ cups} \\ 1000\text{ milliliters (ml)} = 1\text{ liter (L)} & \end{array}$$

*Fluid ounces are a capacity measurement for liquids. 1 fluid ounce ≈ 1 ounce (weight) for water only.

Example 5

A bicycle is traveling at 15 miles per hour. How many feet will it cover in 20 seconds?

Solution

To answer this question, we need to convert 20 seconds into hours. If we know the speed of the bicycle in feet per second, this question would be simpler. Since we don't, we will need to do additional unit conversions. We will need to know that 5280 ft = 1 mile. We might start by converting the 20 seconds into hours:

$$20 \cancel{\text{ seconds}} \cdot \frac{1 \cancel{\text{ minute}}}{60 \cancel{\text{ seconds}}} \cdot \frac{1 \text{ hour}}{60 \cancel{\text{ minutes}}} = \frac{1}{180} \text{ hour} \quad \text{Now we can multiply by the 15 miles/hr}$$

$$\frac{1}{180} \cancel{\text{ hour}} \cdot \frac{15 \text{ miles}}{\cancel{1 \text{ hour}}} = \frac{1}{12} \text{ mile} \quad \text{Now we can convert to feet}$$

$$\frac{1}{12} \cancel{\text{ mile}} \cdot \frac{5280 \text{ feet}}{1 \cancel{\text{ mile}}} = 440 \text{ feet}$$

We could have also done this entire calculation in one long set of products:

$$20 \cancel{\text{ seconds}} \cdot \frac{1 \cancel{\text{ minute}}}{60 \cancel{\text{ seconds}}} \cdot \frac{1 \cancel{\text{ hour}}}{60 \cancel{\text{ minutes}}} \cdot \frac{15 \cancel{\text{ miles}}}{1 \cancel{\text{ hour}}} \cdot \frac{5280 \text{ feet}}{1 \cancel{\text{ mile}}} = 440 \text{ feet}$$

Try it Now 1

A 1000 foot spool of bare 12-gauge copper wire weighs 19.8 pounds. How much will 18 inches of the wire weigh, in ounces?

Answer

$$18 \text{ inches} \cdot \frac{1 \text{ foot}}{12 \text{ inches}} \cdot \frac{19.8 \text{ pounds}}{1000 \text{ feet}} \cdot \frac{16 \text{ ounces}}{1 \text{ pound}} \approx 0.475 \text{ ounces}$$

Notice that with the miles per gallon example, if we double the miles driven, we double the gas used. Likewise, with the map distance example, if the map distance doubles, the real-life distance doubles. This is a key feature of proportional relationships, and one we must confirm before assuming two things are related proportionally.

Example 6

Suppose you're tiling the floor of a 10 ft by 10 ft room, and find that 100 tiles will be needed. How many tiles will be needed to tile the floor of a 20 ft by 20 ft room?

Solution

In this case, while the width the room has doubled, the area has quadrupled. Since the number of tiles needed corresponds with the area of the floor, not the width, 400 tiles will be needed. We could find this using a proportion based on the areas of the rooms:

$$\frac{100 \text{ tiles}}{100 \text{ft}^2} = \frac{n \text{ tiles}}{400 \text{ft}^2}$$

Other quantities just don't scale proportionally at all.

Example 7

Suppose a small company spends $1000 on an advertising campaign, and gains 100 new customers from it. How many new customers should they expect if they spend $10,000?

Solution

While it is tempting to say that they will gain 1000 new customers, it is likely that additional advertising will be less effective than the initial advertising. For example, if the company is a hot tub store, there are likely only a fixed number of people interested in buying a hot tub, so there might not even be 1000 people in the town who would be potential customers.

Sometimes when working with rates, proportions, and percents, the process can be made more challenging by the magnitude of the numbers involved. Sometimes, large numbers are just difficult to comprehend.

Example 8

Compare the 2010 U.S. military budget of $683.7 billion to other quantities.

Here we have a very large number, about $683,700,000,000 written out. Of course, imagining a billion dollars is very difficult, so it can help to compare it to other quantities.

If that amount of money was used to pay the salaries of the 1.4 million Walmart employees in the U.S., each would earn over $488,000.

There are about 300 million people in the U.S. The military budget is about $2,200 per person.

If you were to put $683.7 billion in $100 bills, and count out 1 per second, it would take 216 years to finish counting it.

Example 9

Compare the electricity consumption per capita in China to the rate in Japan.

To address this question, we will first need data. From the CIA[1] website we can find the electricity consumption in 2011 for China was 4,693,000,000,000 KWH (kilowatt-hours), or 4.693 trillion KWH, while the consumption for Japan was 859,700,000,000, or 859.7 billion KWH. To find the rate per capita (per person), we will also need the population of the two countries. From the World Bank[2], we can find the population of China is 1,344,130,000, or 1.344 billion, and the population of Japan is 127,817,277, or 127.8 million.

Solution

Computing the consumption per capita for each country:

China: $\frac{4,693,000,000,000 \mathrm{KWH}}{1,344,130,000 \text{ people}} \approx 3491.5$ KWH per person

Japan: $\frac{859,700,000,000 \mathrm{KWH}}{127,817,277 \text{ people}} \approx 6726$ KWH per person

While China uses more than 5 times the electricity of Japan overall, because the population of Japan is so much smaller, it turns out Japan uses almost twice the electricity per person compared to China.

[1] www.cia.gov/library/publicat.../2042rank.html

[2] http://data.worldbank.org/indicator/SP.POP.TOT

1.3.3: Exercises

Section 1.5.2 Exercises

1. Solve:
   a. $\frac{2}{5}=\frac{6}{x}$
   b. $\frac{2x}{9}=\frac{3}{10}$
   c. $\frac{x}{12}=\frac{2}{3}$
   d. $\frac{4}{9x}=\frac{10}{13}$
   e. $\frac{n}{5}=\frac{16}{20}$.
2. Find a unit rate: You bought 10 pounds of potatoes for $4.
3. Find a unit rate: Joel ran 1500 meters in 4 minutes, 45 seconds.
4. Which is the more economical purchase: 32 oz of detergent for $6.29 or 48 oz of the same detergent for $8.29?
5. Solve the proportion.
   a. $\frac{3}{25}=\frac{x}{3000}$
   b. $\frac{5}{8}=\frac{200}{x}$
6. A student scored 32 out of 40 on a practice exam and 12 out of 15 on the actual exam. Did the student do better on the practice exam or on the actual exam?
7. A soft drink is sold in a six-pack of plastic bottles holding 101.4 fl oz for $2.69. The same soft drink can be purchased in a 12-pack of cans containing 144 fl oz for $3.79. Which is the better buy?

CHAPTER OVERVIEW

2: Module 2 - Finances

2.1: Taxes

2.2 Learning Objectives

- Distinguish among flat tax, progressive tax and regressive tax
- Determine a flat tax amount

Governments collect taxes to pay for the services they provide. In the United States, federal income taxes help fund the military, the environmental protection agency, and thousands of other programs. Property taxes help fund schools. Gasoline taxes help pay for road improvements. While very few people enjoy paying taxes, they are necessary to pay for the services we all depend upon.

Taxes can be computed in a variety of ways, but are typically computed as a percentage of a sale, of one's income, or of one's assets.

Example 1

The sales tax rate in a city is 9.3%. How much sales tax will you pay on a $140 purchase?

Solution

The sales tax will be 9.3% of $140. To compute this, we multiply $140 by the percent written as a decimal: $\$140(0.093) = \$13.02$.

When taxes are not given as a fixed percentage rate, sometimes it is necessary to calculate the **effective rate.**

Effective rate

The effective tax rate is the equivalent percent rate of the tax paid out of the dollar amount the tax is based on.

Example 2

Joan paid $3,200 in property taxes on her house valued at $215,000 last year. What is the effective tax rate?

Solution

We can compute the equivalent percentage: $\frac{3200}{215000} = 0.01488$, or about 1.49% effective rate.

Taxes are often referred to as progressive, regressive, or flat.

Tax categories

A **flat tax**, or proportional tax, charges a constant percentage rate.

A **progressive tax** increases the percent rate as the base amount increases.

A **regressive tax** decreases the percent rate as the base amount increases.

Example 3

The United States federal income tax on earned wages is an example of a progressive tax. People with a higher wage income pay a higher percent tax on their income.

Solution

For a single person in 2011, adjusted gross income (income after deductions) under $8,500 was taxed at 10%. Income over $8,500 but under $34,500 was taxed at 15%.

A person earning $10,000 would pay 10% on the portion of their income under $8,500, and 15% on the income over $8,500, so they'd pay:

$$\begin{array}{ll} 8500(0.10)=850 & 10\% \text{ of } 8500 \\ 1500(0.15)=225 & 15\% \text{ of the remaining } \$1500 \text{ of income} \\ \text{Total tax: } =\$1075 & \end{array}$$

The effective tax rate paid is $\frac{1075}{10000}=10.75\%$

A person earning $30,000 would also pay 10% on the portion of their income under $8,500, and 15% on the income over $8,500, so they'd pay:

$$\begin{array}{ll} 8500(0.10)=850 & 10\% \text{ of } 8500 \\ 21500(0.15)=3225 & 15\% \text{ of the remaining } \$21500 \text{ of income} \\ \text{Total tax: } =\$4075 & \end{array}$$

The effective tax rate paid is $\frac{4075}{30000}=13.58\%$.

Notice that the effective rate has increased with income, showing this is a progressive tax.

Example 4

A gasoline tax is a flat tax when considered in terms of consumption, a tax of, say, $0.30 per gallon is proportional to the amount of gasoline purchased. Someone buying 10 gallons of gas at $4 a gallon would pay $3 in tax, which is $\frac{\$3}{\$40}=7.5\%$. Someone buying 30 gallons of gas at $4 a gallon would pay $9 in tax, which is $\frac{\$9}{\$120}=7.5\%$, the same effective rate.

Solution

However, in terms of income, a gasoline tax is often considered a regressive tax. It is likely that someone earning $30,000 a year and someone earning $60,000 a year will drive about the same amount. If both pay $60 in gasoline taxes over a year, the person earning $30,000 has paid 0.2% of their income, while the person earning $60,000 has paid 0.1% of their income in gas taxes.

Try it Now 1

A sales tax is a fixed percentage tax on a person's purchases. Is this a flat, progressive, or regressive tax?

Answer

While sales tax is a flat percentage rate, it is often considered a regressive tax for the same reasons as the gasoline tax.

2.1.1: Income Taxation

2.2.1 Learning Objectives

- Compare flat tax, modified flat tax, and progressive tax

Many people have proposed various revisions to the income tax collection in the United States. Some, for example, have claimed that a flat tax would be fairer. Others call for revisions to how different types of income are taxed, since currently investment income is taxed at a different rate than wage income.

The following two projects will allow you to explore some of these ideas and draw your own conclusions.

Project 1: Flat tax, Modified Flat Tax, and Progressive Tax.

Imagine the country is made up of 100 households. The federal government needs to collect $800,000 in income taxes to be able to function. The population consists of 6 groups:

Group A: 20 households that earn $12,000 each

Group B: 20 households that earn $29,000 each

Group C: 20 households that earn $50,000 each

Group D: 20 households that earn $79,000 each

Group E: 15 households that earn $129,000 each

Group F: 5 households that earn $295,000 each

This scenario is roughly proportional to the actual United States population and tax needs. We are going to determine new income tax rates.

The first proposal we'll consider is a flat tax – one where every income group is taxed at the same percentage tax rate.

1) Determine the total income for the population (all 100 people together)

2) Determine what flat tax rate would be necessary to collect enough money.

The second proposal we'll consider is a modified flat-tax plan, where everyone only pays taxes on any income over $20,000. So, everyone in group A will pay no taxes. Everyone in group B will pay taxes only on $9,000.

3) Determine the total *taxable* income for the whole population

4) Determine what flat tax rate would be necessary to collect enough money in this modified system

5) Complete this table for both the plans

		Flat Tax Plan		Modified Flat Tax Plan	
Group	**Income per household**	**Income tax per household**	**Income after taxes**	**Income tax per household**	**Income after taxes**
A	$12,000				
B	$29,000				
C	$50,000				
D	$79,000				
E	$129,000				
F	$295,000				

The third proposal we'll consider is a progressive tax, where lower income groups are taxed at a lower percent rate, and higher income groups are taxed at a higher percent rate. For simplicity, we're going to assume that a household is taxed at the same rate on *all* their income.

6) Set progressive tax rates for each income group to bring in enough money. There is no one right answer here – just make sure you bring in enough money!

Group	Income per household	Tax rate (%)	Income tax per household	Total tax collected for all households	Income after taxes per household
A	$12,000				
B	$29,000				
C	$50,000				
D	$79,000				
E	$129,000				
F	$295,000				
				This better total to $800,000	

7) Discretionary income is the income people have left over after paying for necessities like rent, food, transportation, etc. The cost of basic expenses does increase with income, since housing and car costs are higher, however usually not proportionally. For each income group, estimate their essential expenses, and calculate their discretionary income. Then compute the effective tax rate for each plan relative to discretionary income rather than income.

Group	Income per household	Discretionary Income (estimated)	Effective rate, flat	Effective rate, modified	Effective rate, progressive
A	$12,000				
B	$29,000				
C	$50,000				
D	$79,000				
E	$129,000				
F	$295,000				

8) Which plan seems the most fair to you? Which plan seems the least fair to you? Why?

Project 2: Calculating Taxes.

Visit www.irs.gov, and download the most recent version of forms 1040, and schedules A, B, C, and D.

Scenario 1: Calculate the taxes for someone who earned $60,000 in standard wage income (W-2 income), has no dependents, and takes the standard deduction.

Scenario 2: Calculate the taxes for someone who earned $20,000 in standard wage income, $40,000 in qualified dividends, has no dependents, and takes the standard deduction. (Qualified dividends are earnings on certain investments such as stocks.)

Scenario 3: Calculate the taxes for someone who earned $60,000 in small business income, has no dependents, and takes the standard deduction.

Based on these three scenarios, what are your impressions of how the income tax system treats these different forms of income (wage, dividends, and business income)?

Scenario 4: To get a more realistic sense for calculating taxes, you'll need to consider itemized deductions. Calculate the income taxes for someone with the income and expenses listed below.

Married with 2 children, filing jointly

Wage income: $50,000 combined

Paid sales tax in Washington State

Property taxes paid: $3200

Home mortgage interest paid: $4800

Charitable gifts: $1200

2.1.2: Exercises

Section 2.2.2 Exercises

1. Laars earns an annual salary of $60,000. Determine his gross earnings per pay period under each of the following payment frequencies:
   a. Monthly
   b. Semi-monthly
   c. Biweekly
   d. Weekly
2. A city has a sales tax of 10.25%. What would be the sales tax on a $2575.57 purchase? Round to the nearest cent.
3. The tax on gas in California went up to $0.511 cents per gallon in July 2021. If an average gallon of gas costs $4.31 per gallon, what is the effective tax rate of the gas tax? Round the percent to two decimal places.
4. Samantha and Anahi carpool to work to split the cost of gas. They each pay $120 per month to cover the cost of the commute. If Samantha makes $3200 a month and Anahi makes $4100, what is the effective tax rate on their income? Would this be considered a progressive, regressive, or flat tax rate?
5. Alex bought a new gaming system with several games that retailed at $435 before tax. They noticed the tax charged was $42.41. What was the sales tax rate of that city?
6. Use the 2021 Federal Income Tax Brackets at the end of this page for the following questions.
   a. Determine the federal income tax on $23,325 for a single individual.
   b. Determine the federal income tax on $152,904 for married individuals filing jointly.
   c. Determine the federal income tax on $102,350 for a single individual.
   d. Adrian made $40,525 in 2021 and is filing as single. Melany made $40,526 in 2021 and is also filing as single. Melany thinks that since both she and Adrian make approximately the same amount of money, they should be in the same tax bracket. Using the tax bracket chart, determine if Melany's claim is true. Find the difference in the amount of taxes they paid.
   e. Brenda made $41,685 in 2021 and is filing single. Luz made $51,892 filing as Head of Household. Brenda claims to Luz that since she is making less money, that she will pay less in taxes. Using the tax chart below, determine if Brenda's claim is true. Find the difference in the amount of taxes they paid and state who paid more. What is each of their effective tax rates?
7. A lawmaker proposed that everyone will pay the same amount of taxes. Fill in the following table by writing out each rate. Determine if this is an example of a progressive, regressive, or flat tax rate.

Rate	For Single Individuals	Amount of Tax Paid
	$6,500	$2,500
	$9,951	$2,500
	$40,526	$2,500
	$86,376	$2,500
	$164,926	$2,500
	$209,426	$2,500
	$523,601	$2,500

2021 Federal Income Tax Brackets and Rates for Single Filers, Married Couples Filing Jointly, and Heads of Households

Rate	For Single Individuals	For Married Individuals Filing Joint Returns	For Heads of Households
10%	of any amount up to $9,950	of any amount up to $19,900	of any amount up to $14,200
12%	of the portion above $9,950 up to $40,525 plus $995	of the portion above $19,900 up to $81,050 plus $1990	of the portion above $14,200 up to $54,200 plus $1420
22%	of the portion above $40,525 up to $86,375 plus $4664	of the portion above $81,050 up to $172,750 plus $9328	of the portion above $54,200 up to $86,350 plus $6220
24%	of the portion above $86,375 up to $164,925 plus $14751	of the portion above $172,750 to $329,850 plus $29502	of the portion above $86,350 up to $164,900 plus $13296
32%	of the portion above $164,925 up to $209,425 plus $33603	of the portion above $329,850 up to $418,850 plus $ 67206	of the portion above $164,900 up to $209,400 plus $32148
35%	of the portion above $209,425 up to $523,600 plus $47843	of the portion above $418,850 up to $628,300 plus $95686	of the portion above $209,400 up to $523,600 plus $46388
37%	of the portion above $523,600 or more plus $157804	of the portion above $628,300 or more plus $168994	of the portion above $523,600 or more plus 156358

2.2: Modeling with Linear Functions

When modeling scenarios with a linear function and solving problems involving quantities changing linearly, we typically follow the same problem solving strategies that we would use for any type of function:

Problem Solving Strategy

1. Identify changing quantities, and then carefully and clearly define descriptive variables to represent those quantities. When appropriate, sketch a picture or define a coordinate system.
2. Carefully read the problem to identify important information. Look for information giving values for the variables, or values for parts of the functional model, like slope and initial value.
3. Carefully read the problem to identify what we are trying to find, identify, solve, or interpret.
4. Identify a solution pathway from the provided information to what we are trying to find. Often this will involve checking and tracking units, building a table or even finding a formula for the function being used to model the problem.
5. When needed, find a formula for the function.
6. Solve or evaluate using the formula you found for the desired quantities.
7. Reflect on whether your answer is reasonable for the given situation and whether it makes sense mathematically.
8. Clearly convey your result using appropriate units, and answer in full sentences when appropriate.

Example 2.2.1

Emily saved up $3500 for her summer visit to Seattle. She anticipates spending $400 each week on rent, food, and fun. Find and interpret the horizontal intercept and determine a reasonable domain and range for this function.

Solution

In the problem, there are two changing quantities: time and money. The amount of money she has remaining while on vacation depends on how long she stays. We can define our variables, including units.

Output: M, money remaining, in dollars

Input: t, time, in weeks

Reading the problem, we identify two important values. The first, $3500, is the initial value for M. The other value appears to be a rate of change – the units of dollars per week match the units of our output variable divided by our input variable. She is spending money each week, so you should recognize that the amount of money remaining is decreasing each week and the slope is negative.

To answer the first question, looking for the horizontal intercept, it would be helpful to have an equation modeling this scenario. Using the intercept and slope provided in the problem, we can write the equation:

$$M(t) = 3500 - 400t$$

To find the horizontal intercept, we set the output to zero, and solve for the input:

$$\begin{aligned} 0 &= 3500 - 400t \\ t &= \frac{3500}{400} = 8.75 \end{aligned}$$

The horizontal intercept is 8.75 weeks. Since this represents the input value where the output will be zero, interpreting this, we could say: Emily will have no money left after 8.75 weeks.

When modeling any real life scenario with functions, there is typically a limited domain over which that model will be valid – almost no trend continues indefinitely. In this case, it certainly doesn't make sense to talk about input values less than zero. It is also likely that this model is not valid after the horizontal intercept (unless Emily's going to start using a credit card and go into debt).

The domain represents the set of input values and so the reasonable domain for this function is $0 \leq t \leq 8.75$.

However, in a real world scenario, the rental might be weekly or nightly. She may not be able to stay a partial week and so all options should be considered. Emily could stay in Seattle for 0 to 8 full weeks (and a couple of days), but would have to go into debt to stay 9 full weeks, so restricted to whole weeks, a reasonable domain without going in to debt would be $0 \leq t \leq 8$, or $0 \leq t \leq 9$ if she went into debt to finish out the last week.

The range represents the set of output values and she starts with \$3500 and ends with \$0 after 8.75 weeks so the corresponding range is $0 \leq M(t) \leq 3500$. If we limit the rental to whole weeks, however, the range would change. If she left after 8 weeks because she didn't have enough to stay for a full 9 weeks, she would have $M(8) = 3500 - 400(8) = \$300$ dollars left after 8 weeks, giving a range of $300 \leq M(t) \leq 3500$. If she wanted to stay the full 9 weeks she would be \$100 in debt giving a range of $-100 \leq M(t) \leq 3500$.

Most importantly remember that domain and range are tied together, and what ever you decide is most appropriate for the domain (the independent variable) will dictate the requirements for the range (the dependent variable).

Exercise 2.2.1

A database manager is loading a large table from backups. Getting impatient, she notices 1.2 million rows had been loaded. Ten minutes later, 2.5 million rows had been loaded. How much longer will she have to wait for all 80 million rows to load?

Answer

Letting t be the number of minutes since she got impatient, and N be the number rows loaded, in millions, we have two points: (0, 1.2) and (10, 2.5).

The slope is $m = \frac{2.5 - 1.2}{10 - 0} = \frac{1.3}{10} = 0.13$ million rows per minute.

We know the N intercept, so we can write the equation:

$$N = 0.13t + 1.2$$

To determine how long she will have to wait, we need to solve for when $N = 80$.

$$N = 0.13t + 1.2 = 80$$

$$0.13t = 78.8$$

$$t = \frac{78.8}{0.13} = 606$$

. She'll have to wait another 606 minutes, about 10 hours.

Example 2.2.2

Jamal is choosing between two moving companies. The first, U-Haul, charges an up-front fee of \$20, then 59 cents a mile. The second, Budget, charges an up-front fee of \$16, then 63 cents a mile(Rates retrieved Aug 2, 2010 from www.budgettruck.com and http://www.uhaul.com/). When will U-Haul be the better choice for Jamal?

Solution

The two important quantities in this problem are the cost, and the number of miles that are driven. Since we have two companies to consider, we will define two functions:

Input: m, miles driven

Outputs:

$Y(m)$: cost, in dollars, for renting from U-Haul

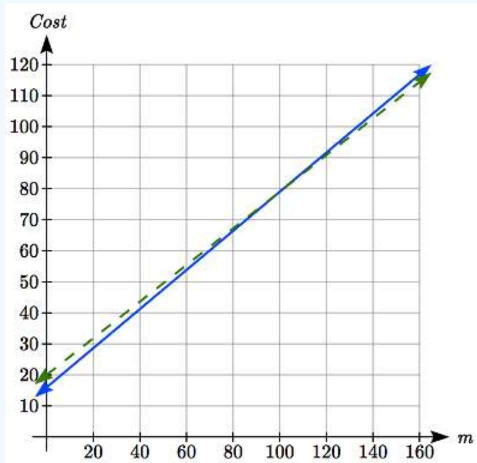
$B(m)$: cost, in dollars, for renting from Budget

Reading the problem carefully, it appears that we were given an initial cost and a rate of change for each company. Since our outputs are measured in dollars but the costs per mile given in the problem are in cents, we will need to convert these quantities to match our desired units: \$0.59 a mile for U-Haul, and \$0.63 a mile for Budget.

Looking to what we're trying to find, we want to know when U-Haul will be the better choice. Since all we have to make that decision from is the costs, we are looking for when U-Haul will cost less, or when $Y(m) < B(m)$. The solution pathway will lead us to find the equations for the two functions, find the intersection, then look to see where the $Y(m)$ function is smaller. Using the rates of change and initial charges, we can write the equations:

$$Y(m) = 20 + 0.59m$$

$$B(m) = 16 + 0.63m$$



These graphs are sketched to the right, with Y(m) drawn dashed.

To find the intersection, we set the equations equal and solve:

$$\begin{aligned} Y(m) &= B(m) \\ 20 + 0.59m &= 16 + 0.63m \\ 4 &= 0.04m \\ m &= 100 \end{aligned}$$

This tells us that the cost from the two companies will be the same if 100 miles are driven. Either by looking at the graph, or noting that $Y(m)$ is growing at a slower rate, we can conclude that U-Haul will be the cheaper price when more than 100 miles are driven.

Example 2.2.3

A town's population has been growing linearly. In 2004 the population was 6,200. By 2009 the population had grown to 8,100. If this trend continues,

a. Predict the population in 2013
b. When will the population reach 15000?

Solution

The two changing quantities are the population and time. While we could use the actual year value as the input quantity, doing so tends to lead to very ugly equations, since the vertical intercept would correspond to the year 0, more than 2000 years ago!

To make things a little nicer, and to make our lives easier too, we will define our input as years since 2004:

Input: t, years since 2004

Output: $P(t)$, the town's population

The problem gives us two input-output pairs. Converting them to match our defined variables, the year 2004 would correspond to $t = 0$, giving the point (0, 6200). Notice that through our clever choice of variable definition, we have "given" ourselves the vertical intercept of the function. The year 2009 would correspond to $t = 5$, giving the point (5, 8100).

To predict the population in 2013 ($t = 9$), we would need an equation for the population. Likewise, to find when the population would reach 15000, we would need to solve for the input that would provide an output of 15000. Either way, we need an equation. To find it, we start by calculating the rate of change:

$$m = \frac{8100 - 6200}{5 - 0} = \frac{1900}{5} = 380 \text{ people per year}$$

Since we already know the vertical intercept of the line, we can immediately write the equation:

$$P(t) = 6200 + 380t$$

To predict the population in 2013, we evaluate our function at $t = 9$

$$P(9) = 6200 + 380(9) = 9620$$

If the trend continues, our model predicts a population of 9,620 in 2013.

To find when the population will reach 15,000, we can set $P(t) = 15000$ and solve for t.

$$\begin{aligned} 15000 &= 6200 + 380t \\ 8800 &= 380t \\ t &\approx 23.158 \end{aligned}$$

Our model predicts the population will reach 15,000 in a little more than 23 years after 2004, or somewhere around the year 2027.

Example 2.2.4

Anna and Emanuel start at the same intersection. Anna walks east at 4 miles per hour while Emanuel walks south at 3 miles per hour. They are communicating with a two-way radio with a range of 2 miles. How long after they start walking will they fall out of radio contact?

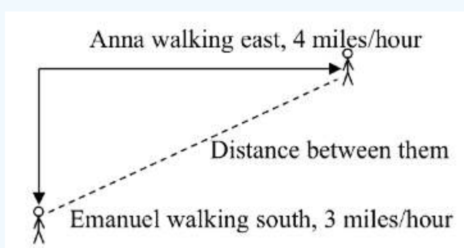
Solution

In essence, we can partially answer this question by saying they will fall out of radio contact when they are 2 miles apart, which leads us to ask a new question: how long will it take them to be 2 miles apart?

In this problem, our changing quantities are time and the two peoples' positions, but ultimately we need to know how long will it take for them to be 2 miles apart. We can see that time will be our input variable, so we'll define

Input: t, time in hours.

Since it is not obvious how to define our output variables, we'll start by drawing a picture.

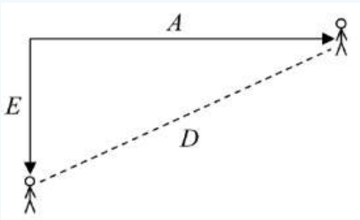


Because of the complexity of this question, it may be helpful to introduce some intermediary variables. These are quantities that we aren't directly interested in, but seem important to the problem. For this problem, Anna's and Emanuel's distances from the starting point seem important. To notate these, we are going to define a coordinate system, putting the "starting point" at the intersection where they both started, then we're going to introduce a variable, A, to represent Anna's position, and define it to be a measurement from the starting point, in the eastward direction. Likewise, we'll introduce a variable, E, to represent

Emanuel's position, measured from the starting point in the southward direction. Note that in defining the coordinate system we specified both the origin, or starting point, of the measurement, as well as the direction of measure.

While we're at it, we'll define a third variable, D, to be the measurement of the distance between Anna and Emanuel. Showing the variables on the picture is often helpful:

Looking at the variables on the picture, we remember we need to know how long it takes for D, the distance between them, to equal 2 miles.



Seeing this picture we remember that in order to find the distance between the two, we can use the Pythagorean Theorem, a property of right triangles.

From here, we can now look back at the problem for relevant information. Anna is walking 4 miles per hour, and Emanuel is walking 3 miles per hour, which are rates of change. Using those, we can write formulas for the distance each has walked.

They both start at the same intersection and so when $t = 0$, the distance travelled by each person should also be 0, so given the rate for each, and the initial value for each, we get:

$$A(t) = 4t$$

$$E(t) = 3t$$

Using the Pythagorean theorem we get:

$$D(t)^2 = A(t)^2 + E(t)^2$$

substitute in the function formulas

$$D(t)^2 = (4t)^2 + (3t)^2 = 16t^2 + 9t^2 = 25t^2$$

solve for $D(t)$ using the square root

$$D(t) = \pm\sqrt{25t^2} = \pm 5|t|$$

Since in this scenario we are only considering positive values of t and our distance $D(t)$ will always be positive, we can simplify this answer to $D(t) = 5t$

Interestingly, the distance between them is also a linear function. Using it, we can now answer the question of when the distance between them will reach 2 miles:

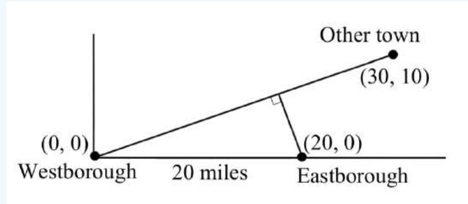
$$\begin{aligned} D(t) &= 2 \\ 5t &= 2 \\ t &= \frac{2}{5} = 0.4 \end{aligned}$$

They will fall out of radio contact in 0.4 hours, or 24 minutes.

Example 2.2.5

There is currently a straight road leading from the town of Westborough to a town 30 miles east and 10 miles north. Partway down this road, it junctions with a second road, perpendicular to the first, leading to the town of Eastborough. If the town of Eastborough is located 20 miles directly east of the town of Westborough, how far is the road junction from Westborough?

Solution



It might help here to draw a picture of the situation. It would then be helpful to introduce a coordinate system. While we could place the origin anywhere, placing it at Westborough seems convenient. This puts the other town at coordinates (30, 10), and Eastborough at (20, 0).

Using this point along with the origin, we can find the slope of the line from Westborough to the other town: $m = \frac{10-0}{30-0} = \frac{1}{3}$. This gives the equation of the road from Westborough to the other town to be $W(x) = \frac{1}{3}x$.

From this, we can determine the perpendicular road to Eastborough will have slope $m = -3$. Since the town of Eastborough is at the point (20, 0), we can find the equation:

$$E(x) = -3x + b$$

plug in the point (20, 0)

$$0 = -3(20) + b$$

$$b = 60$$

$$E(x) = -3x + 60$$

We can now find the coordinates of the junction of the roads by finding the intersection of these lines. Setting them equal,

$$\frac{1}{3}x = -3x + 60$$

$$\frac{10}{3}x = 60$$

$$10x = 180$$

$$x = 18$$

Substituting this back into $W(x)$

$$y = W(18) = \frac{1}{3}(18) = 6$$

The roads intersect at the point (18, 6). Using the distance formula, we can now find the distance from Westborough to the junction:

$$dist = \sqrt{(18-0)^2 + (6-0)^2} \approx 18.934 \text{ miles}$$

Important Topics of this Section

The problem solving process

1. Identify changing quantities, and then carefully and clearly define descriptive variables to represent those quantities. When appropriate, sketch a picture or define a coordinate system.
2. Carefully read the problem to identify important information. Look for information giving values for the variables, or values for parts of the functional model, like slope and initial value.

3. Carefully read the problem to identify what we are trying to find, identify, solve, or interpret.
4. Identify a solution pathway from the provided information to what we are trying to find. Often this will involve checking and tracking units, building a table or even finding a formula for the function being used to model the problem.
5. When needed, find a formula for the function.
6. Solve or evaluate using the formula you found for the desired quantities.
7. Reflect on whether your answer is reasonable for the given situation and whether it makes sense mathematically.
8. Clearly convey your result using appropriate units, and answer in full sentences when appropriate.

2.2E: Modeling with Linear Functions (Exercises)

section 2.3 exercise

1. In 2004, a school population was 1001. By 2008 the population had grown to 1697. Assume the population is changing linearly.

   a. How much did the population grow between the year 2004 and 2008?
   b. How long did it take the population to grow from 1001 students to 1697 students?
   c. What is the average population growth per year?
   d. What was the population in the year 2000?
   e. Find an equation for the population, P, of the school t years after 2000.
   f. Using your equation, predict the population of the school in 2011.

2. In 2003, a town's population was 1431. By 2007 the population had grown to 2134. Assume the population is changing linearly.

   a. How much did the population grow between the year 2003 and 2007?
   b. How long did it take the population to grow from 1431 people to 2134?
   c. What is the average population growth per year?
   d. What was the population in the year 2000?
   e. Find an equation for the population, P, of the town t years after 2000.
   f. Using your equation, predict the population of the town in 2014.

3. A phone company has a monthly cellular plan where a customer pays a flat monthly fee and then a certain amount of money per minute used on the phone. If a customer uses 410 minutes, the monthly cost will be $71.50. If the customer uses 720 minutes, the monthly cost will be $118.

   a. Find a linear equation for the monthly cost of the cell plan as a function of x, the number of monthly minutes used.
   b. Interpret the slope and vertical intercept of the equation.
   c. Use your equation to find the total monthly cost if 687 minutes are used.

4. A phone company has a monthly cellular data plan where a customer pays a flat monthly fee and then a certain amount of money per megabyte (MB) of data used on the phone. If a customer uses 20 MB, the monthly cost will be $11.20. If the customer uses 130 MB, the monthly cost will be $17.80.

   a. Find a linear equation for the monthly cost of the data plan as a function of x, the number of MB used.
   b. Interpret the slope and vertical intercept of the equation.
   c. Use your equation to find the total monthly cost if 250 MB are used.

5. In 1991, the moose population in a park was measured to be 4360. By 1999, the population was measured again to be 5880. If the population continues to change linearly,

   a. Find a formula for the moose population, P.
   b. What does your model predict the moose population to be in 2003?

6. In 2003, the owl population in a park was measured to be 340. By 2007, the population was measured again to be 285. If the population continues to change linearly,

   a. Find a formula for the owl population, P.
   b. What does your model predict the owl population to be in 2012?

7. The Federal Helium Reserve held about 16 billion cubic feet of helium in 2010, and is being depleted by about 2.1 billion cubic feet each year.

   a. Give a linear equation for the remaining federal helium reserves, R, in terms of t, the number of years since 2010.
   b. In 2015, what will the helium reserves be?
   c. If the rate of depletion doesn't change, when will the Federal Helium Reserve be depleted?

8. Suppose the world’s current oil reserves are 1820 billion barrels. If, on average, the total reserves is decreasing by 25 billion barrels of oil each year:

a. Give a linear equation for the remaining oil reserves, R, in terms of t, the number of years since now.
b. Seven years from now, what will the oil reserves be?
c. If the rate of depletion isn’t change, when will the world’s oil reserves be depleted?

9. You are choosing between two different prepaid cell phone plans. The first plan charges a rate of 26 cents per minute. The second plan charges a monthly fee of $19.95 $plus$ 11 cents per minute. How many minutes would you have to use in a month in order for the second plan to be preferable?

10. You are choosing between two different window washing companies. The first charges $5 per window. The second charges a base fee of $40 plus $3 per window. How many windows would you need to have for the second company to be preferable?

11. When hired at a new job selling jewelry, you are given two pay options:

Option A: Base salary of $17,000 a year, with a commission of 12% of your sales
Option B: Base salary of $20,000 a year, with a commission of 5% of your sales

How much jewelry would you need to sell for option A to produce a larger income?

12. When hired at a new job selling electronics, you are given two pay options:

Option A: Base salary of $14,000 a year, with a commission of 10% of your sales
Option B: Base salary of $19,000 a year, with a commission of 4% of your sales

How much electronics would you need to sell for option A to produce a larger income?

13. Find the area of a triangle bounded by the y axis, the line $f(x) = 9 - \frac{6}{7}x$, and the line perpendicular to $f(x)$ that passes through the origin.

14. Find the area of a triangle bounded by the x axis, the line $f(x) = 12 - \frac{1}{3}x$, and the line perpendicular to $f(x)$ that passes through the origin.

15. Find the area of a parallelogram bounded by the y axis, the line $x = 3$, the line $f(x) = 1 + 2x$, and the line parallel to $f(x)$ passing through (2, 7)

16. Find the area of a parallelogram bounded by the x axis, the line $g(x) = 2$, the line $f(x) = 3x$, and the line parallel to $f(x)$ passing through (6, 1)

17. If $b > 0$ and $m < 0$, then the line $f(x) = b + mx$ cuts off a triangle from the first quadrant. Express the area of that triangle in terms of m and b. [UW]

18. Find the value of m so the lines $f(x) = mx + 5$ and $g(x) = x$ and the y-axis form a triangle with an area of 10. [UW]

19. The median home values in Mississippi and Hawaii (adjusted for inflation) are shown below. If we assume that the house values are changing linearly,

Year	Mississippi	Hawaii
1950	25200	74400
2000	71400	272700

a. In which state have home values increased at a higher rate?
b. If these trends were to continue, what would be the median home value in Mississippi in 2010?
c. If we assume the linear trend existed before 1950 and continues after 2000, the two states’ median house values will be (or were) equal in what year? (The answer might be absurd)

20. The median home value ins Indiana and Alabama (adjusted for inflation) are shown below. If we assume that the house values are changing linearly,

Year	Indiana	Alabama
1950	37700	27100
2000	94300	85100

a. In which state have home values increased at a higher rate?
b. If these trends were to continue, what would be the median home value in Indiana in 2010?
c. If we assume the linear trend existed before 1950 and continues after 2000, the two states' median house values will be (or were) equal in what year? (The answer might be absurd)

21. Pam is taking a train from the town of Rome to the town of Florence. Rome is located 30 miles due West of the town of Paris. Florence is 25 miles East, and 45 miles North of Rome. On her trip, how close does Pam get to Paris? [UW]

22. You're flying from Joint Base Lewis-McChord (JBLM) to an undisclosed location 226 km south and 230 km east. Mt. Rainier is located approximately 56 km east and 40 km south of JBLM. If you are flying at a constant speed of 800 km/hr, how long after you depart JBLM will you be the closest to Mt. Rainier?

Answer

1a. 696 people
b. 4 years
c. 174 people per year
d. 305 people
e. $P(t) = 305 + 174t$
f. 2219 people.

3a. $C(x) = 0.15x + 10$
b. The flat monthly fee is $10 and there is an additional $0.15 fee for each additional minute used
c. $113.05

5a. $P(t) = 190t + 4170$
b. 6640 moose

7a. $R(t) = 16 - 2.1t$
b. 5.5 billion cubic feet
c. During the year 2017

9. More than 133 minutes

11. More than $42857.14 worth of jewelry

13. 20.012 square units

15. 6 square units

17. $A = -\frac{b^2}{2m}$

19a. Hawaii
b. $80640
c. During the year 1933

21. 26.225 miles

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2.3: Simple Interest

2.3 Learning Objectives

- Find the simple interest for a loan

Discussing interest starts with the **principal**, or amount your account starts with. This could be a starting investment, or the starting amount of a loan. Interest, in its most simple form, is calculated as a percent of the principal. For example, if you borrowed $100 from a friend and agree to repay it with 5% interest, then the amount of interest you would pay would just be 5% of 100: $\$100(0.05) = \$5$. The total amount you would repay would be $105, the original principal plus the interest.

Simple One-time Interest

$$I = P_0 r \tag{2.3.1}$$

$$A = P_0 + I = P_0 + P_0 r = P_0(1+r) \tag{2.3.2}$$

where

- I is the interest
- A is the end amount: principal plus interest
- P_0 is the principal (starting amount)
- r is the interest rate (in decimal form. Example: $5\% = 0.05$)

Example 1

A friend asks to borrow $300 and agrees to repay it in 30 days with 3% interest. How much interest will you earn?

Solution

$$\begin{array}{ll} P_0 = \$300 & \text{the principal} \\ r = 0.03 & 3\% \text{ rate} \\ I = \$300(0.03) = \$9. & \text{You will earn \$9 interest.} \end{array}$$

One-time simple interest is only common for extremely short-term loans. For longer term loans, it is common for interest to be paid on a daily, monthly, quarterly, or annual basis. In that case, interest would be earned regularly. For example, bonds are essentially a loan made to the bond issuer (a company or government) by you, the bond holder. In return for the loan, the issuer agrees to pay interest, often annually. Bonds have a maturity date, at which time the issuer pays back the original bond value.

Example 2

Suppose your city is building a new park, and issues bonds to raise the money to build it. You obtain a $1,000 bond that pays 5% interest annually that matures in 5 years. How much interest will you earn?

Solution

Each year, you would earn 5% interest: $\$1000(0.05) = \$50$ in interest. So over the course of five years, you would earn a total of $250 in interest. When the bond matures, you would receive back the $1,000 you originally paid, leaving you with a total of $1,250.

We can generalize this idea of simple interest over time.

Simple Interest over Time

$I = P_0 rt$

$A = P_0 + I = P_0 + P_0 rt = P_0(1+rt)$

where

- I is the interest

- A is the end amount: principal plus interest
- P_0 is the principal (starting amount)
- r is the interest rate in decimal form
- t is time

The units of measurement (years, months, etc.) for the time should match the time period for the interest rate.

Definition: Annual Percentage Rate

Interest rates are usually given as an **annual percentage rate (APR)** – the total interest that will be paid in the year. If the interest is paid in smaller time increments, the APR will be divided up.

For example, a 6% APR paid monthly would be divided into twelve 0.5% payments.
A 4% annual rate paid quarterly would be divided into four 1% payments.

Example 3 Treasury Notes (T-notes) are bonds issued by the federal government to cover its expenses. Suppose you obtain a $1,000 T-note with a 4% annual rate, paid semi-annually, with a maturity in 4 years. How much interest will you earn?

Solution

Since interest is being paid semi-annually (twice a year), the 4% interest will be divided into two 2% payments.

$$\begin{array}{ll} P_0 = \$1000 & \text{the principal} \\ r = 0.02 & 2\% \text{ rate} \\ t = 8 & 4 \text{ years} = 8 \text{ half-years} \\ I = \$1000(0.02)(8) = \$160. & \text{You will earn } \$160 \text{ interest total over the four years.} \end{array}$$

Try it Now 1

A loan company charges $30 interest for a one month loan of $500. Find the annual interest rate they are charging.

Answer

$I = \$30$ of interest

$P_0 = \$500$ principal

$r =$ unknown (annual)

$t = 1$ month (1/12 of a year)

2.3.1: Exercises

Section 2.3.1 Exercises

1. You borrow \$350 from your aunt and promise to pay her back in one month with 3% interest. How much extra will she get when you pay her back?
2. You were charged \$487.50 in interest at the end of one year for money you borrowed from a bank that charged 6.5% interest. How much did you borrow?
3. You had to pay \$618.12 in interest on a loan for \$10,750 over the first year. What was the interest rate used? Round the percent to two decimal places.
4. A friend lends you \$200 for a week, which you agree to repay with 5% one-time interest. How much will you have to repay?
5. Your brother deposits \$1200 in a savings account that pays 4% simple interest. How much will he have in the account in 4 years if he doesn't touch it.
6. Suppose you obtain a \$3,000 T-note with a 3% annual rate, paid quarterly, with maturity in 5 years. How much interest will you earn?
7. Suppose you take a cash advance of \$750 on your credit card. The service charge is \$5.50. You pay off the cash advance after 45 days. Your credit card statement shows the interest on the cash advance was \$29.59. For the interest alone, what is the APR?
8. If you buy a 3-year CD that pays 2.4% simple interest for \$75,000, what will it be worth at the end of its term?
9. If you invest \$1500 at 4.25% simple interest and the investment is later worth \$1553.13, what was the term?
10. A T-bill is a type of bond that is sold at a discount over the face value. For example, suppose you buy a 13-week T-bill with a face value of \$10,000 for \$9,800. This means that in 13 weeks, the government will give you the face value, earning you \$200. What annual interest rate have you earned?
11. Suppose you are looking to buy a \$5000 face value 26-week T-bill. If you want to earn at least 1% annual interest, what is the most you should pay for the T-bill?

2.4: Modeling with Exponential Functions

India is the second most populous country in the world, with a population in 2008 of about 1.14 billion people. The population is growing by about 1.34% each year (World Bank, World Development Indicators, as reported on http://www.google.com/publicdata, retrieved August 20, 2010). We might ask if we can find a formula to model the population, P, as a function of time, t, in years after 2008, if the population continues to grow at this rate.

In linear growth, we had a constant rate of change – a constant $number$ that the output increased for each increase in input. For example, in the equation $f(x)=3x+4$, the slope tells us the output increases by three each time the input increases by one. This population scenario is different – we have a $percent$ rate of change rather than a constant number of people as our rate of change.

To see the significance of this difference consider these two companies:

- Company A has 100 stores, and expands by opening 50 new stores a year
- Company B has 100 stores, and expands by increasing the number of stores by 50% of their total each year.

Looking at a few years of growth for these companies:

Year	Stores, company A		Stores, company B
0	100	Starting with 100 each	100
1	100 + 50 = 150	They both grow by 50 stores in the first year.	100 + 50% of 100 100 + 0.50(100)= 150
2	150 + 50 = 200	Store A grows by 50, Store B grows by 75	150 + 50% of 150 150 + 0.50(150) = 225
3	200 + 50 = 250	Store A grows by 50, Store B grows by 112.5	225 + 50% of 225 225 + 0.50(225) = 337.5

Notice that with the percent growth, each year the company it grows by 50% of the current year's total, so as the company grows larger, the number of stores added in a year grows as well.

To try to simplify the calculations, notice that after 1 year the number of stores for company B was:

$$100+0.50$$

or equivalently by factoring

$$100(1+0.50)=150$$

We can think of this as "the new number of stores is the original 100% plus another 50%".

After 2 years, the number of stores was:

$$150+0.50$$

or equivalently by factoring

$$150(1+0.50)$$

now recall the 150 came from 100(1+0.50). Substituting that,

$$100(1+0.50)(1+0.50)=100(1+0.50)^2=225$$

After 3 years, the number of stores was:

$$225+0.50$$

or equivalently by factoring

$$225(1+0.50)$$

now recall the 225 came from $100(1+0.50)^2$. Substituting that,

$$100(1+0.50)^2(1+0.50) = 100(1+0.50)^3 = 225$$

From this, we can generalize, noticing that to show a 50% increase, each year we multiply by a factor of (1+0.50), so after n years, our equation would be

$$B(n) = 100(1+0.50)^n$$

In this equation, the 100 represented the initial quantity, and the 0.50 was the percent growth rate. Generalizing further, we arrive at the general form of exponential functions.

Definition: Exponential Function

An **exponential growth or decay function** is a function that grows or shrinks at a constant percent growth rate. The equation can be written in the form

$$f(x) = a(1+r)^x$$

or

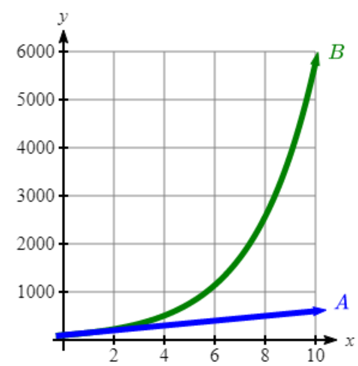
$$f(x) = ab^x$$

where $b = 1 + r$ and

- a is the initial or starting value of the function
- r is the percent growth or decay rate, written as a decimal
- b is the growth factor or growth multiplier. Since powers of negative numbers behave strangely, we limit b to positive values.

To see more clearly the difference between exponential and linear growth, compare the two tables and graphs below, which illustrate the growth of company A and B described above over a longer time frame if the growth patterns were to continue.

years	Company A	Company B
2	200	225
4	300	506
6	400	1139
8	500	2563
10	600	5767



Example 2.4.1

Write an exponential function for India's population, and use it to predict the population in 2020.

Solution

At the beginning of the chapter we were given India's population of 1.14 billion in the year 2008 and a percent growth rate of 1.34%. Using 2008 as our starting time (t = 0), our initial population will be 1.14 billion. Since the percent growth rate was

1.34%, our value for r is 0.0134.

Using the basic formula for exponential growth $f(x) = a(1 + r)^x$ we can write the formula, $f(t) = 1.14(1 + 0.0134)^t$

To estimate the population in 2020, we evaluate the function at t = 12, since 2020 is 12 years after 2008.

$$f = 1.14(1 + 0.0134)^{12} \approx 1.337 \text{ billion people in 2020}$$

Exercise 2.4.1

Given the three statements below, identify which represent exponential functions.

a. The cost of living allowance for state employees increases salaries by 3.1% each year.
b. State employees can expect a $300 raise each year they work for the state.
c. Tuition costs have increased by 2.8% each year for the last 3 years.

Answer

a & c are exponential functions, they grow by a % not a constant number.

Example 2.4.2

A certificate of deposit (CD) is a type of savings account offered by banks, typically offering a higher interest rate in return for a fixed length of time you will leave your money invested. If a bank offers a 24 month CD with an annual interest rate of 1.2% compounded monthly, how much will a $1000 investment grow to over those 24 months?

Solution

First, we must notice that the interest rate is an annual rate, but is compounded monthly, meaning interest is calculated and added to the account monthly. To find the monthly interest rate, we divide the annual rate of 1.2% by 12 since there are 12 months in a year: 1.2%/12 = 0.1%. Each month we will earn 0.1% interest. From this, we can set up an exponential function, with our initial amount of $1000 and a growth rate of r = 0.001, and our input m measured in months.

$$f(m) = 1000\left(1 + \frac{.012}{12}\right)^m$$

$$f(m) = 1000(1 + 0.001)^m$$

After 24 months, the account will have grown to $f(24) = 1000(1 + 0.001)^{24} = \$1024.28$

Exercise 2.4.2

Looking at these two equations that represent the balance in two different savings accounts, which account is growing faster, and which account will have a higher balance after 3 years?

- $A(t) = 1000\,(1.05)^t$
- $B(t) = 900\,(1.075)^t$

Answer

$B(t)$ is growing faster (r = 0.075 > 0.05), but after 3 years A(t) still has a higher account balance

In all the preceding examples, we saw exponential growth. Exponential functions can also be used to model quantities that are decreasing at a constant percent rate. An example of this is radioactive decay, a process in which radioactive isotopes of certain atoms transform to an atom of a different type, causing a percentage decrease of the original material over time.

Example 2.4.3

Bismuth-210 is an isotope that radioactively decays by about 13% each day, meaning 13% of the remaining Bismuth-210 transforms into another atom (polonium-210 in this case) each day. If you begin with 100 mg of Bismuth-210, how much remains after one week?

Solution

With radioactive decay, instead of the quantity increasing at a percent rate, the quantity is decreasing at a percent rate. Our initial quantity is a = 100 mg, and our growth rate will be negative 13%, since we are decreasing: r = -0.13. This gives the equation:

$$Q(d) = 100(1-0.13)^d = 100(0.87)^d$$

This can also be explained by recognizing that if 13% decays, then 87 % remains.

After one week, 7 days, the quantity remaining would be

$$Q(7) = 100(0.87)^7 = 37.73 \text{ mg of Bismuth-210 remains.}$$

Exercise 2.4.3

A population of 1000 is decreasing 3% each year. Find the population in 30 years.

Answer

$$P(t) = 1000(1-0.03)^t = 1000(0.97)^t$$

$$P(30) = 1000(0.97)^{30} = 401.0071$$

Example 2.4.4

$T(q)$ represents the total number of Android smart phone contracts, in thousands, held by a certain Verizon store region measured quarterly since January 1, 2016, interpret all the parts of the equation $T(2) = 86(1.64)^2 = 231.3056$.

Solution

Interpreting this from the basic exponential form, we know that 86 is our initial value. This means that on Jan. 1, 2016 this region had 86,000 Android smart phone contracts. Since $b = 1 + r = 1.64$, we know that every quarter the number of smart phone contracts grows by 64%. $T(2) = 231.3056$ means that in the 2^{nd} quarter (or at the end of the second quarter) there were approximately 231,306 Android smart phone contracts.

Finding Equations of Exponential Functions

Exponential Functions:Finding Equations

In the previous examples, we were able to write equations for exponential functions since we knew the initial quantity and the growth rate. If we do not know the growth rate, but instead know only some input and output pairs of values, we can still construct an exponential function.

Example 2.4.5

In 2009, 80 deer were reintroduced into a wildlife refuge area from which the population had previously been hunted to elimination. By 2015, the population had grown to 180 deer. If this population grows exponentially, find a formula for the function.

Solution

By defining our input variable to be t, years after 2009, the information listed can be written as two input-output pairs: (0,80) and (6,180). Notice that by choosing our input variable to be measured as years after the first year value provided, we have effectively “given” ourselves the initial value for the function: a = 80. This gives us an equation of the form

$$f(t) = 80b^t$$

Substituting in our second input-output pair allows us to solve for b:

$$180 = 80b^6$$

Divide by 80

$$b^6 = \frac{180}{80} = \frac{9}{4}$$

Take the 6^{th} root of both sides.

$$b = \sqrt[6]{\frac{9}{4}} = 1.1447$$

This gives us our equation for the population:

$$f(t) = 80(1.1447)^t$$

Recall that since $b = 1 + r$, we can interpret this to mean that the population growth rate is r = 0.1447, and so the population is growing by about 14.47% each year.

In this example, you could also have used (9/4)^(1/6) to evaluate the 6^{th} root if your calculator doesn't have an n^{th} root button.

In the previous example, we chose to use the $f(x) = ab^x$ form of the exponential function rather than the $f(x) = a(1+r)^x$ form. This choice was entirely arbitrary – either form would be fine to use.

When finding equations, the value for b or r will usually have to be rounded to be written easily. To preserve accuracy, it is important to not over-round these values. Typically, you want to be sure to preserve at least 3 significant digits in the growth rate. For example, if your value for b was 1.00317643, you would want to round this no further than to 1.00318.

In the previous example, we were able to "give" ourselves the initial value by clever definition of our input variable. Next, we consider a situation where we can't do this.

Example $2.4.6$

Find a formula for an exponential function passing through the points (-2,6) and (2,1).

Solution

Since we don't have the initial value, we will take a general approach that will work for any function form with unknown parameters: we will substitute in both given input-output pairs in the function form $f(x) = ab^x$ and solve for the unknown values, a and b.

Substituting in (-2, 6) gives $6 = ab^{-2}$
Substituting in (2, 1) gives $1 = ab^2$

We now solve these as a system of equations. To do so, we could try a substitution approach, solving one equation for a variable, then substituting that expression into the second equation.

Solving $6 = ab^{-2}$ for a:

$$a = \frac{6}{b^{-2}} = 6b^2$$

In the second equation, $1 = ab^2$, we substitute the expression above for a:

$$\begin{aligned} 1 &= (6b^2)b^2 \\ 1 &= 6b^4 \\ \frac{1}{6} &= b^4 \\ b &= \sqrt[4]{\frac{1}{6}} \approx 0.6389 \end{aligned}$$

Going back to the equation $a=6b^2$ lets us find a:

$$a=6b^2=6(0.6389)^2=2.4492$$

Putting this together gives the equation $f(x)=2.4492(0.6389)^x$

Exercise 2.4.4

Given the two points (1, 3) and (2, 4.5) find the equation of an exponential function that passes through these two points.

Answer

$3=ab^1$, so $a=\frac{3}{b}$,

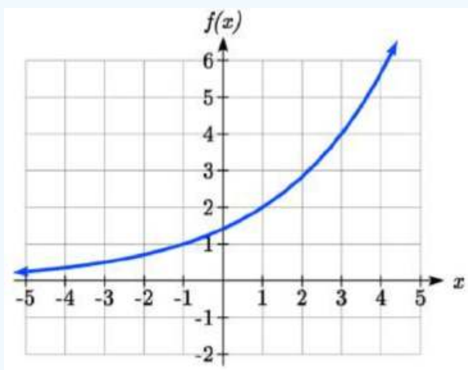
$4.5=ab^2$, so $4.5=\frac{3}{b}b^2$. $4.5=3b$

$b=1.5$. $a=\frac{3}{1.5}=2$

$f(x)=2\,(1.5)^x$

Example 2.4.7

Find an equation for the exponential function graphed.



Solution

The initial value for the function is not clear in this graph, so we will instead work using two clearer points. There are three clear points: (-1, 1), (1, 2), and (3, 4). As we saw in the last example, two points are sufficient to find the equation for a standard exponential, so we will use the latter two points.

Substituting in (1,2) gives $2=ab^1$
Substituting in (3,4) gives $4=ab^3$

Solving the first equation for a gives $a=\frac{2}{b}$.

Substituting this expression for a into the second equation:

$$4=ab^3$$

$$4=\frac{2}{b}\quad b^3=\frac{2b^3}{b}$$

Simplify the right-hand side

$$4=2b^2$$

$$2 = b^2$$

$$b = \pm\sqrt{2}$$

Since we restrict ourselves to positive values of b, we will use $b = \sqrt{2}$. We can then go back and find a:

$$a = \frac{2}{b} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

This gives us a final equation of $f(x) = \sqrt{2}(\sqrt{2})^x$.

Compound Interest

In the bank certificate of deposit (CD) example earlier in the section, we encountered compound interest. Typically bank accounts and other savings instruments in which earnings are reinvested, such as mutual funds and retirement accounts, utilize compound interest. The term *compounding* comes from the behavior that interest is earned not on the original value, but on the accumulated value of the account.

In the example from earlier, the interest was compounded monthly, so we took the annual interest rate, usually called the **nominal rate** or **annual percentage rate (APR)** and divided by 12, the number of compounds in a year, to find the monthly interest. The exponent was then measured in months.

Generalizing this, we can form a general formula for compound interest. If the APR is written in decimal form as r, and there are k compounding periods per year, then the interest per compounding period will be r/k. Likewise, if we are interested in the value after t years, then there will be kt compounding periods in that time.

Definition: Compound Interest Formula

Compound Interest can be calculated using the formula

$$A(t) = a\left(1 + \frac{r}{k}\right)^{kt} \tag{2.4.1}$$

Where $A(t)$ is the account value

- t is measured in years
- a is the starting amount of the account, often called the principal
- r is the annual percentage rate (APR), also called the nominal rate
- k is the number of compounding periods in one year

Example 2.4.8

If you invest \$3,000 in an investment account paying 3% interest compounded quarterly, how much will the account be worth in 10 years?

Solution

Since we are starting with \$3000, $a = 3000$

Our interest rate is 3%, so $r = 0.03$

Since we are compounding quarterly, we are compounding 4 times per year, so $k = 4$

We want to know the value of the account in 10 years, so we are looking for $A(10)$, the value when $t = 10$.

$$A(10) = 3000\left(1 + \frac{0.03}{4}\right)^{4(10)} = \$4045.05$$

The account will be worth \$4045.05 in 10 years.

Example 2.4.9

A 529 plan is a college savings plan in which a relative can invest money to pay for a child's later college tuition, and the account grows tax free. If Lily wants to set up a 529 account for her new granddaughter, wants the account to grow to \$40,000 over 18 years, and she believes the account will earn 6% compounded semi-annually (twice a year), how much will Lily need to invest in the account now?

Solution

Since the account is earning 6%, $r = 0.06$

Since interest is compounded twice a year, $k = 2$

In this problem, we don't know how much we are starting with, so we will be solving for a, the initial amount needed. We do know we want the end amount to be \$40,000, so we will be looking for the value of a so that $A(18) = 40,000$.

$$\begin{aligned}
&40,000 = A(18) = a\left(1+\frac{0.06}{2}\right)^{2(18)} \\
&40,000 = a(2.8983) \\
&a = \frac{40,000}{2.8983} \approx \$13,801
\end{aligned}$$

Lily will need to invest \$13,801 to have \$40,000 in 18 years.

Exercise 2.4.5

Recalculate example 2 from above with quarterly compounding.

Answer

24 months = 2 years. $1000\left(1+\frac{.012}{4}\right)^{4(2)}$ = \$1024.25

Because of compounding throughout the year, with compound interest the actual increase in a year is *more* than the annual percentage rate. If \$1,000 were invested at 10%, the table below shows the value after 1 year at different compounding frequencies:

Frequency	Value after 1 year
Annually	\$1100
Semiannually	\$1102.50
Quarterly	\$1103.81
Monthly	\$1104.71
Daily	\$1105.16

If we were to compute the actual percentage increase for the daily compounding, there was an increase of \$105.16 from an original amount of \$1,000, for a percentage increase of $\frac{105.16}{1000} = 0.10516$= 10.516% increase. This quantity is called the **annual percentage yield (APY)**.

Notice that given any starting amount, the amount after 1 year would be

$A(1) = a\left(1+\frac{r}{k}\right)^k$. To find the total change, we would subtract the original amount, then to find the percentage change we would divide that by the original amount:

$$\frac{a\left(1+\frac{r}{k}\right)^k - a}{a} = \left(1+\frac{r}{k}\right)^k - 1 \tag{2.4.2}$$

Annual Percentage Yield

The **annual percentage yield** (APY) is the actual percent a quantity increases in one year. It can be calculated as

$$APY = \left(1 + \frac{r}{k}\right)^k - 1 \tag{2.4.3}$$

This is equivalent to finding the value of $1 after 1 year, and subtracting the original dollar.

Example 2.4.10

Bank A offers an account paying 1.2% compounded quarterly. Bank B offers an account paying 1.1% compounded monthly. Which is offering a better rate?

Solution

We can compare these rates using the annual percentage yield – the actual percent increase in a year.

- Bank A: $APY = \left(1 + \frac{0.012}{4}\right)^4 - 1 = 0.012054$ = 1.2054%
- Bank B: $APY = \left(1 + \frac{0.011}{12}\right)^{12} - 1 = 0.011056$ = 1.1056%

Bank B's monthly compounding is not enough to catch up with Bank A's better APR. Bank A offers a better rate.

A Limit to Compounding

As we saw earlier, the amount we earn increases as we increase the compounding frequency. The table, though, shows that the increase from annual to semi-annual compounding is larger than the increase from monthly to daily compounding. This might lead us to believe that although increasing the frequency of compounding will increase our result, there is an upper limit to this process.

To see this, let us examine the value of $1 invested at 100% interest for 1 year.

Frequency	Value
Annual	$2
Quarterly	$2.441406
Monthly	$2.613035
Daily	$2.714567
Hourly	$2.718127
Once per minute	$2.718279
Once per second	$2.718282

These values do indeed appear to be approaching an upper limit. This value ends up being so important that it gets represented by its own letter, much like how π represents a number.

Definition: Euler's number: e

e is the letter used to represent the value that $\left(1 + \frac{1}{k}\right)^k$ approaches as k gets big.

$$e \approx 2.718282$$

Because e is often used as the base of an exponential, most scientific and graphing calculators have a button that can calculate powers of e, usually labeled e^x. Some computer software instead defines a function $exp(x)$, where $exp(x) = e^x$.

Because e arises when the time between compounds becomes very small, e allows us to define **continuous growth** and allows us to define a new toolkit function, $f(x)=e^x$.

Definition: Continuous Growth Formula

Continuous Growth can be calculated using the formula

$$f(x)=ae^{rx}$$

where

a is the starting amount

r is the continuous growth rate

This type of equation is commonly used when describing quantities that change more or less continuously, like chemical reactions, growth of large populations, and radioactive decay.

Example 2.4.11

Radon-222 decays at a continuous rate of 17.3% per day. How much will 100mg of Radon-222 decay to in 3 days?

Solution

Since we are given a continuous decay rate, we use the continuous growth formula. Since the substance is decaying, we know the growth rate will be negative: r = -0.173

$f(3)=100e^{-0.173(3)}\approx 59.512$ mg of Radon-222 will remain.

Exercise 2.4.6

Interpret the following: $S(t)=20e^{0.12t}$ if $S(t)$ represents the growth of a substance in grams, and time is measured in days.

Answer

An initial substance weighing 20g is growing at a continuous rate of 12% per day.

Continuous growth is also often applied to compound interest, allowing us to talk about continuous compounding.

Example 2.4.12

If \$1000 is invested in an account earning 10% compounded continuously, find the value after 1 year.

Solution

Here, the continuous growth rate is 10%, so r = 0.10. We start with \$1000, so a = 1000.

To find the value after 1 year,

$$f(1)=1000e^{0.10(1)}\approx \$1105.17$$

Notice this is a \$105.17 increase for the year. As a percent increase, this is $\frac{105.17}{1000}=0.10517=10.517\%$ increase over the original \$1000.

Notice that this value is slightly larger than the amount generated by daily compounding in the table computed earlier.

The continuous growth rate is like the nominal growth rate (or APR) – it reflects the growth rate before compounding takes effect. This is different than the annual growth rate used in the formula $f(x)=a(1+r)^x$, which is like the annual percentage yield – it reflects the *actual* amount the output grows in a year.

While the continuous growth rate in the example above was 10%, the actual annual yield was 10.517%. This means we could write two different looking but equivalent formulas for this account's growth:

$f(t) = 1000e^{0.10t}$ using the 10% continuous growth rate

$f(t) = 1000(1.10517)^t$ using the 10.517% actual annual yield rate.

Important Topics of this Section

- Percent growth
- Exponential functions
- Finding formulas
- Interpreting equations
- Graphs
- Exponential Growth & Decay
- Compound interest
- Annual Percent Yield
- Continuous Growth

2.4E: Exponential Functions (Exercises)

Section 4.1 Exercise

For each table below, could the table represent a function that is linear, exponential, or neither?

1.

x	1	2	3	4
$f(x)$	70	40	10	-20

2.

x	1	2	3	4
$g(x)$	40	32	26	22

3.

x	1	2	3	4
$h(x)$	70	49	34.3	24.01

4.

x	1	2	3	4
$k(x)$	90	80	70	60

5.

x	1	2	3	4
$m(x)$	80	61	42.9	25.61

6.

x	1	2	3	4
$n(x)$	90	81	72.9	65.61

7. A population numbers 11,000 organisms initially and grows by 8.5% each year. Write an exponential model for the population.

8. A population is currently 6,000 and has been increasing by 1.2% each day. Write an exponential model for the population.

9. The fox population in a certain region has an annual growth rate of 9 percent per year. It is estimated that the population in the year 2010 was 23,900. Estimate the fox population in the year 2018.

10. The amount of area covered by blackberry bushes in a park has been growing by 12% each year. It is estimated that the area covered in 2009 was 4,500 square feet. Estimate the area that will be covered in 2020.

11. A vehicle purchased for $32,500 depreciates at a constant rate of 5% each year. Determine the approximate value of the vehicle 12 years after purchase.

12. A business purchases $125,000 of office furniture which depreciates at a constant rate of 12% each year. Find the residual value of the furniture 6 years after purchase.

Find a formula for an exponential function passing through the two points.

13. $(0,\ 6)$, $(3,\ 750)$

14. $(0,\ 3)$, $(2,\ 75)$

15. $(0,\ 2000)$, $(2,\ 20)$

16. $(0,\ 9000)$, $(3,\ 72)$

17. $\left(-1, \frac{3}{2}\right)$, $(3,\ 24)$

18. $\left(-1, \frac{2}{5}\right)$, $(1, 10)$

19. $(-2, 6)$, $(3, 1)$

20. $(-3, 4)$, $(3,\ 2)$

21. $(3,\ 1)$, $(5,\ 4)$

22. $(2, 5)$, $(6,\ 9)$

23. A radioactive substance decays exponentially. A scientist begins with 100 milligrams of a radioactive substance. After 35 hours, 50 mg of the substance remains. How many milligrams will remain after 54 hours?

24. A radioactive substance decays exponentially. A scientist begins with 110 milligrams of a radioactive substance. After 31 hours, 55 mg of the substance remains. How many milligrams will remain after 42 hours?

25. A house was valued at \$110,000 in the year 1985. The value appreciated to \$145,000 by the year 2005. What was the annual growth rate between 1985 and 2005? Assume that the house value continues to grow by the same percentage. What did the value equal in the year 2010?

26. An investment was valued at \$11,000 in the year 1995. The value appreciated to \$14,000 by the year 2008. What was the annual growth rate between 1995 and 2008? Assume that the value continues to grow by the same percentage. What did the value equal in the year 2012?

27. A car was valued at \$38,000 in the year 2003. The value depreciated to \$11,000 by the year 2009. Assume that the car value continues to drop by the same percentage. What was the value in the year 2013?

28. A car was valued at \$24,000 in the year 2006. The value depreciated to \$20,000 by the year 2009. Assume that the car value continues to drop by the same percentage. What was the value in the year 2014?

29. If \$4,000 is invested in a bank account at an interest rate of 7 per cent per year, find the amount in the bank after 9 years if interest is compounded annually, quarterly, monthly, and continuously.

30. If \$6,000 is invested in a bank account at an interest rate of 9 per cent per year, find the amount in the bank after 5 years if interest is compounded annually, quarterly, monthly, and continuously.

31. Find the annual percentage yield (APY) for a savings account with annual percentage rate of 3% compounded quarterly.

32. Find the annual percentage yield (APY) for a savings account with annual percentage rate of 5% compounded monthly.

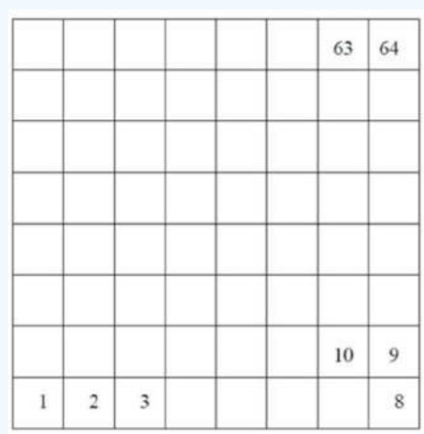
33. A population of bacteria is growing according to the equation $P(t) = 1600e^{0.21\,t}$, with t measured in years. Estimate when the population will exceed 7569.

34. A population of bacteria is growing according to the equation $P(t) = 1200e^{0.17\,t}$, with t measured in years. Estimate when the population will exceed 3443.

35. In 1968, the U.S. minimum wage was \$1.60 per hour. In 1976, the minimum wage was \$2.30 per hour. Assume the minimum wage grows according to an exponential model $w(t)$, where t represents the time in years after 1960. [UW]

a. Find a formula for $w(t)$.
b. What does the model predict for the minimum wage in 1960?
c. If the minimum wage was \$5.15 in 1996, is this above, below or equal to what the model predicts?

36. In 1989, research scientists published a model for predicting the cumulative number of AIDS cases (in thousands) reported in the United States: $a(t) = 155\left(\frac{t-1980}{10}\right)^3$, where t is the year. This paper was considered a "relief", since there was a fear the correct model would be of exponential type. Pick two data points predicted by the research model $a(t)$ to construct a new exponential model $b(t)$ for the number of cumulative AIDS cases. Discuss how the two models differ and explain the use of the word "relief." [UW]



37. You have a chess board as pictured, with squares numbered 1 through 64. You also have a huge change jar with an unlimited number of dimes. On the first square you place one dime. On the second square you stack 2 dimes. Then you continue, always doubling the number from the previous square. [UW]

a. How many dimes will you have stacked on the 10th square?
b. How many dimes will you have stacked on the nth square?
c. How many dimes will you have stacked on the 64th square?
d. Assuming a dime is 1 mm thick, how high will this last pile be?
e. The distance from the earth to the sun is approximately 150 million km. Relate the height of the last pile of dimes to this distance.

Answer

1. Linear

3. Exponential

5. Neither

7. $P(t)=11,000(1.085)^{t}$

9. 47622 Fox

11. $17561.70

13. $y=.6(5)^{x}$

15. $y=2000(0.1)^{x}$

17. $y=3(2)^{x}$

19. $y=(\frac{1}{6})^{-\frac{3}{5}}(\frac{1}{6})^{\frac{x}{5}}=2.93(0.699)^{x}$

21. $y=\frac{1}{8}(2)^{x}$

23. 34.32 mg

25. 1.39%; $155,368.09

27. $4,813.55

29. Annual ≈ $7353.84
Quarterly ≈ $7,469.63
Monthly ≈ $7,496.71
Continuously ≈ $7,510.44

31. 3.03%

33. 7.4 years

35a. $w(t)=(1.113)(1.046)^{t}$
b. $1.11
c. Below what the model predicts ≈ $5.70

2.5: Compound Interest

2.4 Learning Objectives

- Use the compound interest formula to compute beginning or ending balance in an account

With simple interest, we were assuming that we pocketed the interest when we received it. In a standard bank account, any interest we earn is automatically added to our balance, and we earn interest on that interest in future years. This reinvestment of interest is called **compounding**.

Suppose that we deposit $1000 in a bank account offering 3% interest, compounded monthly. How will our money grow?

The 3% interest is an annual percentage rate (APR) – the total interest to be paid during the year. Since interest is being paid monthly, each month, we will earn $\frac{3\%}{12} = 0.25\%$ per month.

In the first month,

$P_0 = \$1000$

$r = 0.0025(0.25\%)$

$I = \$1000(0.0025) = \$2.50$

$A = \$1000 + \$2.50 = \$1002.50$

In the first month, we will earn $2.50 in interest, raising our account balance to $1002.50.

In the second month,

$P_0 = \$1002.50$

$I = \$1002.50(0.0025) = \$2.51$ (rounded)

$A = \$1000 + \$2.50 = \$1002.50$

Notice that in the second month we earned more interest than we did in the first month. This is because we earned interest not only on the original $1000 we deposited, but we also earned interest on the $2.50 of interest we earned the first month. This is the key advantage that **compounding** of interest gives us.

Calculating out a few more months:

Month	Starting balance	Interest earned	Ending Balance
1	1000.00	2.50	1002.50
2	1002.50	2.51	1005.01
3	1005.01	2.51	1007.52
4	1007.52	2.52	1010.04
5	1010.04	2.53	1012.57
6	1012.57	2.53	1015.10
7	1015.10	2.54	1017.64
8	1017.64	2.54	1020.18
9	1020.18	2.55	1022.73
10	1022.73	2.56	1025.29
11	1025.29	2.56	1027.85
12	1027.85	2.57	1030.42

To find an equation to represent this, if P_m represents the amount of money after m months, then we could write the recursive equation:

$P_0 = \$1000$

$P_m = (1 + 0.0025)P_{m-1}$

You probably recognize this as the recursive form of exponential growth. If not, we could go through the steps to build an explicit equation for the growth:

$P_0 = \$1000$

$P_1 = 1.0025P_0 = 1.0025(1000)$

$P_2 = 1.0025P_1 = 1.0025(1.0025(1000)) = 1.0025^2(1000)$

$P_3 = 1.0025P_2 = 1.0025\left(1.0025^2(1000)\right) = 1.0025^3(1000)$

$P_4 = 1.0025P_3 = 1.0025\left(1.0025^3(1000)\right) = 1.0025^4(1000)$

Observing a pattern, we could conclude

$P_m = (1.0025)^m(\$1000)$

Notice that the $1000 in the equation was P_0, the starting amount. We found 1.0025 by adding one to the growth rate divided by 12, since we were compounding 12 times per year.

Generalizing our result, we could write

$A = P_0\left(1+\frac{r}{n}\right)^m$

In this formula:

m is the number of compounding periods (months in our example)

r is the annual interest rate

n is the number of compounds per year.

While this formula works fine, it is more common to use a formula that involves the number of years, rather than the number of compounding periods. If t is the number of years, then $m = tn$. Making this change gives us the standard formula for compound interest.

Compound Interest

$A = P_0\left(1+\frac{r}{n}\right)^{tn}$

A is the balance in the account after N years.

P_0 is the starting balance of the account (also called initial deposit, or principal)

r is the annual interest rate in decimal form

n is the number of compounding periods in one year.

If the compounding is done annually (once a year), $n = 1$.

If the compounding is done quarterly, $n = 4$.

If the compounding is done monthly, $n = 12$.

If the compounding is done daily, $n = 365$.

The most important thing to remember about using this formula is that it assumes that we put money in the account once and let it sit there earning interest.

Example 1

A certificate of deposit (CD) is a savings instrument that many banks offer. It usually gives a higher interest rate, but you cannot access your investment for a specified length of time. Suppose you deposit $3000 in a CD paying 6% interest, compounded monthly. How much will you have in the account after 20 years?

Solution

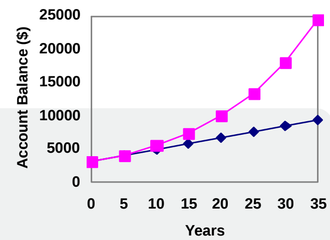
In this example,

$P_0 = \$3000$	the initial deposit
$r = 0.06$	6% annual rate
$n = 12$	12 months in 1 year
$t = 20$	since we're looking for how much we'll have after 20 years

So $A = 3000\left(1+\frac{0.06}{12}\right)^{20\times 12} = \$9930.61$ (round your answer to the nearest penny)

Let us compare the amount of money earned from compounding against the amount you would earn from simple interest

Years	Simple Interest (\$15 per month)	6% compounded monthly $=0.5\%$ each month.
5	\$3900	\$4046.55
10	\$4800	\$5458.19
15	\$5700	\$7362.28
20	\$6600	\$9930.61
25	\$7500	\$13394.91
30	\$8400	\$18067.73
35	\$9300	\$24370.65



As you can see, over a long period of time, compounding makes a large difference in the account balance. You may recognize this as the difference between linear growth and exponential growth.

Evaluating exponents on the calculator

When we need to calculate something like 5^3 it is easy enough to just multiply $5\cdot 5\cdot 5=125$. But when we need to calculate something like 1.005^{240}, it would be very tedious to calculate this by multiplying 1.005 by itself 240 times! So to make things easier, we can harness the power of our scientific calculators.

Most scientific calculators have a button for exponents. It is typically either labeled like:

[∧], $[y^x]$, or $[x^y]$

To evaluate 1.005^{240} we'd type 1.005 [∧] 240, or 1.005 $[y^x]$ 240. Try it out - you should get something around 3.3102044758.

Example 2

You know that you will need \$40,000 for your child's education in 18 years. If your account earns 4% compounded quarterly, how much would you need to deposit now to reach your goal?

Solution

We're looking for P_0.

$r = 0.04$	4%
$n = 4$	4 quarters in 1 year
$t = 18$	Since we know the balance in 18 years
$A = \$40,000$	The amount we have in 18 years

In this case, we're going to have to set up the equation, and solve for P_0.

$$40000 = P_0\left(1+\frac{0.04}{4}\right)^{18\times 4}$$

$$40000 = P_0(2.0471)$$

$$P_0 = \frac{40000}{2.0471} = \$19539.84$$

So you would need to deposit \$19,539.84 now to have \$40,000 in 18 years.

Rounding

It is important to be very careful about rounding when calculating things with exponents. In general, you want to keep as many decimals during calculations as you can. Be sure to **keep at least 3 significant digits** (numbers after any leading zeros). Rounding 0.00012345 to 0.000123 will usually give you a "close enough" answer, but keeping more digits is always better.

Example 3

To see why not over-rounding is so important, suppose you were investing $1000 at 5% interest compounded monthly for 30 years.

Solution

$P_0 = \$1000$ the initial deposit
$r = 0.05$ 5%
$n = 12$ 12 months in 1 year
$t = 30$ since we're looking for the amount after 30 years

If we first compute $\frac{r}{n}$, we find $\frac{0.05}{12} = 0.00416666666667$

Here is the effect of rounding this to different values:

r/n rounded to:	Gives P_{30} to be:	Error
0.004	$4208.59	$259.15
0.0042	$4521.45	$53.71
0.00417	$4473.09	$5.35
0.004167	$4468.28	$0.54
0.0041667	$4467.80	$0.06
no rounding	$4467.74	

If you're working in a bank, of course you wouldn't round at all. For our purposes, the answer we got by rounding to 0.00417, three significant digits, is close enough - $5 off of $4500 isn't too bad. Certainly keeping that fourth decimal place wouldn't have hurt.

Using your calculator

In many cases, you can avoid rounding completely by how you enter things in your calculator. For example, in the example above, we needed to calculate

$$A = 1000\left(1 + \frac{0.05}{12}\right)^{12\times 30}$$

We can quickly calculate $12 \times 30 = 360$, giving $A = 1000\left(1 + \frac{0.05}{12}\right)^{360}$.

Now we can use the calculator.

Type this	Calculator shows
0.05[÷]12[=]	0.00416666666667
[+]11[=]	1.00416666666667
[y^x]360[=]	4.46774431400613
[×]1000[=]	4467.74431400613

Using your calculator continued

The previous steps were assuming you have a "one operation at a time" calculator; a more advanced calculator will often allow you to type in the entire expression to be evaluated. If you have a calculator like this, you will probably just need to enter:

1000 [×] (1 [+] 0.05 [÷] 12) [y^x] 360 [=].

2.5.1: Exercises

Section 2.4.1 Exercises

1. You deposit $300 in an account earning 5% interest compounded annually. How much will you have in the account in 10 years?
2. How much will $1000 deposited in an account earning 7% interest compounded annually be worth in 20 years?
3. You deposit $2000 in an account earning 3% interest compounded monthly.
   a. How much will you have in the account in 20 years?
   b. How much interest will you earn?
4. You deposit $10,000 in an account earning 4% interest compounded monthly.
   a. How much will you have in the account in 25 years?
   b. How much interest will you earn?
5. How much would you need to deposit in an account now in order to have $6,000 in the account in 8 years? Assume the account earns 6% interest compounded monthly.
6. How much would you need to deposit in an account now in order to have $20,000 in the account in 4 years? Assume the account earns 5% interest compounded weekly.

2.6: Annuities

2.5 Learning Objectives

- Use the annuity formula to determine the balance in an account
- Use the annuity formula to determine what regular deposits are needed to achieve a goal
- Determine when to use the compound interest formula and when to use the annuity formula

For most of us, we aren't able to put a large sum of money in the bank today. Instead, we save for the future by depositing a smaller amount of money from each paycheck into the bank. This idea is called a **savings annuity**. Most retirement plans like 401k plans or IRA plans are examples of savings annuities.

An annuity can be described recursively in a fairly simple way. Recall that basic compound interest follows from the relationship

$$P_m=\left(1+\frac{r}{n}\right)P_{m-1}$$

For a savings annuity, we simply need to add a deposit, d, to the account with each compounding period:

$$P_m=\left(1+\frac{r}{n}\right)P_{m-1}+d$$

Taking this equation from recursive form to explicit form is a bit trickier than with compound interest. It will be easiest to see by working with an example rather than working in general.

Suppose we will deposit \$100 each month into an account paying 6% interest. We assume that the account is compounded with the same frequency as we make deposits unless stated otherwise. In this example:

$r=0.06$ (6%)

$n=12$ (12 compounds/deposits per year)

$d=\$100$ (our deposit per month)

Writing out the recursive equation gives

$$P_m=\left(1+\frac{0.06}{12}\right)P_{m-1}+100=(1.005)P_{m-1}+100$$

Assuming we start with an empty account, we can begin using this relationship:

$P_0=0$

$P_1=(1.005)P_0+100=100$

$P_2=(1.005)P_1+100=(1.005)(100)+100=100(1.005)+100$

$P_3=(1.005)P_2+100=(1.005)(100(1.005)+100)+100=100(1.005)^2+100(1.005)+100$

Continuing this pattern, after m deposits, we'd have saved:

$$P_m=100(1.005)^{m-1}+100(1.005)^{m-2}+\cdots+100(1.005)+100$$

In other words, after m months, the first deposit will have earned compound interest for $m-1$ months. The second deposit will have earned interest for $m\text{-}-2$ months. Last months deposit would have earned only one month worth of interest. The most recent deposit will have earned no interest yet.

This equation leaves a lot to be desired, though – it doesn't make calculating the ending balance any easier! To simplify things, multiply both sides of the equation by 1.005:

$$1.005P_m=1.005\left(100(1.005)^{m-1}+100(1.005)^{m-2}+\cdots+100(1.005)+100\right)$$

Distributing on the right side of the equation gives

$$1.005P_m=100(1.005)^m+100(1.005)^{m-1}+\cdots+100(1.005)^2+100(1.005)$$

Now we'll line this up with like terms from our original equation, and subtract each side

$$\begin{aligned} 1.005P_m &= 100(1.005)^m + 100(1.005)^{m-1} + \cdots + \quad 100(1.005) \\ P_m &= \quad 100(1.005)^{m-1} + \cdots + \quad 100(1.005) + 100 \end{aligned}$$

Almost all the terms cancel on the right hand side when we subtract, leaving

$$1.005P_m - P_m = 100(1.005)^m - 100$$

Solving for P_m

$$0.005P_m = 100\left((1.005)^m - 1\right)$$

$$P_m = \frac{100\left((1.005)^m - 1\right)}{0.005}$$

Replacing m months with $12t$, where t is measured in years, gives

$$P_t = \frac{100\left((1.005)^{12t} - 1\right)}{0.005}$$

Recall 0.005 was $\frac{r}{k}$ and 100 was the deposit d. 12 was k, the number of deposit each year. Generalizing this result, we get the saving annuity formula.

Annuity Formula

$$A = \frac{d\left[\left(1+\frac{r}{n}\right)^{tn} - 1\right]}{\left(\frac{r}{n}\right)}$$

A is the balance in the account after t years.

d is the regular deposit (the amount you deposit each year, each month, etc.)

r is the annual interest rate in decimal form.

n is the number of compounding periods in one year.

If the compounding frequency is not explicitly stated, assume there are the same number of compounds in a year as there are deposits made in a year.

For example, if the compounding frequency isn't stated:

If you make your deposits every month, use monthly compounding, $n = 12$.

If you make your deposits every year, use yearly compounding, $n = 1$.

If you make your deposits every quarter, use quarterly compounding, $n = 4$.

Etc.

When do you use this

Annuities assume that you put money in the account on a regular schedule (every month, year, quarter, etc.) and let it sit there earning interest.

Compound interest assumes that you put money in the account once and let it sit there earning interest.

Compound interest: One deposit

Annuity: Many deposits.

Example 1

A traditional individual retirement account (IRA) is a special type of retirement account in which the money you invest is exempt from income taxes until you withdraw it. If you deposit $100 each month into an IRA earning 6% interest, how much will you have in the account after 20 years?

Solution

In this example,

$$\begin{array}{ll} d=\$100 & \text{the monthly deposit} \\ r=0.06 & 6\% \text{ annual rate} \\ n=12 & \text{since we're doing monthly deposits, we'll compound monthly} \\ t=20 & \text{we want the amount after 20 years} \end{array}$$

Putting this into the equation:

$$A=\frac{100\left[\left(1+\frac{0.06}{12}\right)^{20(12)}-1\right]}{\left(\frac{0.06}{12}\right)}$$

$$A=\frac{100\left((1.005)^{240}-1\right)}{(0.005)}$$

$$A=\frac{100(3.310-1)}{(0.005)}$$

$$A=\frac{100(2.310)}{(0.005)}=\$46200$$

The account will grow to $46,200 after 20 years.

Notice that you deposited into the account a total of $24,000 ($100 a month for 240 months). The difference between what you end up with and how much you put in is the interest earned. In this case it is $\$46,200-\$24,000=\$22,200$.

Example 2

You want to have $200,000 in your account when you retire in 30 years. Your retirement account earns 8% interest. How much do you need to deposit each month to meet your retirement goal?

Solution

In this example,

We're looking for d.

$$\begin{array}{ll} r=0.08 & 8\% \text{ annual rate} \\ n=12 & \text{since we're doing monthly deposits, we'll compound monthly} \\ t=30 & 30 \text{ years} \\ A=\$200,000 & \text{The amount we want to have in 30 years} \end{array}$$

In this case, we're going to have to set up the equation, and solve for d.

$$200,000=\frac{d\left[\left(1+\frac{0.08}{12}\right)^{30(12)}-1\right]}{\left(\frac{0.08}{12}\right)}$$

$$200,000=\frac{d\left((1.00667)^{360}-1\right)}{(0.00667)}$$

$$200,000=d(1491.57)$$

$$d=\frac{200,000}{1491.57}=\$134.09$$

So you would need to deposit $\$134.09$ each month to have $\$200,000$ in 30 years if your account earns 8% interest

Try it Now 1

A more conservative investment account pays 3% interest. If you deposit $5 a day into this account, how much will you have after 10 years? How much is from interest?

Answer

$$
\begin{array}{ll}
d=\$5 & \text{the daily deposit} \\
r=0.03 & 3\% \text{ annual rate} \\
n=365 & \text{since we're doing daily deposits, we'll compound daily} \\
t=10 & \text{we want the amount after 10 years}
\end{array}
$$

$$A=\frac{5\left[\left(1+\frac{0.03}{365}\right)^{365\times 10}-1\right]}{\frac{0.03}{365}}=\$21,282.07$$

We would have deposited a total of $\$5\cdot 365\cdot 10=\$18,250$, so \$3,032.07 is from interest

2.6.1: Exercises

Section 2.5.1 Exercises

1. You wish to have \$3000 in 2 years to buy a fancy new stereo system. How much should you deposit each quarter into an account paying 8% compounded quarterly?
2. You deposit \$200 each month into an account earning 3% interest compounded monthly.
   a. How much will you have in the account in 30 years?
   b. How much total money will you put into the account?
   c. How much total interest will you earn?
3. You deposit \$1000 each year into an account earning 8% compounded annually.
   a. How much will you have in the account in 10 years?
   b. How much total money will you put into the account?
   c. How much total interest will you earn?
4. Raymond has determined he needs to have \$800,000 for retirement in 30 years. His account earns 6% interest.
   a. How much would he need to deposit in the account each month?
   b. How much total money will he put into the account?
   c. How much total interest will he earn?

2.7: Payout Annuities

2.6 Learning Objectives

- Use the payout annuity formula to determine the regular withdrawal amount
- Determine the amount needed in an account to allow for a given payout amount
- Determine when to use the annuity formula and when to use the payout formula

In the last section you learned about annuities. In an annuity, you start with nothing, put money into an account on a regular basis, and end up with money in your account.

In this section, we will learn about a variation called a **Payout Annuity**. With a payout annuity, you start with money in the account, and pull money out of the account on a regular basis. Any remaining money in the account earns interest. After a fixed amount of time, the account will end up empty.

Payout annuities are typically used after retirement. Perhaps you have saved $500,000 for retirement, and want to take money out of the account each month to live on. You want the money to last you 20 years. This is a payout annuity. The formula is derived in a similar way as we did for savings annuities. The details are omitted here.

Payout Annuity Formula

$$P_0=\frac{d\left[1-\left(1+\frac{r}{n}\right)^{-tn}\right]}{\left(\frac{r}{n}\right)}$$

P_0 is the balance in the account at the beginning (starting amount, or principal).

d is the regular withdrawal (the amount you take out each year, each month, etc.)

r is the annual interest rate (in decimal form. Example: $5\% = 0.05$)

n is the number of compounding periods in one year.

t is the number of years we plan to take withdrawals

Like with annuities, the compounding frequency is not always explicitly given, but is determined by how often you take the withdrawals.

When do you use this

Payout annuities assume that you take money from the account on a regular schedule (every month, year, quarter, etc.) and let the rest sit there earning interest.

Compound interest: One deposit

Annuity: Many deposits.

Payout Annuity: Many withdrawals

Example 1

After retiring, you want to be able to take $1000 every month for a total of 20 years from your retirement account. The account earns 6% interest. How much will you need in your account when you retire?

Solution

In this example,

$$\begin{array}{ll} d=\$1000 & \text{the monthly withdrawal} \\ r=0.06 & 6\% \text{ annual rate} \\ n=12 & \text{since we're doing monthly withdrawals, we'll compound monthly} \\ t=20 & \text{ since we're taking withdrawals for 20 years} \end{array}$$

We're looking for P_0; how much money needs to be in the account at the beginning.

Putting this into the equation:

$$P_0=\frac{1000\left[1-\left(1+\frac{0.06}{12}\right)^{-20(12)}\right]}{\left(\frac{0.06}{12}\right)}$$

$$P_0=\frac{1000\times\left(1-(1.005)^{-240}\right)}{(0.005)}$$

$$P_0=\frac{1000\times(1-0.302)}{(0.005)}=\$139,600$$

You will need to have $139,600 in your account when you retire.

Notice that you withdrew a total of $240,000 ($1000 a month for 240 months). The difference between what you pulled out and what you started with is the interest earned. In this case it is $\$240,000-\$139,600=\$100,400$ in interest.

Evaluating negative exponents on your calculator

With these problems, you need to raise numbers to negative powers. Most calculators have a separate button for negating a number that is different than the subtraction button. Some calculators label this [(-)], some with [+/-] . The button is often near the = key or the decimal point.

If your calculator displays operations on it (typically a calculator with multiline display), to calculate 1.005^{-240}you'd type something like: $1.005[\wedge][(-)]240$

If your calculator only shows one value at a time, then usually you hit the (-) key after a number to negate it, so you'd hit: $1.005[y^x]240[(-)]=$

Give it a try - you should get $1.005^{-240}=0.302096$

Example 2

You know you will have $500,000 in your account when you retire. You want to be able to take monthly withdrawals from the account for a total of 30 years. Your retirement account earns 8% interest. How much will you be able to withdraw each month?

Solution

In this example,

We're looking for d.

$$\begin{array}{ll} r=0.08 & 8\% \text{ annual rate} \\ n=12 & \text{since we're doing monthly withdrawals} \\ t=30 & \text{ since were taking withdrawals for 30 years} \\ P_0=\$500,000 & \text{we are beginning with } \$500,000 \end{array}$$

In this case, we're going to have to set up the equation, and solve for d.

$$500,000=\frac{d\left[1-\left(1+\frac{0.08}{12}\right)^{-30(12)}\right]}{\left(\frac{0.08}{12}\right)}$$

$$500,000=\frac{d\left(1-(1.00667)^{-360}\right)}{(0.00667)}$$

$$500,000 = d(136.232)$$

$$d = \frac{500,000}{136.232} = \$3670.21$$

You would be able to withdraw $3,670.21 each month for 30 years.

Try it Now 1

A donor gives $100,000 to a university, and specifies that it is to be used to give annual scholarships for the next 20 years. If the university can earn 4% interest, how much can they give in scholarships each year?

Answer

$$\begin{array}{ll} d = \text{ unknown} & \\ r = 0.04 & 4\% \text{ annual rate} \\ n = 1 & \text{since we're doing annual scholarships} \\ t = 20 & \text{ since were taking withdrawals for 20 years} \\ P_0 = \$100,000 & \text{we are starting with } \$100,000 \end{array}$$

$$100,000 = \frac{d\left[1-\left(1+\frac{0.04}{1}\right)^{-20\times 1}\right]}{\frac{0.04}{1}}$$

Solving for d gives $7,358.18 each year that they can give in scholarships.

It is worth noting that usually donors instead specify that only interest is to be used for scholarship, which makes the original donation last indefinitely. If this donor had specified that, $\$100,000(0.04) = \$4,000$ a year would have been available.

2.7.1: Exercises

Section 2.6.1 Exercises

1. You want to be able to withdraw $30,000 each year for 25 years. Your account earns 8% interest compounded annually.
   a. How much do you need in your account at the beginning?
   b. How much total money will you pull out of the account?
   c. How much of that money is interest?
2. How much money will I need to have at retirement so I can withdraw $60,000 a year for 20 years from an account earning 8% compounded annually?
   a. How much do you need in your account at the beginning
   b. How much total money will you pull out of the account?
   c. How much of that money is interest?
3. You have $500,000 saved for retirement. Your account earns 6% interest compounded monthly. How much will you be able to pull out each month, if you want to be able to take withdrawals for 20 years?
4. Loren already knows that he will have $500,000 when he retires. If he sets up a payout annuity for 30 years in an account paying 10% interest compounded monthly, how much could the annuity provide each month?

2.8: Installment Loans

2.7 Learning Objectives

- Determine the payment on an installment loan
- Determine amount that can be borrowed on an installment loan given the amount of the loan payment

In the last section, you learned about payout annuities.

In this section, you will learn about conventional loans (also called amortized loans or installment loans). Examples include auto loans and home mortgages. These techniques do not apply to payday loans, add-on loans, or other loan types where the interest is calculated up front.

One great thing about loans is that they use exactly the same formula as a payout annuity. To see why, imagine that you had $10,000 invested at a bank, and started taking out payments while earning interest as part of a payout annuity, and after 5 years your balance was zero. Flip that around, and imagine that you are acting as the bank, and a car lender is acting as you. The car lender invests $10,000 in you. Since you're acting as the bank, you pay interest. The car lender takes payments until the balance is zero.

Loans Formula

$$P_0 = \frac{d\left[1-\left(1+\frac{r}{n}\right)^{-tn}\right]}{\left(\frac{r}{n}\right)}$$

P_0 is the balance in the account at the beginning (the principal, or amount of the loan).

d is your loan payment (your monthly payment, annual payment, etc)

r is the annual interest rate in decimal form.

n is the number of compounding periods in one year.

t is the length of the loan, in years

Like before, the compounding frequency is not always explicitly given, but is determined by how often you make payments.

When do you use this

The loan formula assumes that you make loan payments on a regular schedule (every month, year, quarter, etc.) and are paying interest on the loan.

Compound interest: One deposit

Annuity: Many deposits.

Payout Annuity: Many withdrawals

Loans: Many payments

Example 1

You can afford $200 per month as a car payment. If you can get an auto loan at 3% interest for 60 months (5 years), how expensive of a car can you afford? In other words, what amount loan can you pay off with $200 per month?

Solution

In this example,

$$\begin{array}{ll} d=\$200 & \text{the monthly loan payment} \\ r=0.03 & 3\% \text{ annual rate} \\ n=12 & \text{since we're doing monthly payments, we'll compound monthly} \\ t=5 & \text{since we're making monthly payments for 5 years} \end{array}$$

We're looking for P_0, the starting amount of the loan.

$$P_0=\frac{200\left[1-\left(1+\frac{0.03}{12}\right)^{-5(2)}\right]}{\left(\frac{0.03}{12}\right)}$$

$$P_0=\frac{200\left(1-(1.0025)^{-60}\right)}{(0.0025)}$$

$$P_0=\frac{200(1-0.861)}{(0.0025)}=\$11,120$$

You can afford a $\$11,120$ loan.

You will pay a total of $12,000 ($200 per month for 60 months) to the loan company. The difference between the amount you pay and the amount of the loan is the interest paid. In this case, you're paying $\$12,000-\$11,120=\$880$ interest total.

Example 2

You want to take out a $140,000 mortgage (home loan). The interest rate on the loan is 6%, and the loan is for 30 years. How much will your monthly payments be?

Solution

In this example,

We're looking for d.

$$\begin{array}{ll} r=0.06 & 6\% \text{ annual rate} \\ n=12 & \text{since we're doing monthly payments, we'll compound monthly} \\ t=30 & \text{since we're making monthly payments for 30 years} \\ P_0=\$140,000 & \text{the starting loan amount} \end{array}$$

In this case, we're going to have to set up the equation, and solve for d.

$$140,000=\frac{d\left[1-\left(1+\frac{0.06}{12}\right)^{-30(12)}\right]}{\left(\frac{0.06}{12}\right)}$$

$$140,000=\frac{d\left(1-(1.005)^{-360}\right)}{(0.005)}$$

$$140,000=d(166.792)$$

$$d=\frac{140,000}{166.792}=\$839.37$$

You will make payments of $839.37 per month for 30 years.

You're paying a total of $\$302,173.20$ to the loan company: $\$839.37$ per month for 360 months. You are paying a total of $\$302,173.20-\$140,000=\$162,173.20$ in interest over the life of the loan.

Try it Now 1

Janine bought $3,000 of new furniture on credit. Because her credit score isn't very good, the store is charging her a fairly high interest rate on the loan: 16%. If she agreed to pay off the furniture over 2 years, how much will she have to pay each month?

Answer

$$\begin{array}{ll} d = & \text{unknown} \\ r = 0.16 & 16\% \text{ annual rate} \\ n = 12 & \text{since we're doing monthly payments, we'll compound monthly} \\ t = 2 & 2 \text{ year to repay} \\ P_0 = 3,000 & \text{the starting loan amount } \$3,000 \text{ loan} \end{array}$$

$$3,000 = \frac{d\left[1-\left(1+\frac{0.16}{12}\right)^{-2\times 12}\right]}{\frac{0.16}{12}}$$

Solving for d gives $\$146.89$ as monthly payments.

In total, she will pay $\$3,525.36$ to the store, meaning she will pay $\$525.36$ in interest over the two years.

2.8.1: Exercises

Section 2.7.1 Exercises

1. You can afford a $700 per month mortgage payment. You've found a 30 year loan at 5% interest.
    a. How big of a loan can you afford?
    b. What would be the total payments made on the loan?
    c. How much of that money is interest?
2. Marie can afford a $250 per month car payment. She's found a 5 year loan at 7% interest.
    a. How expensive of a car can she afford?
    b. What would be the total payments made on the loan?
    c. How much of that money is interest?
3. You want to buy a $25,000 car. The company is offering a 2% interest rate for 48 months (4 years). What will your monthly payments be?
4. You decide finance a $12,000 car at 3% compounded monthly for 4 years. What will your monthly payments be? How much interest will you pay over the life of the loan?
5. You want to buy a $200,000 home. You plan to pay 10% as a down payment, and take out a 30 year loan for the rest.
    a. How much is the loan amount going to be?
    b. What will your monthly payments be if the interest rate is 5%?
    c. What will your monthly payments be if the interest rate is 6%?
6. Lynn bought a $300,000 house, paying 10% down, and financing the rest at 6% interest for 30 years.
    a. Find her monthly payments.
    b. How much interest will she pay over the life of the loan?
7. Emile bought a car for $24,000 three years ago. The loan had a 5 year term at 3% interest rate, making monthly payments. How much does he still owe on the car?
8. A friend bought a house 15 years ago, taking out a $120,000 mortgage at 6% for 30 years, making monthly payments. How much does she still owe on the mortgage?

CHAPTER OVERVIEW

3: Module 3- Probability and Statistics

3.1: Basics of Sets

An art collector might own a collection of paintings, while a music lover might keep a collection of CDs. Any collection of items can form a **set**.

Set

A **set** is a collection of distinct objects, called **elements** of the set

A set can be defined by describing the contents, or by listing the elements of the set, enclosed in curly brackets.

Example 1

Some examples of sets defined by describing the contents:

a. The set of all even numbers
b. The set of all books written about travel to Chile

Some examples of sets defined by listing the elements of the set:

a. {1, 3, 9, 12}
b. {red, orange, yellow, green, blue, indigo, purple}

A set simply specifies the contents; order is not important. The set represented by {1, 2, 3} is equivalent to the set {3, 1, 2}.

Notation

Commonly, we will use a variable to represent a set, to make it easier to refer to that set later.

The symbol $\in$ means "is an element of".

A set that contains no elements, {}, is called the **empty set** and is notated $\emptyset$

Example 2

Let $A = \{1, 2, 3, 4\}$

To notate that 2 is element of the set, we'd write $2 \in A$

Sometimes a collection might not contain all the elements of a set. For example, Chris owns three Madonna albums. While Chris's collection is a set, we can also say it is a **subset** of the larger set of all Madonna albums.

Subset

A **subset** of a set A is another set that contains only elements from the set A, but may not contain all the elements of A.

If B is a subset of A, we write $B \subseteq A$

A **proper subset** is a subset that is not identical to the original set – it contains fewer elements.

If B is a proper subset of A, we write $B \subset A$

Example 3

Consider these three sets

$A =$ the set of all even numbers $\quad B = \{2, 4, 6\} \quad C = \{2, 3, 4, 6\}$

Here $B \subset A$ since every element of B is also an even number, so is an element of A.

More formally, we could say $B \subset A$ since if $x \in B$, then $x \in A$

It is also true that $B \subset C$.

C is not a subset of A, since C contains an element, 3 , that is not contained in A

✓ Example 4

Suppose a set contains the plays "Much Ado About Nothing", "MacBeth", and "A Midsummer's Night Dream". What is a larger set this might be a subset of?

Solution

There are many possible answers here. One would be the set of plays by Shakespeare. This is also a subset of the set of all plays ever written. It is also a subset of all British literature.

? Try it Now 1

The set $A = \{1, 3, 5\}$. What is a larger set this might be a subset of?

Answer

There are several answers: The set of all odd numbers less than 10. The set of all odd numbers. The set of all integers. The set of all real numbers.

3.2: Union, Intersection, and Complement

Commonly sets interact. For example, you and a new roommate decide to have a house party, and you both invite your circle of friends. At this party, two sets are being combined, though it might turn out that there are some friends that were in both sets.

Union, Interection, and Complement

The **union** of two sets contains all the elements contained in either set (or both sets).

The union is notated $A \cup B$

More formally, $x \in A \cup B$ if $x \in A$ or $x \in B$ (or both)

The **intersection** of two sets contains only the elements that are in both sets.

The intersection is notated $A \cap B$

More formally, $x \in A \cap B$ if $x \in A$ and $x \in B$

The **complement** of a set A contains everything that is *not* in the set A.

The complement is notated A, or A^c, or sometimes $\sim A$.

Example 5

Consider the sets:

$$A = \{\text{ red, green, blue }\} \quad B = \{\text{ red, yellow, orange }\} \quad C = \{\text{ red, orange, yellow, green, blue, purple }\}$$

a. Find $A \cup B$
b. Find $A \cap B$
c. Find $A^c \cap C$

Solution

a) The union contains all the elements in either set: $A \cup B = \{\text{ red, green, blue, yellow, orange }\}$

Notice we only list red once.

b) The intersection contains all the elements in both sets: $A \cap B = \{\text{ red }\}$

c) Here we're looking for all the elements that are not in set A and are also in C.

$A^c \cap C = \{\text{ orange, yellow, purple }\}$

Try it Now 2

Using the sets from the previous example, find $A \cup C$ and $B^c \cap A$

Answer

$A \cup C = \{\text{ red, orange, yellow, green, blue purple }\}$

$B^c \cap A = \{\text{ green, blue }\}$

Notice that in the example above, it would be hard to just ask for A^c, since everything from the color fuchsia to puppies and peanut butter are included in the complement of the set. For this reason, complements are usually only used with intersections, or when we have a universal set in place.

Universal Set

A **universal set** is a set that contains all the elements we are interested in. This would have to be defined by the context.

A complement is relative to the universal set, so A^C contains all the elements in the universal set that are not in A.

Example 6

a. If we were discussing searching for books, the universal set might be all the books in the library.
b. If we were grouping your Facebook friends, the universal set would be all your Facebook friends.
c. If you were working with sets of numbers, the universal set might be all whole numbers, all integers, or all real numbers

Example 7

Suppose the universal set is $U =$ all whole numbers from 1 to 9. If $A = \{1, 2, 4\}$, then

$A^c = \{3, 5, 6, 7, 8, 9\}$

As we saw earlier with the expression $A^c \cap C$, set operations can be grouped together. Grouping symbols can be used like they are with arithmetic - to force an order of operations.

Example 8

Suppose

$H = \{\text{ cat, dog, rabbit, mouse }\}, F = \{\text{ dog, cow, duck, pig, rabbit }\} \quad W = \{\text{ duck, rabbit, deer, frog, mouse }\}$

a. Find $(H \cap P) \cup W$
b. Find $H \cap (F \cup W)$
c. Find $(H \cap P) \cap W$

Solution

a) We start with the intersection: $H \cap F = \{\text{ dog, rabbit }\}$

Now we union that result with W : $(H \cap F) \cup W = \{\text{dog, duck, rabbit, deer, frog, mouse }\}$

b) We start with the union: $F \cup W = \{\text{dog, cow, rabbit, duck, pig, deer, frog, mouse }\}$

Now we intersect that result with H : $H \cap (F \cup W) = \{\text{ dog, rabbit, mouse }\}$

c) We start with the intersection: $H \cap F = \{\text{dog, rabbit }\}$

Now we want to find the elements of W that are not in $\mathrm{H} \cap F$

$(H \cap P)^c \cap W = \{\text{ duck, deer, frog, mouse }\}$

3.3: Venn Diagrams

To visualize the interaction of sets, John Venn in 1880 thought to use overlapping circles, building on a similar idea used by Leonhard Euler in the 18$^{\text{th}}$ century. These illustrations now called **Venn Diagrams**.

Venn Diagram

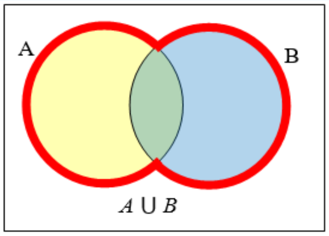
A Venn diagram represents each set by a circle, usually drawn inside of a containing box representing the universal set. Overlapping areas indicate elements common to both sets.

Basic Venn diagrams can illustrate the interaction of two or three sets.

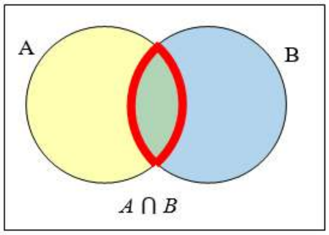
Example 9

Create Venn diagrams to illustrate $A \cup B$, $A \cap B$, and $A^c \cap B$

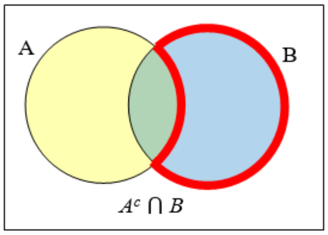
$A \cup B$ contains all elements in either set.



$A \cup B$ contains all elements in either set.



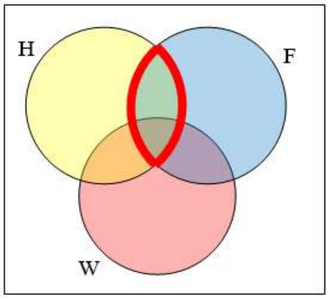
$A \cap B$ contains only those elements in both sets - in the overlap of the circles.



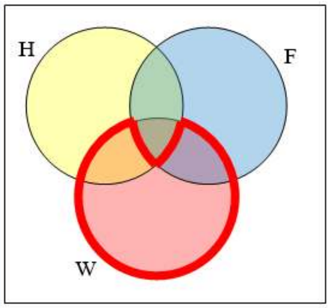
Example 10

Use a Venn diagram to illustrate $(H \cap P)^c \cap W$

We'll start by identifying everything in the set $\mathrm{H} \cap P$



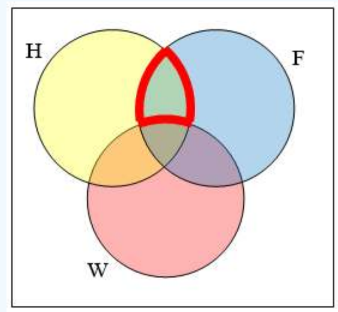
Now, $(H \cap P)^c \cap W$ will contain everything not in the set identified above that is also in set W



✓ Example 11

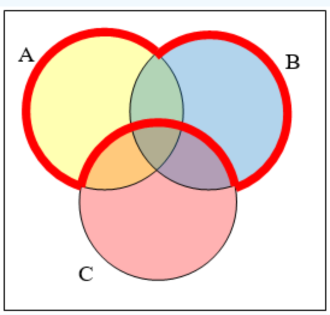
Create an expression to represent the outlined part of the Venn diagram shown.

The elements in the outlined set are in sets H and F, but are not in set W. So we could represent this set as $H \cap F \cap W$



? Try it Now 3

Create an expression to represent the outlined portion of the Venn diagram shown



Answer

$A \cup B \cap C^c$

3.4: Introduction to Probability

4.1 Learning Objectives

- Describe theoretical, empirical, and subjective probability
- Distinguish among the three uses of probability

The probability of a specified event is the chance or likelihood that it will occur. There are several ways of viewing probability.

One would be **experimental** in nature, where we repeatedly conduct an experiment. Suppose we flipped a coin over and over and over again and it came up heads about half of the time; we would expect that in the future whenever we flipped the coin it would turn up heads about half of the time. When a weather reporter says “there is a 10% chance of rain tomorrow,” she is basing that on prior evidence; that out of all days with similar weather patterns, it has rained on 1 out of 10 of those days.

Example 1

Conduct an experiment to determine the probability of the spinner landing on 1?

Solution

Using the experimental method, suppose out of 10,000 spins, 1,711 of those landed on 1. This is normally done on a computer application that can randomly generate outcomes of each spin to avoid someone having to spin the spinner 10,000 times.

To calculate this probability, divide the number of times 1 has occurred which is 1,711 by the number of times the spinner was spun which was 10,000.

The result is 0.1711 or approximately 17% of the time. So 17 times out of 100 spins one would expect the spinner to land on the number 1.

Another view would be **subjective** in nature, in other words an educated guess or opinion. But this is just a guess, with no way to verify its accuracy, and depending upon how educated the educated guesser is, a subjective probability may not be worth very much.

Example 2

Determine the probability that the Seattle Mariners would win their next baseball game.

Solution

It would be impossible to conduct an experiment where the same two teams played each other repeatedly, each time with the same starting lineup and starting pitchers, each starting at the same time of day on the same field under the precisely the same conditions.

Since there are so many variables to take into account, someone familiar with baseball and with the two teams involved might make an educated guess that there is a 75% chance the Mariners will win the game; that is, *if* the same two teams were to play each other repeatedly under identical conditions, the Mariners would win about three out of every four games.

Definition: Probabilities

Empirical Probability uses the results of an experiment to predict the percent chance an event could occur.

Subjective Probability uses intuition or guesswork to predict the percent chance an event could occur.

Theoretical Probability uses the number of possible desired outcomes of an event compared to the number of all possible outcomes of an event to predict the percent chance an event could occur.

Theoretical Probability is defined mathematically as follows:

Suppose there is a situation with n equally likely possible outcomes and that m of those n outcomes correspond to a particular event; then the **probability** of that event is defined as $\frac{m}{n}$.

We will return to the empirical and subjective probabilities from time to time, but in this course we will mostly be concerned with **theoretical** probability.

3.4.1: Exercises

1. In a sample of 50 people who were stating their favorite fruit, 21 liked apples the most, 18 people like peaches the most, 5 people liked bananas the most and 6 liked pineapple the most. The probability that all people like bananas the most is 10%. What type of probability is this?
2. A person wants to determine the probability of winning a raffle. One winner is selected out of 84 entrants. What type of probability is used to answer this question?
3. A mother thinks there is a 100% chance that her son will love the jelly donut because he loves jelly, and he loves donuts. What kind of probability was used here?

3.5: Basic Concepts

4.2 Learning Objectives

- Find the probability of an event
- Find the probabilities certain and impossible events

If you roll a die, pick a card from deck of playing cards, or randomly select a person and observe their hair color, you are executing an experiment or procedure. In probability, we look at the likelihood of different outcomes. We begin with some terminology.

Definition: Events and Outcomes

The result of an experiment is called an **outcome**.

An **event** is any particular outcome or group of outcomes.

A **simple event** is an event that cannot be broken down further

The **sample space** is the set of all possible simple events.

Example 1

If we roll a standard 6-sided die, describe the sample space and some simple events.

The sample space is the set of all possible simple events: $\{1, 2, 3, 4, 5, 6\}$

Some examples of simple events:

We roll a 1

We roll a 5

Some compound events:

We roll a number bigger than 4

We roll an even number


Two dice One die

Definition: Basic Probability

Given that all outcomes are equally likely, we can compute the probability of an event E using this formula:

$$P(E) = \frac{\text{Number of outcomes corresponding to the event E}}{\text{Total number of equally likely outcomes}} \tag{3.5.1}$$

Example 2

If we roll a 6-sided die, calculate

a. P(rolling a 1)
b. P(rolling a number bigger than 4)

Solution

Recall that the sample space is $\{1, 2, 3, 4, 5, 6\}$

a. There is one outcome corresponding to "rolling a 1", so the probability is $\frac{1}{6}$
b. There are two outcomes bigger than a 4, so the probability is $\frac{2}{6} = \frac{1}{3}$

Probabilities are essentially fractions, and can be simplified to lower terms like fractions. Probabilities are also written as decimals or percents.

Example 3

Let's say you have a bag with 20 cherries, 14 sweet and 6 sour. If you pick a cherry at random, what is the probability that it will be sweet?

Solution

There are 20 possible cherries that could be picked, so the number of possible outcomes is 20. Of these 20 possible outcomes, 14 are favorable (sweet), so the probability that the cherry will be sweet is $\frac{14}{20} = \frac{7}{10}$.

There is one potential complication to this example, however. It must be assumed that the probability of picking any of the cherries is the same as the probability of picking any other. This wouldn't be true if (let us imagine) the sweet cherries are smaller than the sour ones. (The sour cherries would come to hand more readily when you sampled from the bag.) Let us keep in mind, therefore, that when we assess probabilities in terms of the ratio of favorable to all potential cases, we rely heavily on the assumption of equal probability for all outcomes.

Try it Now 1

At some random moment, you look at your clock and note the minutes showing.

a. What is probability the minutes reading is 15?
b. What is the probability the minutes reading is 15 or less?

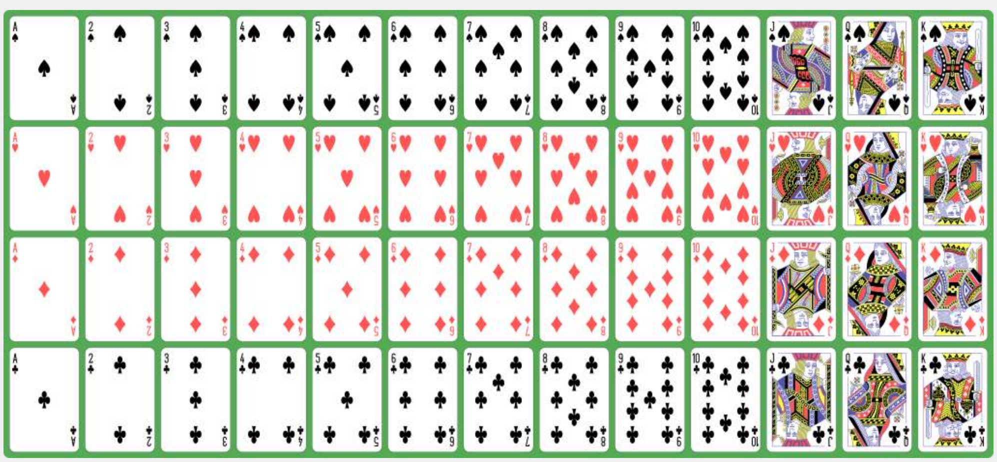
Answer

There are 60 possible readings, from 00 to 59.

a. $\frac{1}{60}$
b. $\frac{16}{60}$ (counting 00 through 15)

Cards

A standard deck of 52 playing cards consists of four **suits** (hearts, spades, diamonds and clubs). Spades and clubs are black while hearts and diamonds are red. Each suit contains 13 cards, each of a different **rank**: an Ace (which in many games functions as both a low card and a high card), cards numbered 2 through 10, a Jack, a Queen and a King.



Example 4

Compute the probability of randomly drawing one card from a deck and getting an Ace.

Solution

There are 52 cards in the deck and 4 Aces so

$$P(Ace) = \frac{4}{52} = \frac{1}{13} \approx 0.0769$$

We can also think of probabilities as percents: There is a 7.69% chance that a randomly selected card will be an Ace.

Notice that the smallest possible probability is 0 – if there are no outcomes that correspond with the event. The largest possible probability is 1 – if all possible outcomes correspond with the event.

Definition: Certain and Impossible Events

An **impossible event** has a probability of 0.

A **certain event** has a probability of 1.

The probability of any event must be $0 \le P(E) \le 1$.

In the course of this chapter, *if you compute a probability and get an answer that is negative or greater than 1, you have made a mistake and should check your work.*

3.5.1: Exercises

1. A ball is drawn randomly from a jar that contains 6 red balls, 2 white balls, and 5 yellow balls. Find the probability of the given event.
   a. A red ball is drawn
   b. A white ball is drawn
2. Suppose you write each letter of the alphabet on a different slip of paper and put the slips into a hat. If you draw one slip of paper from the hat at random, what is the probability of getting:
   a. A consonant?
   b. A vowel?
3. A group of people were asked if they had run a red light in the last year. 150 responded "yes", and 185 responded "no". Find the probability that if a person is chosen at random, they have run a red light in the last year.
4. In a survey, 205 people indicated they prefer cats, 160 indicated they prefer dogs, and 40 indicated they don't enjoy either pet. Find the probability that if a person is chosen at random, they prefer cats.
5. Compute the probability of tossing a six-sided die (with sides numbered 1 through 6) and getting a 5.
6. Compute the probability of tossing a six-sided die and getting a 7.
7. Giving a test to a group of students, the grades and gender are summarized below. If one student was chosen at random, find the probability that the student was female.

	A	B	C	Total
Male	8	18	13	39
Female	10	4	12	26
Total	18	22	25	65

8. The table below shows the number of credit cards owned by a group of individuals. If one person was chosen at random, find the probability that the person had no credit cards.

	Zero	One	Two or more	Total
Male	9	5	19	33
Female	18	10	20	48
Total	27	15	39	81

9. Compute the probability of tossing a six-sided die and getting an even number.
10. Compute the probability of tossing a six-sided die and getting a number less than 3.
11. If you pick one card at random from a standard deck of cards, what is the probability it will be a King?
12. If you pick one card at random from a standard deck of cards, what is the probability it will be a Diamond

3.6: Working with Events

4.3 Learning Objectives

- Find the probability that an event does not happen
- Determine whether or not two events are independent
- Find P(A and B) for independent events
- Determine whether or not two events are mutually exclusive
- Find P(A or B)
- Find P(A|B)
- Determine probabilities from two-ways tables
- Determine odds for and against an event

Complementary Events

Now let us examine the probability that an event does **not** happen. As in the previous section, consider the situation of rolling a six-sided die and first compute the probability of rolling a six: the answer is $P(\text{six})=\frac{1}{6}$. Now consider the probability that we do *not* roll a six: there are 5 outcomes that are not a six, so the answer is $P(\text{not a six})=\frac{5}{6}$. Notice that

$$P(\text{six})+P(\text{not a six})=\frac{1}{6}+\frac{5}{6}=\frac{6}{6}=1 \tag{3.6.1}$$

This is not a coincidence. Consider a generic situation with n possible outcomes and an event E that corresponds to m of these outcomes. Then the remaining $n-m$ outcomes correspond to E not happening, thus

$$P(\text{not } E)=\frac{n-m}{n}=\frac{n}{n}-\frac{m}{n}=1-\frac{m}{n}=1-P(E) \tag{3.6.2}$$

Complement of an Event

The **complement** of an event is the event “E doesn’t happen”

The notation $\bar{E}$ is used for the complement of event E.

We can compute the probability of the complement using $P(\bar{E})=1-P(E)$

Notice also that $P(E)=1-P(\bar{E})$

Example 1

If you pull a random card from a deck of playing cards, what is the probability it is not a heart?

Solution

There are 13 hearts in the deck, so

$$P(\text{heart})=\frac{13}{52}=\frac{1}{4}.$$

The probability of *not* drawing a heart is the complement:

$$P(\text{ not heart })=1-P(\text{ heart })=1-\frac{1}{4}=\frac{3}{4}$$

Probability of Two Independent Events

Example 2

Suppose we flipped a coin and rolled a die, and wanted to know the probability of getting a head on the coin and a 6 on the die.

Solution

We could list all possible outcomes: $\{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$.

Notice there are $2 \cdot 6 = 12$ total outcomes. Out of these, only 1 is the desired outcome, so the probability is $\frac{1}{12}$.

The prior example was looking at two independent events.

Independent Events

Events A and B are **independent events** if the probability of Event B occurring is the same whether or not Event A occurs.

Example 3

Are these events independent?

a. A fair coin is tossed two times. The two events are (1) first toss is a head and (2) second toss is a head.
b. The two events (1) "It will rain tomorrow in Houston" and (2) "It will rain tomorrow in Galveston" (a city near Houston).
c. You draw a red card from a deck, then draw a second red card without replacing the first.

Solution

a. The probability that a head comes up on the second toss is $\frac{1}{2}$ regardless of whether or not a head came up on the first toss, so these events are independent.
b. These events are not independent because it is more likely that it will rain in Galveston on days it rains in Houston than on days it does not.
c. The probability of the second card being red depends on whether the first card is red or not, so these events are not independent.

When two events are independent, the probability of both occurring is the product of the probabilities of the individual events.

$P(A \text{ and } B)$ for Independent Events

If events A and B are independent, then the probability of both A and B occurring is

$$P(A \text{ and } B) = P(A) \cdot P(B) \tag{3.6.3}$$

where $P(A \text{ and } B)$ is the probability of events A and B both occurring, $P(A)$ is the probability of event A occurring, and $P(B)$ is the probability of event B occurring

If you look back at the coin and die example from earlier, you can see how the number of outcomes of the first event multiplied by the number of outcomes in the second event multiplied to equal the total number of possible outcomes in the combined event.

Example 4

In your drawer you have 10 pairs of socks, 6 of which are white, and 7 tee shirts, 3 of which are white. If you randomly reach in and pull out a pair of socks and a tee shirt, what is the probability both are white?

Solution

The probability of choosing a white pair of socks is $\frac{6}{10}$.

The probability of choosing a white tee shirt is $\frac{3}{7}$.

The probability of both being white is $\frac{6}{10} \cdot \frac{3}{7} = \frac{18}{70} = \frac{9}{35}$

Try it Now 1

A card is pulled a deck of cards and noted. The card is then replaced, the deck is shuffled, and a second card is removed and noted. What is the probability that both cards are Aces?

Answer

since the second draw is made after replacing the first card, these events are independent The probability of an ace on each draw is $\frac{4}{52} = \frac{1}{13}$, so the probability of an Ace on both draws is $\frac{1}{13} \cdot \frac{1}{13} = \frac{1}{169}$

The previous examples looked at the probability of *both* events occurring. Now we will look at the probability of *either* event occurring.

Example 5

Suppose we flipped a coin and rolled a die, and wanted to know the probability of getting a head on the coin *or* a 6 on the die.

Solution

Here, there are still 12 possible outcomes:

$\{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$

By simply counting, we can see that 7 of the outcomes have a head on the coin *or* a 6 on the die *or* both – we use *or* inclusively here (these 7 outcomes are $H1, H2, H3, H4, H5, H6, T6$), so the probability is $\frac{7}{12}$. How could we have found this from the individual probabilities?

As we would expect, $\frac{1}{2}$ of these outcomes have a head, and $\frac{1}{6}$ of these outcomes have a 6 on the die. If we add these, $\frac{1}{2} + \frac{1}{6} = \frac{6}{12} + \frac{2}{12} = \frac{8}{12}$, which is not the correct probability. Looking at the outcomes we can see why: the outcome H6 would have been counted twice, since it contains both a head and a 6; the probability of both a head *and* rolling a 6 is $\frac{1}{12}$.

If we subtract out this double count, we have the correct probability: $\frac{8}{12} - \frac{1}{12} = \frac{7}{12}$.

$P(A \text{ or } B)$

Inclusive OR: Either event could occur or both could occur at the same time.

The probability of either A or B occurring (or both) is

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) \tag{3.6.4}$$

Mutually Exclusive OR: Either event could occur but they could not occur at the same time.

The probability of either A or B occurring (or both) is

$$P(A \text{ or } B) = P(A) + P(B) \tag{3.6.5}$$

Example 6

Suppose we draw one card from a standard deck. What is the probability that we get a Queen or a King?

Solution

There are 4 Queens and 4 Kings in the deck, hence 8 outcomes corresponding to a Queen or King out of 52 possible outcomes.

Note that in this case, there are no cards that are both a Queen and a King, so $P(\text{King and Queen}) = 0$ so this is a mutually exclusive event. Using our probability rule, the probability of drawing a Queen or a King is:

$$P(\text{ King or Queen}) = P(\text{ King }) + P(\text{ Queen}) - P(\text{ King and Queen}) = \frac{4}{52} + \frac{4}{52} - 0 = \frac{8}{52}$$

Example 7

Suppose we draw one card from a standard deck. What is the probability that we get a red card or a King? Simplify your answers.

Solution

- Half the cards are red, so $P(\text{red}) = \frac{26}{52}$.
- There are four kings, so $P(\text{ King }) = \frac{4}{52}$.
- There are two red kings, so $P(\text{ Red and King}) = \frac{2}{52}$.

We can then calculate:

$$P(\text{Red or King}) = P(\text{Red}) + P(\text{King}) - P(\text{Red and King}) = \frac{26}{52} + \frac{4}{52} - \frac{2}{52} = \frac{28}{52} = \frac{7}{13}$$

Try it Now 2

In your drawer you have 10 pairs of socks, 6 of which are white, and 7 tee shirts, 3 of which are white. If you reach in and randomly grab a pair of socks and a tee shirt, what the probability at least one is white?

Answer

$$P(\text{ white socks and white tee }) = \left(\frac{6}{10}\right)\left(\frac{3}{7}\right) = \frac{9}{35}$$

$$P(\text{ white socks or white tee }) = \frac{6}{10} + \frac{3}{7} - \frac{9}{35} = \frac{27}{35}$$

Example 8

The table below shows the number of survey subjects who have received and not received a speeding ticket in the last year, and the color of their car. Find the probability that a randomly chosen person:

a. Has a red car *and* got a speeding ticket
b. Has a red car *or* got a speeding ticket.

	Speeding ticket	No speeding ticket	Total
Red car	15	135	150
Not red car	45	470	515
Total	60	605	665

Solution

We can see that 15 people of the 665 surveyed had both a red car and got a speeding ticket, so the probability is $\frac{15}{665} \approx 0.0226$.

Notice that having a red car and getting a speeding ticket are not independent events, so the probability of both of them occurring is not simply the product of probabilities of each one occurring.

We could answer this question by simply adding up the numbers: 15 people with red cars and speeding tickets + 135 with red cars but no ticket + 45 with a ticket but no red car = 195 people. So the probability is $\frac{195}{665} \approx 0.2932$.

We also could have found this probability by:

$$\mathrm{P}(\text{had a red car}) + \mathrm{P}(\text{got a speeding ticket}) - \mathrm{P}(\text{had a red car and got a speeding ticket}) = \frac{150}{665} + \frac{60}{665} - \frac{15}{665} = \frac{195}{665}$$

Conditional Probability

Often it is required to compute the probability of an event given that another event has occurred.

Example 9

What is the probability that two cards drawn at random from a deck of playing cards without replacement will both be aces?

Solution

It might seem that you could use the formula for the probability of two independent events and simply multiply $\frac{4}{52} \cdot \frac{4}{52} = \frac{1}{169}$. This would be incorrect, however, because the two events are not independent. If the first card drawn is an ace, then the probability that the second card is also an ace would be lower because there would only be three aces left in the deck.

Once the first card chosen is an ace, the probability that the second card chosen is also an ace is called the **conditional probability** of drawing an ace. In this case the "condition" is that the first card is an ace. Symbolically, we write this as:

P(ace on second draw | an ace on the first draw).

The vertical bar "|" is read as "given," so the above expression is short for "The probability that an ace is drawn on the second draw given that an ace was drawn on the first draw." What is this probability? After an ace is drawn on the first draw, there are 3 aces out of 51 total cards left. This means that the conditional probability of drawing an ace after one ace has already been drawn is $\frac{3}{51} = \frac{1}{17}$.

Thus, the probability of both cards being aces is $\frac{4}{52} \cdot \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}$.

Conditional Probability

The probability the event B occurs, given that event A has happened, is represented as

$$P(B|A) = \frac{\text{number of outcomes common to A and B}}{\text{number of outcomes in A}} \tag{3.6.6}$$

This is read as “the probability of B given A”

Example 10

Find the probability that a die rolled shows a 6, given that a flipped coin shows a head.

Solution

These are two independent events, so the probability of the die rolling a 6 is $\frac{1}{6}$, regardless of the result of the coin flip.

Example 11

The table below shows the number of survey subjects who have received and not received a speeding ticket in the last year, and the color of their car. Find the probability that a randomly chosen person:

a) Has a speeding ticket *given* they have a red car

b) Has a red car *given* they have a speeding ticket

	Speeding ticket	No speeding ticket	Total
Red car	15	135	150
Not red car	45	470	515
Total	60	605	665

Solution

a) Since we know the person has a red car, we are only considering the 150 people in the first row of the table. Of those, 15 have a speeding ticket, so

P(ticket | red car) = $\frac{15}{150} = \frac{1}{10} = 0.1$

b) Since we know the person has a speeding ticket, we are only considering the 60 people in the first column of the table. Of those, 15 have a red car, so

P(red car | ticket) = $\frac{15}{60} = \frac{1}{4} = 0.25$.

Notice from the last example that $P(B|A)$ is **not** equal to $P(A|B)$.

These kinds of conditional probabilities are what insurance companies use to determine your insurance rates. They look at the conditional probability of you having accident, given your age, your car, your car color, your driving history, etc., and price your policy based on that likelihood.

Conditional Probability Formula

If events A and B are not independent, then

$$P(A \text{ and } B) = P(A)P(B|A) \tag{3.6.7}$$

Example 12

If you pull 2 cards out of a deck, what is the probability that both are spades?

Solution

The probability that the first card is a spade is $\frac{13}{52}$.

The probability that the second card is a spade, given the first was a spade, is $\frac{12}{51}$, since there is one less spade in the deck, and one less total cards.

The probability that both cards are spades is $\left(\frac{13}{52}\right)\left(\frac{12}{51}\right) = \frac{156}{2652} \approx 0.0588$

Example 13

If you draw two cards from a deck, what is the probability that you will get the Ace of Diamonds and a black card?

Solution

You can satisfy this condition by having Case A or Case B, as follows:

Case A) you can get the Ace of Diamonds first and then a black card or

Case B) you can get a black card first and then the Ace of Diamonds.

Let's calculate the probability of **Case A**. The probability that the first card is the Ace of Diamonds is $\frac{1}{52}$. The probability that the second card is black given that the first card is the Ace of Diamonds is $\frac{26}{51}$ because 26 of the remaining 51 cards are black. The probability is therefore $\left(\frac{1}{52}\right)\left(\frac{26}{51}\right) = \frac{1}{102}$.

Now for **Case B:** the probability that the first card is black is $\frac{26}{52} = \frac{1}{2}$. The probability that the second card is the Ace of Diamonds given that the first card is black is $\frac{1}{51}$. The probability of Case B is therefore $\left(\frac{1}{2}\right)\left(\frac{1}{51}\right) = \frac{1}{102}$, the same as the probability of Case 1.

Recall that the probability of A or B is $P(\mathrm{A}) + P(\mathrm{B}) - P(\mathrm{A \text{ and } B})$. In this problem, $P(\mathrm{A \text{ and } B}) = 0$ since the first card cannot be the Ace of Diamonds and be a black card. Therefore, the probability of Case A or Case B is $\frac{1}{102} + \frac{1}{102} = \frac{2}{102} = \frac{1}{51}$. The probability that you will get the Ace of Diamonds and a black card when drawing two cards from a deck is $\frac{1}{51}$.

Try it Now 3

In your drawer you have 10 pairs of socks, 6 of which are white. If you reach in and randomly grab two pairs of socks, what is the probability that both are white?

Answer

$$\left(\frac{6}{10}\right)\left(\frac{5}{9}\right)=\frac{30}{90}=\frac{1}{3}$$

Example 14

A home pregnancy test was given to women, then pregnancy was verified through blood tests. The following table shows the home pregnancy test results. Find

a) P(not pregnant | positive test result)

b) P(positive test result | not pregnant)

	Positive test	Negative test	Total
Pregnant	70	4	74
Not Pregnant	5	14	19
Total	75	18	93

Solution

a) Since we know the test result was positive, we're limited to the 75 women in the first column, of which 5 were not pregnant. P(not pregnant | positive test result) = $\frac{5}{75} \approx 0.067$.

b) Since we know the woman is not pregnant, we are limited to the 19 women in the second row, of which 5 had a positive test. P(positive test result | not pregnant) = $\frac{5}{19} \approx 0.263$.

The second result is what is usually called a false positive: A positive result when the woman is not actually pregnant.

Odds

If we know the probability, we can determine the odds in favor of or against an event happening.

Probability to Odds

For any event, E

To convert from probability to odds *for* an event,

$$\text{Odds for E} = \frac{P(E)}{P(\bar{E})} \tag{3.6.8}$$

To convert from probability to odd *against* an event,

$$\text{Odds against E} = \frac{P(\bar{E})}{P(E)} \tag{3.6.9}$$

Odd can be written as

a to b, a:b, or $\frac{a}{b}$

Example 15

If you roll two die, what is the probability you will roll a sum of 10?

Solution

First determine the probability of the event happening. Since there are 36 possible rolls and only 3 ways to roll a sum of 10, the probability for the event happening is $\frac{3}{36}=\frac{1}{12}$

Next determine the probability of the event *not* happening. Since the probability of the event happening is $\frac{1}{12}$. The probability of an event *not* happening is $1-\frac{1}{12}=\frac{12}{12}-\frac{1}{12}=\frac{11}{12}$

Now, use the given formula to determine the odds.

Since the probability is *for* rolling a sum a 10, we will use 4.3.8

$$\text{Odds for sum of } 10=\frac{\frac{1}{12}}{\frac{11}{12}}=\frac{1}{11}$$

So the odds you will roll a sum of 10 are 1 to 11. This can be read as, the odds are that out of 12 rolls 1 will be a sum of 10 and the other 11 will not. We can also say 1 out of 12 rolls will be a sum a 10.

Odds to Probability

If the odds in favor of an event E are x to y, then the probability of the event occurring is given by

$$P(E)=\frac{x}{x+y} \tag{3.6.10}$$

Example 16

The horse named Bolt has 10:2 odds against winning at the race track. What is the probability Bolt will win the race?

Solution

Referring to the formula 4.3.10 determine what the value for x and y is. The odds 10:2 can be rewrittens as 10 to 2 against Bolt winning. This means the odds in favor of Bolt winning would be 2 to 10. Therefore, $x=2$ and $y=10$. The probability Bolt wins would be

$$P(\text{Bolt wins})=\frac{2}{2+10}=\frac{2}{12}=\frac{1}{6}=0.1667\times 100=16.67$$

3.6.1: Exercises

1. Compute the probability of rolling a 12-sided die and getting a number other than 8.
2. If you pick one card at random from a standard deck of cards, what is the probability it is not the Ace of Spades?
3. Referring to the grade table below, what is the probability that a student chosen at random did NOT earn a C?

	A	B	C	Total
Male	8	18	13	39
Female	10	4	12	26
Total	18	22	25	65

4. Referring to the credit card table below, what is the probability that a person chosen at random has at least one credit card?

	Zero	One	Two or more	Total
Male	9	5	19	33
Female	18	10	20	48
Total	27	15	39	81

5. A six-sided die is rolled twice. What is the probability of showing a 6 on both rolls?
6. A fair coin is flipped twice. What is the probability of showing heads on both flips?
7. A six-sided die is rolled twice. What is the probability of showing a 5 on the first roll and an even number on the second roll?
8. Suppose that 21% of people own dogs. If you pick two people at random, what is the probability that they both own a dog?
9. Suppose a jar contains 17 red marbles and 32 blue marbles. If you reach in the jar and pull out 2 marbles at random without replacement, find the probability that both are red.
10. Suppose you write each letter of the alphabet on a different slip of paper and put the slips into a hat. If you pull out two slips at random, find the probability that both are vowels.
11. Bert and Ernie each have a well-shuffled standard deck of 52 cards. They each draw one card from their own deck. Compute the probability that:
    a. Bert and Ernie both draw an Ace.
    b. Bert draws an Ace but Ernie does not.
    c. neither Bert nor Ernie draws an Ace.
    d. Bert and Ernie both draw a heart.
    e. Bert gets a card that is not a Jack and Ernie draws a card that is not a heart.
12. Bert has a well-shuffled standard deck of 52 cards, from which he draws one card; Ernie has a 12-sided die, which he rolls at the same time Bert draws a card. Compute the probability that:
    a. Bert gets a Jack and Ernie rolls a five.
    b. Bert gets a heart and Ernie rolls a number less than six.
    c. Bert gets a face card (Jack, Queen or King) and Ernie rolls an even number.
    d. Bert gets a red card and Ernie rolls a fifteen.
    e. Bert gets a card that is not a Jack and Ernie rolls a number that is not twelve.
13. Compute the probability of drawing a King from a deck of cards and then drawing a Queen without replacement.
14. Compute the probability of drawing two spades from a deck of cards without replacement.
15. A math class consists of 25 students, 14 freshmen and 11 sophomores. Two students are selected at random to participate in a probability experiment. Compute the probability that
    a. a sophomore is selected, then a freshman.
    b. a freshman is selected, then a sophomore.
    c. two sophomores are selected.
    d. two freshmen are selected.
    e. no sophomores are selected.
16. A math class consists of 25 students, 14 freshmen and 11 sophomores. Three students are selected at random to participate in a probability experiment. Compute the probability that
    a. a sophomore is selected, then two freshmen.

b. a freshmen is selected, then two sophomores.
c. two freshmen are selected, then one sophomore.
d. three sophomores are selected.
e. three freshmen are selected.

17. Giving a test to a group of students, the grades and class are summarized below. If one student was chosen at random, find the probability that the student was a freshman and earned an A.

	A	B	C	Total
Sophomore	8	18	13	39
Freshman	10	4	12	26
Total	18	22	25	65

18. The table below shows the number of credit cards owned by a group of individuals. If one person was chosen at random, find the probability that the person was male and had two or more credit cards.

	Zero	One	Two or more	Total
Male	9	5	19	33
Female	18	10	20	48
Total	27	15	39	81

19. A jar contains 6 red marbles numbered 1 to 6 and 8 blue marbles numbered 1 to 8. A marble is drawn at random from the jar. Find the probability the marble is red or odd-numbered.
20. A jar contains 4 red marbles numbered 1 to 4 and 10 blue marbles numbered 1 to 10. A marble is drawn at random from the jar. Find the probability the marble is blue or even-numbered.
21. Referring to the grade table below, find the probability that a student chosen at random is a freshman or earned a B.

	A	B	C	Total
Male	8	18	13	39
Female	10	4	12	26
Total	18	22	25	65

22. Referring to the table below, find the probability that a person chosen at random is male or has no credit cards.

	Zero	One	Two or more	Total
Male	9	5	19	33
Female	18	10	20	48
Total	27	15	39	81

23. Compute the probability of drawing the King of hearts or a Queen from a deck of cards.
24. Compute the probability of drawing a King or a heart from a deck of cards.
25. A jar contains 5 red marbles numbered 1 to 5 and 8 blue marbles numbered 1 to 8. A marble is drawn at random from the jar. Find the probability the marble is

    a. Even-numbered given that the marble is red.
    b. Red given that the marble is even-numbered.

26. A jar contains 4 red marbles numbered 1 to 4 and 8 blue marbles numbered 1 to 8. A marble is drawn at random from the jar. Find the probability the marble is

    a. Odd-numbered given that the marble is blue.
    b. Blue given that the marble is odd-numbered.

27. Compute the probability of flipping a coin and getting heads, given that the previous flip was tails.
28. Find the probability of rolling a “1” on a fair die, given that the last 3 rolls were all ones.
29. Suppose a math class contains 25 students, 14 females (three of whom speak French) and 11 males (two of whom speak French). Compute the probability that a randomly selected student speaks French, given that the student is female.
30. Suppose a math class contains 25 students, 14 females (three of whom speak French) and 11 males (two of whom speak French). Compute the probability that a randomly selected student is male, given that the student speaks French.

31. A certain virus infects one in every 400 people. A test used to detect the virus in a person is positive 90% of the time if the person has the virus and 10% of the time if the person does not have the virus. Let A be the event "the person is infected" and B be the event "the person tests positive".
    a. Find the probability that a person has the virus given that they have tested positive, i.e. find $P(A|B)$.
    b. Find the probability that a person does not have the virus given that they test negative, i.e. find $P(\text{ not } A| \text{ not } B)$.
32. A certain virus infects one in every 2000 people. A test used to detect the virus in a person is positive 96% of the time if the person has the virus and 4% of the time if the person does not have the virus. Let A be the event "the person is infected" and B be the event "the person tests positive".
    a. Find the probability that a person has the virus given that they have tested positive, i.e. find $P(A|B)$.
    b. Find the probability that a person does not have the virus given that they test negative, i.e. find $P(\text{ not } A| \text{ not } B)$.

3.7: Two-Way Tables

4.4 Learning Objectives

- Use two-way tables to find conditional probabilities

In this section we concentrate on the more complex conditional probability problems we began to look at in the last section.

Example 1

Suppose a certain disease has an incidence rate of 0.1% (that is, it afflicts 0.1% of the population). A test has been devised to detect this disease. The test does not produce false negatives (that is, anyone who has the disease will test positive for it), but the false positive rate is 5% (that is, about 5% of people who take the test will test positive, even though they do not have the disease). Suppose a randomly selected person takes the test and tests positive. What is the probability that this person actually has the disease?

Solution

Let's break down the information in the problem piece by piece, using a two-way table to record our results.

Suppose a certain disease has an incidence rate of 0.1% (that is, it afflicts 0.1% of the population). The percentage 0.1% can be converted to a decimal number by moving the decimal place two places to the left, to get 0.001. In turn, 0.001 can be rewritten as a fraction: 1/1000. This tells us that about 1 in every 1000 people has the disease. (If we wanted we could write P(disease)=0.001.)

A test has been devised to detect this disease. The test does not produce false negatives (that is, anyone who has the disease will test positive for it). This part is fairly straightforward: everyone who has the disease will test positive, or alternatively everyone who tests negative does not have the disease. (We could also say P (positive | disease)=1.)

The false positive rate is 5% (that is, about 5% of people who take the test will test positive, even though they do not have the disease). This is even more straightforward. Another way of looking at it is that of every 100 people who are tested and do not have the disease, 5 will test positive even though they do not have the disease. (We could also say that P(positive | no disease)=0.05.)

Suppose a randomly selected person takes the test and tests positive. What is the probability that this person actually has the disease? Here we want to compute P(disease|positive). We already know that P(positive|disease)=1, but remember that conditional probabilities are not equal if the conditions are switched.

Rather than thinking in terms of all these probabilities we have developed, let's create a hypothetical situation and apply the facts as set out above. First, suppose we randomly select 1000 people and administer the test. How many do we expect to have the disease? Since about 1/1000 of all people are afflicted with the disease, $\frac{1}{1000}$ of 1000 people is 1. (Now you know why we chose 1000.) Only 1 of 1000 test subjects actually has the disease; the other 999 do not.

	Has Disease	Does Not Have Disease	Total
Tests Positive			
Tests Negative			
Total	1	999	1000

We also know that 5% of all people who do not have the disease will test positive. There are 999 disease-free people, so we would expect $(0.05)(999) = 49.95$ (so, about 50) people to test positive who do not have the disease.

	Has Disease	Does Not Have Disease	Total
Tests Positive	1	50	
Tests Negative	0	949	
Total	1	999	1000

Now back to the original question, computing P (disease|positive). There are 51 people who test positive in our example (the one unfortunate person who actually has the disease, plus the 50 people who tested positive but don't). Only one of these people has the disease, so

P(disease | positive) $\approx \frac{1}{51} \approx 0.0196$

or less than 2%. This means that of all people who test positive, over 98% *do not have the disease*. Does this surprise you?

The answer we got was slightly approximate, since we rounded 49.95 to 50.

But back to the surprising result. *Of all people who test positive, over 98% do not have the disease.* If your guess for the probability a person who tests positive has the disease was wildly different from the right answer (2%), don't feel bad. The exact same problem was posed to doctors and medical students at the Harvard Medical School 25 years ago and the results revealed in a 1978 *New England Journal of Medicine* article. Only about 18% of the participants got the right answer. Most of the rest thought the answer was closer to 95% (perhaps they were misled by the false positive rate of 5%).

The significance of this finding and similar results from other studies in the intervening years lies in the possibly catastrophic consequences it might have for patient care. If a doctor thinks the chances that a positive test result nearly guarantees that a patient has a disease, they might begin an unnecessary and possibly harmful treatment regimen on a healthy patient. Or worse, as in the early days of the AIDS crisis when being HIV-positive was often equated with a death sentence, the patient might take a drastic action and commit suicide.

As we have seen in this hypothetical example, the most responsible course of action for treating a patient who tests positive would be to counsel the patient that they most likely do *not* have the disease and to order further, more reliable, tests to verify the diagnosis.

Example 2

A certain disease has an incidence rate of 2%. If the false negative rate is 10% and the false positive rate is 1%, compute the probability that a person who tests positive actually has the disease.

Solution

Imagine 10,000 people who are tested. Of these 10,000, 200 will have the disease since the disease as an incident rate of 2%.

	Has Disease	Does Not Have Disease	Total
Tests Positive			
Tests Negative			
Total	200	9800	10000

Of those who have the disease, 10% of them, or 20, will test negative and the remaining 180 will test positive. Of the 9800 who do not have the disease, 1% of them, or 98, will test positive. So of the 278 total people who test positive, 180 will have the disease.

	Has Disease	Does Not Have Disease	Total
Tests Positive	180	98	278
Tests Negative	20	9702	
Total	200	9800	10000

Thus, $P(\text{ disease } \mid \text{ positive }) = \frac{180}{278} \approx 0.647$.

So, about 65% of the people who test positive will have the disease.

Try it Now 1

A certain disease has an incidence rate of 0.5%. If there are no false negatives and if the false positive rate is 3%, compute the probability that a person who tests positive actually has the disease.

Answer

Out of 100,000 people, 500 would have the disease since the disease as an incidence rate of 0.5%.

	Has Disease	Does Not Have Disease	Total
Tests Positive			
Tests Negative			
Total	500	99500	100000

Of those, all 500 would test positive since there are no false negatives. Of the 99,500 without the disease 3%, or 2,985, would falsely test positive and the other 96,515 would test negative.

	Has Disease	Does Not Have Disease	Total
Tests Positive	500	2985	3485
Tests Negative	0	96515	96515
Total	500	99500	100000

$P(\text{ disease } \mid \text{ positive }) = \frac{500}{500+2985} = \frac{500}{3485} \approx 14.3\%$

3.7.1: Exercises

1. A certain disease has an incidence rate of 0.3%. If the false negative rate is 6% and the false positive rate is 4%. Set up a two way table with a sample size of 10,000 people.
   a. Compute the probability that a person who tests positive actually has the disease.
   b. Compute the probability that a person who tests negative actually has the disease.
   c. Compute the probability that a person who has the disease tests negative.
   d. Compute the probability that a person who does not have the disease tests negative.
2. A certain disease has an incidence rate of 0.1%. If the false negative rate is 8% and the false positive rate is 3%. Set up a two way table with a sample size of 10,000 people.
   a. Compute the probability that a person who tests positive does NOT have the disease.
   b. Compute the probability that a person who tests negative does NOT have the disease.
   c. Compute the probability that a person who has the disease tests positive.
   d. Compute the probability that a person who does not have the disease test positive.
3. A certain group of symptom-free women between the ages of 40 and 50 are randomly selected to participate in mammography screening. The incidence rate of breast cancer among such women is 0.8%. The false negative rate for the mammogram is 10%. The false positive rate is 7%. If a the mammogram results for a particular woman are positive (indicating that she has breast cancer), what is the probability that she actually has breast cancer?
4. About 0.01% of men with no known risk behavior are infected with HIV. The false negative rate for the standard HIV test 0.01% and the false positive rate is also 0.01%. If a randomly selected man with no known risk behavior tests positive for HIV, what is the probability that he is actually infected with HIV?

3.8: Counting

4.5 Learning Objectives

- Use the Fundamental Counting Rule to determine the number of outcomes
- Use permutations to determine the number of outcomes
- Use combinations to determine the number of outcomes
- Determine whether a counting situation is permutations or combinations
- Use counting techniques to find probabilities

Counting? You already know how to count or you wouldn't be taking a college-level math class, right? Well yes, but what we'll really be investigating here are ways of counting *efficiently*. When we get to the probability situations a bit later in this chapter we will need to count some *very* large numbers, like the number of possible winning lottery tickets. One way to do this would be to write down every possible set of numbers that might show up on a lottery ticket, but believe me: you don't want to do this.

Basic Counting

We will start, however, with some more reasonable sorts of counting problems in order to develop the ideas that we will soon need.

Example 1

Suppose at a particular restaurant you have three choices for an appetizer (soup, salad or breadsticks) and five choices for a main course (hamburger, sandwich, quiche, fajita or pizza). If you are allowed to choose exactly one item from each category for your meal, how many different meal options do you have?

Solution

Solution 1: One way to solve this problem would be to systematically list each possible meal:

soup + hamburger	soup + sandwich	soup + quiche
soup + fajita	soup + pizza	salad + hamburger
salad + sandwich	salad + quiche	salad + fajita
salad + pizza	breadsticks + hamburger	breadsticks + sandwich
breadsticks + quiche	breadsticks + fajita	breadsticks + pizza

Assuming that we did this systematically and that we neither missed any possibilities nor listed any possibility more than once, the answer would be 15. Thus you could go to the restaurant 15 nights in a row and have a different meal each night.

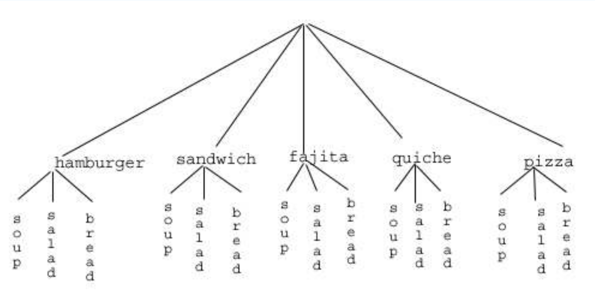
Solution 2: Another way to solve this problem would be to list all the possibilities in a table:

	hamburger	**sandwich**	**quiche**	**fajita**	**pizza**
soup	Soup + burger				
salad	Salad + burger				
bread	*etc.*				

In each of the cells in the table we could list the corresponding meal: soup + hamburger in the upper left corner, salad + hamburger below it, etc. But if we didn't really care *what* the possible meals are, only *how many* possible meals there are, we could just count the number of cells and arrive at an answer of 15, which matches our answer from the first solution. (It's always good when you solve a problem two different ways and get the same answer!)

Solution 3: We already have two perfectly good solutions. Why do we need a third? The first method was not very systematic, and we might easily have made an omission. The second method was better, but suppose that in addition to the appetizer and the main course we further complicated the problem by adding desserts to the menu: we've used the rows of the table for the appetizers and the columns for the main courses—where will the desserts go? We would need a third dimension, and since drawing 3-D tables on a 2-D page or computer screen isn't terribly easy, we need a better way in case we have three categories to choose form instead of just two.

So, back to the problem in the example. What else can we do? Let's draw a **tree diagram**:



This is called a "tree" diagram because at each stage we branch out, like the branches on a tree. In this case, we first drew five branches (one for each main course) and then for each of those branches we drew three more branches (one for each appetizer). We count the number of branches at the final level and get (surprise, surprise!) 15.

If we wanted, we could instead draw three branches at the first stage for the three appetizers and then five branches (one for each main course) branching out of each of those three branches.

OK, so now we know how to count possibilities using tables and tree diagrams. These methods will continue to be useful in certain cases, but imagine a game where you have two decks of cards (with 52 cards in each deck) and you select one card from each deck. Would you really want to draw a table or tree diagram to determine the number of outcomes of this game?

Let's go back to the previous example that involved selecting a meal from three appetizers and five main courses, and look at the second solution that used a table. Notice that one way to count the number of possible meals is simply to number each of the appropriate cells in the table, as we have done above. But another way to count the number of cells in the table would be multiply the number of rows (3) by the number of columns (5) to get 15. Notice that we could have arrived at the same result without making a table at all by simply multiplying the number of choices for the appetizer (3) by the number of choices for the main course (5). We generalize this technique as the *fundamental counting rule*:

Fundamental Counting Rule

If we are asked to choose one item from each of two separate categories where there are m items in the first category and n items in the second category, then the total number of available choices is

$$\text{number of choices} = m \cdot n \tag{3.8.1}$$

This is sometimes called the multiplication counting rule.

Example 2

There are 21 novels and 18 volumes of poetry on a reading list for a college English course. How many different ways can a student select one novel and one volume of poetry to read during the quarter?

Solution

There are 21 choices from the first category and 18 for the second, so there are $21 \cdot 18 = 378$ possibilities.

The Fundamental Counting Rule can be extended when there are more than two categories by applying it repeatedly, as we see in the next example.

Example 3

Suppose at a particular restaurant you have three choices for an appetizer (soup, salad or breadsticks), five choices for a main course (hamburger, sandwich, quiche, fajita or pasta) and two choices for dessert (pie or ice cream). If you are allowed to choose exactly one item from each category for your meal, how many different meal options do you have?

Solution

There are 3 choices for an appetizer, 5 for the main course and 2 for dessert, so there are $3 \cdot 5 \cdot 2 = 30$ possibilities.

Example 4

A quiz consists of 3 true-or-false questions. In how many ways can a student answer the quiz?

Solution

There are 3 questions. Each question has 2 possible answers (true or false), so the quiz may be answered in $2 \cdot 2 \cdot 2 = 8$ different ways. Recall that another way to write $2 \cdot 2 \cdot 2$ is 2^3 which is much more compact.

Try it Now 1

Suppose at a particular restaurant you have eight choices for an appetizer, eleven choices for a main course and five choices for dessert. If you are allowed to choose exactly one item from each category for your meal, how many different meal options do you have?

Answer

$8 \cdot 11 \cdot 5 = 440$ menu combinations

Permutations

In this section we will develop an even faster way to solve some of the problems we have already learned to solve by other means. Let's start with a couple examples.

Example 5

How many different ways can the letters of the word MATH be rearranged to form a four-letter code word?

Solution

This problem is a bit different. Instead of choosing one item from each of several different categories, we are repeatedly choosing items from the *same* category (the category is: the letters of the word MATH) and each time we choose an item we *do not replace* it, so there is one fewer choice at the next stage: we have 4 choices for the first letter (say we choose A), then 3 choices for the second (M, T and H; say we choose H), then 2 choices for the next letter (M and T; say we choose M) and only one choice at the last stage (T). Thus there are $4 \cdot 3 \cdot 2 \cdot 1 = 24$ ways to spell a code word with the letters of the word MATH.

In example 5, we needed to calculate $n \cdot (n-1) \cdot (n-2) \cdots 3 \cdot 2 \cdot 1$. This calculation shows up often in mathematics, and is called the **factorial**, and is notated $n!$

Factorial

$$n! = n \cdot (n-1) \cdot (n-2) \cdots 3 \cdot 2 \cdot 1 \tag{3.8.2}$$

0! = 1 by definition.

Example 6

How many ways can five different door prizes be distributed among five people?

Solution

There are 5 choices of prize for the first person, 4 choices for the second, and so on. The number of ways the prizes can be distributed will be $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ ways.

Now we will consider some slightly different examples.

Example 7

A charity benefit is attended by 25 people and three gift certificates are given away as door prizes: one gift certificate is in the amount of \$100, the second is worth \$25 and the third is worth \$10. Assuming that no person receives more than one prize, how many different ways can the three gift certificates be awarded?

Solution

Using the Fundamental Counting Rule, there are 25 choices for the person who receives the $\$100$ certificate, 24 remaining choices for the $\$25$ certificate and 23 choices for the $\$10$ certificate, so there are $25 \cdot 24 \cdot 23 = 13,800$ ways in which the prizes can be awarded.

Example 8

Eight sprinters have made it to the Olympic finals in the 100-meter race. In how many different ways can the gold, silver and bronze medals be awarded?

Solution

Using the Fundamental Counting Rule, there are 8 choices for the gold medal winner, 7 remaining choices for the silver, and 6 for the bronze, so there are $8 \cdot 7 \cdot 6 = 336$ ways the three medals can be awarded to the 8 runners.

Note that in these preceding examples, the gift certificates and the Olympic medals were awarded *without replacement*; that is, once we have chosen a winner of the first door prize or the gold medal, they are not eligible for the other prizes. Thus, at each succeeding stage of the solution there is one fewer choice (25, then 24, then 23 in the first example; 8, then 7, then 6 in the second). Contrast this with the situation of a multiple choice test, where there might be five possible answers — A, B, C, D or E — for each question on the test.

Note also that *the order of selection was important* in each example: for the three door prizes, being chosen first means that you receive substantially more money; in the Olympics example, coming in first means that you get the gold medal instead of the silver or bronze. In each case, if we had chosen the same three people in a different order there might have been a different person who received the \$100 prize, or a different goldmedalist. (Contrast this with the situation where we might draw three names out of a hat to each receive a \$10 gift certificate; in this case the order of selection is *not* important since each of the three people receive the same prize. Situations where the order is *not* important will be discussed in the next section.)

We can generalize the situation in the two examples above to any problem *without replacement* where the *order of selection is important*. If we are arranging in order r items out of n possibilities (instead of 3 out of 25 or 3 out of 8 as in the previous examples), the number of possible arrangements will be given by

$n \cdot (n-1) \cdot (n-2) \cdots (n-r+1)$

If you don't see why $(n-r+1)$ is the right number to use for the last factor, just think back to the first example in this section, where we calculated $25 \cdot 24 \cdot 23$ to get $13,800$. In this case $n = 25$ and $r = 3$, so $n-r+1 = 25-3+1 = 23$, which is exactly the right number for the final factor.

Now, why would we want to use this complicated formula when it's actually easier to use the Fundamental Counting Rule, as we did in the first two examples? Well, we won't actually use this formula all that often, we only developed it so that we could attach a special notation and a special definition to this situation where we are choosing r items out of n possibilities *without replacement* and where the *order of selection is important*. In this situation we write:

Permutations

$${}_nP_r = n \cdot (n-1) \cdot (n-2) \cdots (n-r+1)$$

We say that there are ${}_nP_r$ **permutations** of size r that may be selected from among n choices *without replacement* when *order matters*.

It turns out that we can express this result more simply using factorials.

$$ {}_nP_r = \frac{n!}{(n-r)!} \tag{3.8.3} $$

In practice, we usually use technology rather than factorials or repeated multiplication to compute permutations.

Example 9

I have nine paintings and have room to display only four of them at a time on my wall. How many different ways could I do this?

Solution

Since we are choosing 4 paintings out of 9 *without replacement* where the *order of selection is important* there are ${}_9P_4 = 9 \cdot 8 \cdot 7 \cdot 6 = 3,024$ permutations.

Example 10

How many ways can a four-person executive committee (president, vice-president, secretary, treasurer) be selected from a 16-member board of directors of a non-profit organization?

Solution

We want to choose 4 people out of 16 without replacement and where the order of selection is important. So the answer is ${}_{16}P_4 = 16 \cdot 15 \cdot 14 \cdot 13 = 43,680$.

Try it Now 2

How many 5 character passwords can be made using the letters A through Z

a. if repeats are allowed
b. if no repeats are allowed

Answer

There are 26 characters.

a. $26^5 = 11,881,376$.
b. ${}_{26}\mathrm{P}_5 = 26 \cdot 25 \cdot 24 \cdot 23 \cdot 22 = 7,893,600$

In some cases, some of the choices available are identical. Suppose someone was given the chance to choose 5 gift cards out of 10 available gift cards and they notice that 3 of the gift cards is from their favorite coffee shop. They might choose those 3 first and then decide on the other two. The number of choices changes when there are duplicate items available.

Permutations with Duplicate Items

The number of permutations of n items when p are identical, q are identical, r are identical, and so on... is given by,

$$ \text{choices} = \frac{n!}{p!q!r!\cdots} \tag{3.8.4} $$

Example 11

How many ways could you arrange the letters in the word MISSISSIPPI?

Solution

Consider first the the total number of items. There are 11 letters in the word MISSISSIPPI. This means $n = 11$

Next consider the items. There are four of the letter I, four of the letter S, and two of the letter P. If we arrange the four S's, it would not matter exactly which S was moved to another position.

$\text{arrangements} = \frac{11!}{4!4!2!\cdots} = 34650$

In comparison, if we were asked to arrange the letters in an 11 letter word with no duplicate letters the answer would be

${}_{11}P_{11} = \frac{11!}{(11-11)!} = 11! = 39,916,800$

Combinations

In the previous section we considered the situation where we chose r items out of n possibilities *without replacement* and where the *order of selection was important*. We now consider a similar situation in which the order of selection is *not* important.

Example 12

A charity benefit is attended by 25 people at which three $50 gift certificates are given away as door prizes. Assuming no person receives more than one prize, how many different ways can the gift certificates be awarded?

Solution

Using the Fundamental Counting Rule, there are 25 choices for the first person, 24 remaining choices for the second person and 23 for the third, so there are $25 \cdot 24 \cdot 23 = 13,800$ ways to choose three people. Suppose for a moment that Abe is chosen first, Bea second and Cindy third; this is one of the 13,800 possible outcomes. Another way to award the prizes would be to choose Abe first, Cindy second and Bea third; this is another of the 13,800 possible outcomes. But either way Abe, Bea and Cindy each get $50, so it doesn't really matter the order in which we select them. In how many different orders can Abe, Bea and Cindy be selected? It turns out there are 6:

ABC ACB BAC BCA CAB CBA

How can we be sure that we have counted them all? We are really just choosing 3 people out of 3, so there are $3 \cdot 2 \cdot 1 = 6$ ways to do this; we didn't really need to list them all, we can just use permutations!

So, out of the 13,800 ways to select 3 people out of 25, six of them involve Abe, Bea and Cindy. The same argument works for any other group of three people (say Abe, Bea and David or Frank, Gloria and Hildy) so each three-person group is counted *six times*. Thus the 13,800 figure is six times too big. The number of distinct three-person groups will be $\frac{13,800}{6} = 2300$.

We can generalize the situation in this example above to any problem of choosing a collection of items *without replacement* where the *order of selection is* ***not*** *important*. If we are choosing r items out of n possibilities (instead of 3 out of 25 as in the previous examples), the number of possible choices will be given by this formula ${}_nC_r = \frac{n!}{(n-r)!r!}$.

Combinations

We say that there are ${}_nC_r$ **combinations** of size r that may be selected from among n choices *without replacement where order doesn't matter*.

We can write the combinations formula in terms of factorials:

$${}_nC_r = \frac{n!}{(n-r)!r!} \tag{3.8.5}$$

Example 13

A group of four students is to be chosen from a 35-member class to represent the class on the student council. How many ways can this be done?

Solution

Since we are choosing 4 people out of 35 *without replacement* where the *order of selection is* ***not*** *important* there are ${}_{35}C_4 = \frac{35 \cdot 34 \cdot 33 \cdot 32}{4 \cdot 3 \cdot 2 \cdot 1} = 52,360$ combinations.

Try it Now 3

Suppose the United States Senate Appropriations Committee consists of 29 members, and the Defense Subcommittee of the Appropriations Committee consists of 19 members. Disregarding party affiliation or any special seats on the Subcommittee, how many different 19-member subcommittees may be chosen from among the 29 Senators on the Appropriations Committee?

Answer

Order does not matter. ${}_{29}\mathrm{C}_{19} = 20,030,010$ possible subcommittees

In the preceding Try it Now problem we assumed that the 19 members of the Defense Subcommittee were chosen without regard to party affiliation. In reality this would never happen: if Republicans are in the majority they would never let a majority of Democrats sit on (and thus control) any subcommittee. (The same of course would be true if the Democrats were in control.) So let's consider the problem again, in a slightly more complicated form:

Example 14

Suppose the United States Senate Appropriations Committee consists of 29 members, 15 Republicans and 14 Democrats, and the Defense Subcommittee consists of 19 members, 10 Republicans and 9 Democrats. How many different ways can the members of the Defense Subcommittee be chosen from among the 29 Senators on the Appropriations Committee?

Solution

In this case we need to choose 10 of the 15 Republicans and 9 of the 14 Democrats. There are ${}_{15}C_{10} = 3003$ ways to choose the 10 Republicans and ${}_{14}C_{9} = 2002$ ways to choose the 9 Democrats. But now what? How do we finish the problem?

Suppose we listed all of the possible 10-member Republican groups on 3003 slips of red paper and all of the possible 9-member Democratic groups on 2002 slips of blue paper. How many ways can we choose one red slip and one blue slip? This is a job for the Fundamental Counting Rule! We are simply making one choice from the first category and one choice from the second category, just like in the restaurant menu problems from earlier.

There must be $3003 \cdot 2002 = 6,012,006$ possible ways of selecting the members of the Defense Subcommittee.

Probability using Permutations and Combinations

We can use permutations and combinations to help us answer more complex probability questions.

Example 15

A 4 digit PIN number is selected. What is the probability that there are no repeated digits?

Solution

There are $10 \text{ possible values for each digit of the PIN } (\text{namely: } 0,1,2,3,4,5,6,7,8,9)$, so there are $10 \cdot 10 \cdot 10 \cdot 10 = 10^4 = 10000$ total possible PIN numbers.

To have no repeated digits, all four digits would have to be different, which is selecting without replacement. We could either compute $10 \cdot 9 \cdot 8 \cdot 7$, or notice that this is the same as the permutation ${}_{10}P_4 = 5040$.

The probability of no repeated digits is the number of 4 digit PIN numbers with no repeated digits divided by the total number of 4 digit PIN numbers. This probability is $\frac{{}_{10}P_4}{10^4} = \frac{5040}{10000} = 0.504$

Example 16

In a certain state's lottery, 48 balls numbered 1 through 48 are placed in a machine and six of them are drawn at random. If the six numbers drawn match the numbers that a player had chosen, the player wins $1,000,000. In this lottery, the order the numbers are drawn in doesn’t matter. Compute the probability that you win the million-dollar prize if you purchase a single lottery ticket.

Solution

In order to compute the probability, we need to count the total number of ways six numbers can be drawn, and the number of ways the six numbers on the player's ticket could match the six numbers drawn from the machine. Since there is no stipulation that the numbers be in any particular order, the number of possible outcomes of the lottery drawing is ${}_{48}C_6 = 12,271,512$. Of these possible outcomes, only one would match all six numbers on the player's ticket, so the probability of winning the grand prize is:

$$\frac{{}_6C_6}{{}_{48}C_6} = \frac{1}{12271512} \approx 0.0000000815$$

Example 17

In the state lottery from the previous example, if five of the six numbers drawn match the numbers that a player has chosen, the player wins a second prize of $1,000. Compute the probability that you win the second prize if you purchase a single lottery ticket.

Solution

As above, the number of possible outcomes of the lottery drawing is ${}_{48}C_6 = 12,271,512$. In order to win the second prize, five of the six numbers on the ticket must match five of the six winning numbers; in other words, we must have chosen five of the six winning numbers and one of the 42 losing numbers. The number of ways to choose 5 out of the 6 winning numbers is given by ${}_6C_5 = 6$ and the number of ways to choose 1 out of the 42 losing numbers is given by ${}_{42}C_1 = 42$. Thus the number of favorable outcomes is then given by the Fundamental Counting Rule: ${}_6C_5 \cdot {}_{42}C_1 = 6 \cdot 42 = 252$. So the probability of winning the second prize is.

$$\frac{({}_6C_5)({}_{42}C_1)}{{}_{48}C_6} = \frac{252}{12271512} \approx 0.0000205$$

Try it Now 4

A multiple-choice question on an economics quiz contains 10 questions with five possible answers each. Compute the probability of randomly guessing the answers and getting 9 questions correct.

Answer

There are $5^{10} = 9,765,625$ different ways the exam can be answered. There are 10 possible locations for the one missed question, and in each of those locations there are 4 wrong answers, so there are 40 ways the test could be answered with one wrong answer.

$\mathrm{P}(9 \text{ answers correct }) = \frac{40}{5^{10}} \approx 0.0000041$ chance

Example 18

Compute the probability of randomly drawing five cards from a deck and getting exactly one Ace.

Solution

In many card games (such as poker) the order in which the cards are drawn is not important (since the player may rearrange the cards in his hand any way he chooses); in the problems that follow, we will assume that this is the case unless otherwise stated. Thus we use combinations to compute the possible number of 5-card hands, ${}_{52}C_5$. This number will go in the denominator of our probability formula, since it is the number of possible outcomes.

For the numerator, we need the number of ways to draw one Ace and four other cards (none of them Aces) from the deck. Since there are four Aces and we want exactly one of them, there will be ${}_4C_1$ ways to select one Ace; since there are 48 non-Aces and we want 4 of them, there will be ${}_{48}C_4$ ways to select the four non-Aces. Now we use the Fundamental Counting Rule to calculate that there will be ${}_4C_1 \cdot {}_{48}C_4$ ways to choose one ace and four non-Aces.

Putting this all together, we have

$$P(\text{one Ace}) = \frac{({}_4C_1)\,({}_{48}C_4)}{{}_{52}C_5} = \frac{778320}{2598960} \approx 0.299$$

Example 19

Compute the probability of randomly drawing five cards from a deck and getting exactly two Aces.

Solution

The solution is similar to the previous example, except now we are choosing 2 Aces out of 4 and 3 non-Aces out of 48; the denominator remains the same:

$P(\text{two Aces}) = \frac{({}_4C_2)({}_{48}C_3)}{{}_{52}C_5} = \frac{103776}{2598960} \approx 0.0399$

It is useful to note that these card problems are remarkably similar to the lottery problems discussed earlier.

Try it Now 5

Compute the probability of randomly drawing five cards from a deck of cards and getting three Aces and two Kings.

Answer

$$P(\text{three Aces and two Kings}) = \frac{({}_4C_3)({}_4C_2)}{{}_{52}C_5} = \frac{24}{2598960} \approx 0.0000092$$

Birthday Problem

Let's take a pause to consider a famous problem in probability theory:

Suppose you have a room full of 30 people. What is the probability that there is at least one shared birthday?

Take a guess at the answer to the above problem. Was your guess fairly low, like around 10%? That seems to be the intuitive answer ($\frac{30}{365}$, perhaps?). Let's see if we should listen to our intuition. Let's start with a simpler problem, however.

Example 20

Suppose three people are in a room. What is the probability that there is at least one shared birthday among these three people?

Solution

There are a lot of ways there could be at least one shared birthday. Fortunately there is an easier way. We ask ourselves “What is the alternative to having at least one shared birthday?” In this case, the alternative is that there are **no** shared birthdays. In other words, the alternative to “at least one” is having **none**. In other words, since this is a complementary event,

$P(\text{at least one}) = 1 - P(\text{none})$

We will start, then, by computing the probability that there is no shared birthday. Let's imagine that you are one of these three people. Your birthday can be anything without conflict, so there are 365 choices out of 365 for your birthday. What is the probability that the second person does not share your birthday? There are 365 days in the year (let's ignore leap years) and removing your birthday from contention, there are 364 choices that will guarantee that you do not share a birthday with this person, so the probability that the second person does not share your birthday is $\frac{364}{365}$. Now we move to the third person. What is the probability that this third person does not have the same birthday as either you or the second person? There are 363 days that will not duplicate your birthday or the second person's, so the probability that the third person does not share a birthday with the first two is $\frac{363}{365}$.

We want the second person not to share a birthday with you *and* the third person not to share a birthday with the first two people, so we use the multiplication rule:

$$P(\text{no shared birthday}) = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \approx 0.9918$$

and then subtract from 1 to get

$P(\text{shared birthday}) = 1 - P(\text{no shared birthday}) = 1 - 0.9918 = 0.0082.$

This is a pretty small number, so maybe it makes sense that the answer to our original problem will be small. Let's make our group a bit bigger.

Example 21

Suppose five people are in a room. What is the probability that there is at least one shared birthday among these five people?

Solution

Continuing the pattern of the previous example, the answer should be

$$P(\text{shared birthday}) = 1 - \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdot \frac{362}{365} \cdot \frac{361}{365} \approx 0.0271$$

Note that we could rewrite this more compactly as

$$P(\text{shared birthday}) = 1 - \frac{{}_{365}P_5}{365^5} \approx 0.0271$$

which makes it a bit easier to type into a calculator or computer, and which suggests a nice formula as we continue to expand the population of our group.

Example 21

Suppose 30 people are in a room. What is the probability that there is at least one shared birthday among these 30 people?

Solution

Here we can calculate

$$P(\text{ shared birthday }) = 1 - \frac{{}_{365}P_{30}}{365^{30}} \approx 0.706$$

which gives us the surprising result that when you are in a room with 30 people there is a 70% chance that there will be at least one shared birthday!

If you like to bet, and if you can convince 30 people to reveal their birthdays, you might be able to win some money by betting a friend that there will be at least two people with the same birthday in the room anytime you are in a room of 30 or more people. (Of course, you would need to make sure your friend hasn't studied probability!) You wouldn't be guaranteed to win, but you should win more than half the time.

This is one of many results in probability theory that is counterintuitive; that is, it goes against our gut instincts. If you still don't believe the math, you can carry out a simulation. Just so you won't have to go around rounding up groups of 30 people, someone has kindly developed a Java applet so that you can conduct a computer simulation. Go to this web page: http://statweb.stanford.edu/~susan/surprise/Birthday.html, and once the applet has loaded, select 30 birthdays and then keep clicking Start and Reset. If you keep track of the number of times that there is a repeated birthday, you should get a repeated birthday about 7 out of every 10 times you run the simulation.

Try it Now 6

Suppose 10 people are in a room. What is the probability that there is at least one shared birthday among these 10 people?

Answer

$$P(\text{shared birthday}) = 1 - \frac{{}_{365}P_{10}}{365^{10}} \approx 0.117$$

3.8.1: Exercises

1. A boy owns 2 pairs of pants, 3 shirts, 8 ties, and 2 jackets. How many different outfits can he wear to school if he must wear one of each item?
2. At a restaurant you can choose from 3 appetizers, 8 entrees, and 2 desserts. How many different three-course meals can you have?
3. How many three-letter "words" can be made from 4 letters "FGHI" if
    a. repetition of letters is allowed
    b. repetition of letters is not allowed
4. How many four-letter "words" can be made from 6 letters "AEBWDP" if
    a. repetition of letters is allowed
    b. repetition of letters is not allowed
5. All of the license plates in a particular state feature three letters followed by three digits (e.g. ABC 123). How many different license plate numbers are available to the state's Department of Motor Vehicles? Letters and digits may be repeated.
6. A computer password must be eight characters long. How many passwords are possible if only the 26 letters of the alphabet are allowed? A letter may not be used more than once.
7. A pianist plans to play 4 pieces at a recital. In how many ways can she arrange these pieces in the program?
8. In how many ways can first, second, and third prizes be awarded in a contest with 210 contestants?
9. Seven Olympic sprinters are eligible to compete in the 4 x 100 m relay race for the USA Olympic team. How many four-person relay teams can be selected from among the seven athletes?
10. A computer user has downloaded 25 songs using an online file-sharing program and wants to create a CD-R with ten songs to use in his portable CD player. If the order that the songs are placed on the CD-R is important to him, how many different CD-Rs could he make from the 25 songs available to him?
11. In western music, an octave is divided into 12 pitches. For the film *Close Encounters of the Third Kind*, director Steven Spielberg asked composer John Williams to write a five-note theme, which aliens would use to communicate with people on Earth. Disregarding rhythm and octave changes, how many five-note themes are possible if no note is repeated?
12. In the early twentieth century, proponents of the Second Viennese School of musical composition (including Arnold Schönberg, Anton Webern and Alban Berg) devised the twelve-tone technique, which utilized a tone row consisting of all 12 pitches from the chromatic scale in any order, but with no pitches repeated in the row. Disregarding rhythm and octave changes, how many tone rows are possible?
13. In how many ways can 4 pizza toppings be chosen from 12 available toppings?
14. At a baby shower 17 guests are in attendance and 5 of them are randomly selected to receive a door prize. If all 5 prizes are identical, in how many ways can the prizes be awarded?
15. In the 6/50 lottery game, a player picks six numbers from 1 to 50. How many different choices does the player have if order doesn’t matter?
16. In a lottery daily game, a player picks three numbers from 0 to 9. How many different choices does the player have if order doesn’t matter?
17. A jury pool consists of 27 people. How many different ways can 11 people be chosen to serve on a jury and one additional person be chosen to serve as the jury foreman?
18. Suppose the United States Senate Committee on Commerce, Science, and Transportation consists of 23 members, 12 Republicans and 11 Democrats and that the Surface Transportation and Merchant Marine Subcommittee consists of 8 Republicans and 7 Democrats. How many ways can members of the Subcommittee be chosen from the Committee?
19. You own 16 vinyl albums. You want to randomly arrange 5 of them in a album rack. What is the probability that the rack ends up in alphabetical order?
20. A jury pool consists of 27 people, 14 men and 13 women. Compute the probability that a randomly selected jury of 12 people is all male.
21. In a lottery game, a player picks six numbers from 1 to 48. If 5 of the 6 numbers match those drawn, they player wins second prize. What is the probability of winning this prize?

22. In a lottery game, a player picks six numbers from 1 to 48. If 4 of the 6 numbers match those drawn, they player wins third prize. What is the probability of winning this prize?
23. Compute the probability that a 5-card poker hand is dealt to you that contains all hearts.
24. Compute the probability that a 5-card poker hand is dealt to you that contains four Aces.

3.9: What is Statistics?

5.1 Learning Objectives

- Describe what statistics is
- Identify and describe two branches of statistics

Like most people, you probably feel that it is important to "take control of your life." But what does this mean? Partly it means being able to properly evaluate the data and claims that bombard you every day. If you cannot distinguish good from faulty reasoning, then you are vulnerable to manipulation and to decisions that are not in your best interest. Statistics provides tools that you need in order to react intelligently to information you hear or read. In this sense, Statistics is one of the most important things that you can study.

To be more specific, here are some claims that we have heard on several occasions. (We are *not* saying that each one of these claims is true!)

- 4 out of 5 dentists recommend Dentyne.
- Almost 85% of lung cancers in men and 45% in women are tobacco-related.
- Condoms are effective 94% of the time.
- Native Americans are significantly more likely to be hit crossing the streets than are people of other ethnicities.
- People tend to be more persuasive when they look others directly in the eye and speak loudly and quickly.
- Women make 75 cents to every dollar a man makes when they work the same job.
- A surprising new study shows that eating egg whites can increase one's life span.
- People predict that it is very unlikely there will ever be another baseball player with a batting average over 400.
- There is an 80% chance that in a room full of 30 people that at least two people will share the same birthday.
- 79.48% of all statistics are made up on the spot.

All of these claims are statistical in character. We suspect that some of them sound familiar; if not, we bet that you have heard other claims like them. Notice how diverse the examples are; they come from psychology, health, law, sports, business, etc. Indeed, data and data-interpretation show up in discourse from virtually every facet of contemporary life.

Definition: Statistics

Statistics is the science of collecting, organizing, analyzing, and interpreting data in order to make a decision.

Statistics are often presented in an effort to add credibility to an argument or advice. You can see this by paying attention to television advertisements. Many of the numbers thrown about in this way do not represent careful statistical analysis. They can be misleading, and push you into decisions that you might find cause to regret. For these reasons, learning about statistics is a long step towards taking control of your life. (It is not, of course, the only step needed for this purpose.) This chapter will help you learn statistical essentials. It will make you into an intelligent consumer of statistical claims.

You can take the first step right away. To be an intelligent consumer of statistics, your first reflex must be to question the statistics that you encounter. The British Prime Minister Benjamin Disraeli famously said, "There are three kinds of lies -- lies, damned lies, and statistics." This quote reminds us why it is so important to understand statistics. So let us invite you to reform your statistical habits from now on. No longer will you blindly accept numbers or findings. Instead, you will begin to think about the numbers, their sources, and most importantly, the procedures used to generate them.

We have put the emphasis on defending ourselves against fraudulent claims wrapped up as statistics. Just as important as detecting the deceptive use of statistics is the appreciation of the proper use of statistics. You must also learn to recognize statistical evidence that supports a stated conclusion. When a research team is testing a new treatment for a disease, statistics allows them to conclude based on a relatively small trial that there is good evidence their drug is effective. Statistics allowed prosecutors in the 1950’s and 60’s to demonstrate racial bias existed in jury panels. Statistics are all around you, sometimes used well, sometimes not. We must learn how to distinguish the two cases.

There are two main branches of statistics; descriptive statistics and inferential statistics.

Definition: Branches of Statistics

Descriptive statistics involves the organization, the summary, and how the data is displayed.

Inferential statistics involves using a sample to draw conclusions about a population.

Example 1

Twenty of the 80 students enrolled in Math 150 were asked if they prefer to do an in-class examination or a take-home project for their second assessment. Fifteen, or 75% of the students polled, indicated their preference is to have a take-home project as a second assessment. What type of statistics was performed?

Solution

This was descriptive statistics since the data collected displays a summary of the results. It is not inferential statistics because we are not extending it to a much larger population.

Example 2

From data collected in the years past elections, it is predicted that 67% of registered voters will vote in the next presidential election. What type of statistics is this?

Solution

This was inferential statistics because data was collected from a sample of previous elections, and used it to draw conclusions for the entire population.

3.9.1: Exercises

Determine the branch of statistics that is being used.

1. A survey conducted to 54 various donut shops found that 38% of their customers like buttermilk glazed donut the most. Which branch of statistics is being used here?
2. A survey conducted to 54 various donut shops found that 38% of their customers like buttermilk glazed donut the most, and then the researcher concludes that 38% of all donut-purchasing customers will love the buttermilk glazed donut the most. Which branch of statistics is being used here?
3. A survey was given to movie goers at 100 movie theaters to estimate what percent of all movie goers arrive later than the published start time to avoid the previews. Which branch of statistics is being used here?
4. A survey was given to students at a local community college to determine how they liked their experience with online registration. They determined the 76% of the students were satisfied with their experience. Which branch of statistics is being used here?
5. A study measured the lifespan of 1000 golden retrievers and used that to predict the lifespan of all golden retrievers. What branch of statistics is being used here?

3.10: Sampling methods

5.4 Learning Objectives

- Identify the sampling method used in a study

There are various types of sampling so it is important to understand the differences. We would not anticipate very accurate results if we drew all of our samples from among the customers at a Starbucks, nor would we expect that a sample drawn entirely from the membership list of the local Elks club would provide a useful picture of district-wide support for our candidate. Drawing samples in this way is called **convenience sampling**.

One way to ensure that the sample has a reasonable chance of mirroring the population we are interested in is to employ *randomness*. The most basic random method is simple random sampling.

A **random sample** is one in which each member of the population has an equal probability of being chosen. A **simple random sample** is one in which every member of the population and any group of members has an equal probability of being chosen.

Example 1

If we could somehow identify all likely voters in the state, put each of their names on a piece of paper, toss the slips into a (very large) hat and draw 1000 slips out of the hat, we would have a simple random sample.

In practice, computers are better suited for this sort of endeavor than millions of slips of paper and extremely large headgear.

It is always possible, however, that even a random sample might end up not being totally representative of the population. If we repeatedly take samples of 1000 people from among the population of likely voters in the state of Washington, some of these samples might tend to have a slightly higher percentage of Democrats (or Republicans) than does the general population; some samples might include more older people and some samples might include more younger people; etc. In most cases, this **sampling variability** is not significant.

Definition: Sampling variability

The natural variation of samples is called **sampling variability**.

This is unavoidable and expected in random sampling, and in most cases is not an issue.

To help account for variability, pollsters might instead use a **stratified sample**.

Definition: Stratified sampling

In **stratified sampling**, a population is divided into a number of subgroups (or strata). Random samples are then taken from each subgroup with sample sizes proportional to the size of the subgroup in the population.

Example 2

Suppose in a particular state that previous data indicated that the electorate was comprised of 39% Democrats, 37% Republicans and 24% independents. In a sample of 1000 people, they would then expect to get about 390 Democrats, 370 Republicans and 240 independents. To accomplish this, they could randomly select 390 people from among those voters known to be Democrats, 370 from those known to be Republicans, and 240 from those with no party affiliation.

Stratified sampling can also be used to select a sample with people in desired age groups, a specified mix ratio of males and females, etc. A variation on this technique is called **quota sampling**.

Definition: Quota Sampling

Quota sampling is a variation on stratified sampling, wherein samples are collected in each subgroup until the desired quota is met.

Example 3

Suppose the pollsters call people at random, but once they have met their quota of 390 Democrats, they only gather people who do not identify themselves as a Democrat.

You may have had the experience of being called by a telephone pollster who started by asking you your age, income, etc. and then thanked you for your time and hung up before asking any "real" questions. Most likely, they already had contacted enough people in your demographic group and were looking for people who were older or younger, richer or poorer, etc. Quota sampling is usually a bit easier than stratified sampling, but also does not ensure the same level of randomness.

Another sampling method is **cluster sampling**, in which the population is divided into groups, and one or more groups are randomly selected to be in the sample.

Cluster sampling

In **cluster sampling**, the population is divided into subgroups (clusters), and a set of subgroups are selected to be in the sample

Example 4

If the college wanted to survey students, since students are already divided into classes, they could randomly select 10 classes and give the survey to all the students in those classes. This would be cluster sampling.

Other sampling methods include **systematic sampling**.

Systematic sampling

In **systematic sampling**, every n^{th} member of the population is selected to be in the sample.

Example 5

To select a sample using systematic sampling, a pollster calls every 100^{th} name in the phone book.

Solution

Systematic sampling is not as random as a simple random sample (if your name is Albert Aardvark and your sister Alexis Aardvark is right after you in the phone book, there is no way you could both end up in the sample) but it can yield acceptable samples.

Perhaps the worst types of sampling methods are **convenience samples** and **voluntary response samples**.

Convenience sampling and voluntary response sampling

Convenience sampling is samples chosen by selecting whoever is convenient.

Voluntary response sampling is allowing the sample to volunteer.

Example 6

A pollster stands on a street corner and interviews the first 100 people who agree to speak to him. This is a convenience sample.

Example 7

A website has a survey asking readers to give their opinion on a tax proposal. This is a self-selected sample, or voluntary response sample, in which respondents volunteer to participate.

Usually voluntary response samples are skewed towards people who have a particularly strong opinion about the subject of the survey or who just have way too much time on their hands and enjoy taking surveys.

Try it Now 1

In each case, indicate what sampling method was used

a. Every 4th person in the class was selected
b. A sample was selected to contain 25 men and 35 women
c. Viewers of a new show are asked to vote on the show’s website
d. A website randomly selects 50 of their customers to send a satisfaction survey to
e. To survey voters in a town, a polling company randomly selects 10 city blocks, and interviews everyone who lives on those blocks.

Answer

a. Systematic
b. Stratified or Quota
c. Voluntary response
d. Simple random
e. Cluster

3.10.1: Exercises

1. In a study, the sample is chosen by separating all cars by size, and selecting 10 of each size grouping. What is the sampling method?
2. In a study, the sample is chosen by writing everyone’s name on a playing card, shuffling the deck, then choosing the top 20 cards. What is the sampling method?
3. A city government needs to know if its residents will support the construction of a new hospital. Which procedure would be the most appropriate for creating a sample of the city's population?
   a. Survey a random sample of doctors who live in the city.
   b. Survey a random sample of persons within each neighborhood of the city.
   c. Survey a random sample of people who live within a 5-mile radius of the proposed hospital site.
   d. Survey every 100th person who visits one of the other hospitals in the city.
4. The math department wants to know students' opinions of how much math homework they should be doing. Which procedures would be the most appropriate for getting an unbiased sample?
   a. Survey a selection of students enrolled at the college.
   b. Survey enrolled students chosen from the first 1000 students listed alphabetically.
   c. Survey a selection of students enrolled in a math course.
   d. Survey a selection of students walking through the Science Building.
5. A city government wants to conduct a survey among the city's homeless to discover their opinion about the recent ban on camping in certain public areas.
   a. They decide to do a random survey of people residing in homeless shelters. Does this seem like a good sample? Why or why not?
   b. They decide to survey all the homeless people contacted by the police. Does this seem like a good sample? Why or why not?
   c. They leave signs at several homeless encampments asking people to text their answers to some questions. Does this seem like a good sample? Why or why not?
   d. Design a better sampling process than any of those already given.
6. A local supermarket is considering the addition of a coffee bar to its premises. They want to find out how much people might patronize the coffee bar. Design a survey for the supermarket to use.

3.11: Presenting Data Graphically

5.7 Learning Objectives

- Create a frequency table from a data set
- Create bar graphs, pareto charts, pie charts, histograms and frequency polygons from data sets
- Describe the shape of a histogram

Categorical, or qualitative, data are pieces of information that allow us to classify the objects under investigation into various categories. We usually begin working with categorical data by summarizing the data into a **frequency table.**

Definition: Frequency Table

A **frequency table** is a table with two columns. One column lists the categories, and another for the frequencies with which the items in the categories occur (how many items fit into each category).

Example 1

An insurance company determines vehicle insurance premiums based on known risk factors. If a person is considered a higher risk, their premiums will be higher. One potential factor is the color of your car. The insurance company believes that people with some color cars are more likely to get in accidents. To research this, they examine police reports for recent total-loss collisions. The data is summarized in the frequency table below.

Color	Frequency
Blue	25
Green	52
Red	41
White	36
Black	39
Grey	23

Sometimes we need an even more intuitive way of displaying data. This is where charts and graphs come in. There are many, many ways of displaying data graphically, but we will concentrate on one very useful type of graph called a bar graph. In this section we will work with bar graphs that display categorical data; the next section will be devoted to bar graphs that display quantitative data.

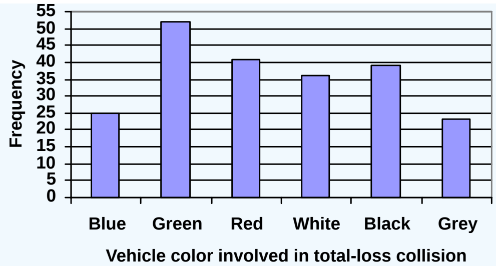
Definition: Bar Graph

A **bar graph** is a graph that displays a bar for each category with the length of each bar indicating the frequency of that category.

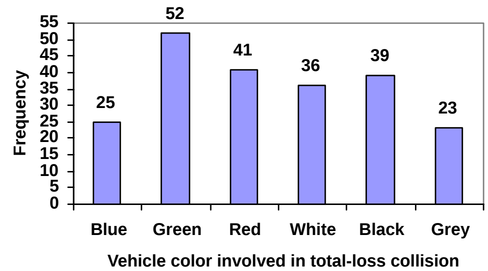
To construct a bar graph, we need to draw a vertical axis and a horizontal axis. The vertical direction will have a scale and measure the frequency of each category; the horizontal axis has no scale in this instance. The construction of a bar chart is most easily described by use of an example.

Example 2

Using our car data from above, note the highest frequency is 52, so our vertical axis needs to go from 0 to 52, but we might as well use 0 to 55, so that we can put a hash mark every 5 units:



Notice that the height of each bar is determined by the frequency of the corresponding color. The horizontal gridlines are a nice touch, but not necessary. In practice, you will find it useful to draw bar graphs using graph paper, so the gridlines will already be in place, or using technology. Instead of gridlines, we might also list the frequencies at the top of each bar, like this:



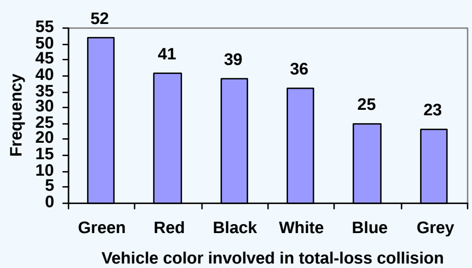
In this case, our chart might benefit from being reordered from largest to smallest frequency values. This arrangement can make it easier to compare similar values in the chart, even without gridlines. When we arrange the categories in decreasing frequency order like this, it is called a **Pareto chart**.

Definition: Pareto Chart

A **Pareto chart** is a bar graph ordered from highest to lowest frequency.

Example 3

Transforming our bar graph from earlier into a Pareto chart, we get:



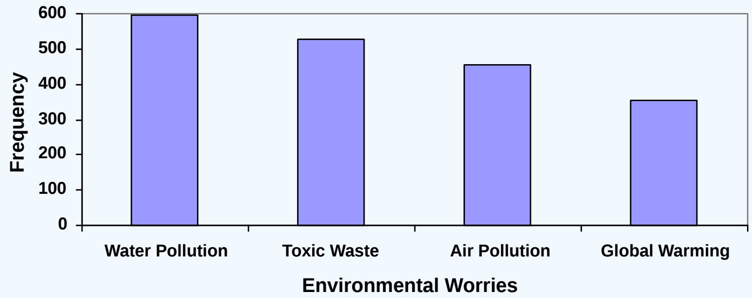
Example 4

In a survey[1], adults were asked whether they personally worried about a variety of environmental concerns. The numbers (out of 1012 surveyed) who indicated that they worried “a great deal” about some selected concerns are summarized below.

Environmental Issue	Frequency
Pollution of drinking water	597
Contamination of soil and water by toxic waste	526
Air pollution	455
Global warming	354

Solution

This data could be shown graphically in a bar graph:



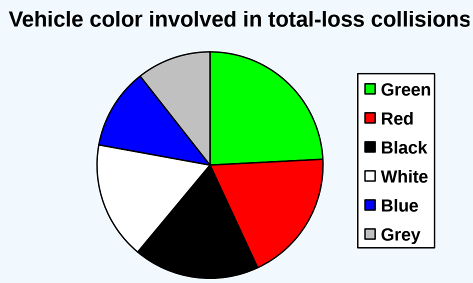
To show relative sizes, it is common to use a pie chart.

Definition: Pie Chart

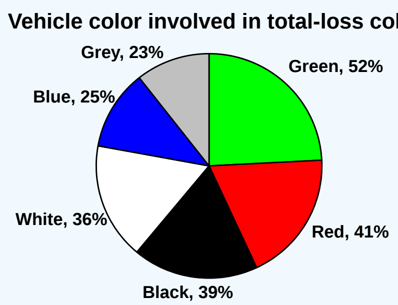
A **pie chart** is a circle with wedges cut of varying sizes marked out like slices of pie or pizza. The relative sizes of the wedges correspond to the relative frequencies of the categories.

Example 5

For our vehicle color data, a pie chart might look like this:



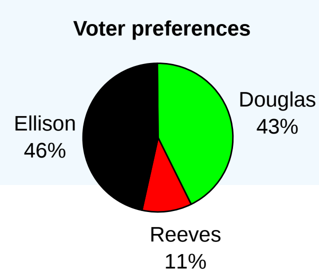
Pie charts can often benefit from including frequencies or relative frequencies (percents) in the chart next to the pie slices. Often having the category names next to the pie slices also makes the chart clearer.



Example 6

The pie chart to the right shows the percentage of voters supporting each candidate running for a local senate seat.

If there are 20,000 voters in the district, the pie chart shows that about 11% of those, about 2,200 voters, support Reeves.



Pie charts look nice, but are harder to draw by hand than bar charts since to draw them accurately we would need to compute the angle each wedge cuts out of the circle, then measure the angle with a protractor. Computers are much better suited to drawing pie charts. Common software programs like Microsoft Word or

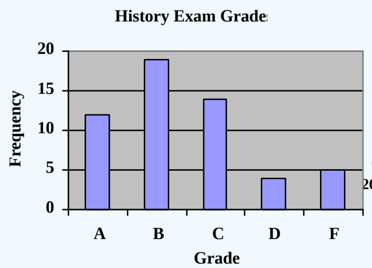
Excel, OpenOffice.org Write or Calc, or Google Docs are able to create bar graphs, pie charts, and other graph types. There are also numerous online tools that can create graphs[2].

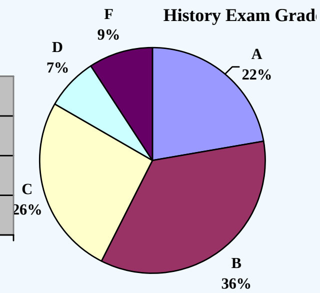
Try it Now 1

Create a bar graph and a pie chart to illustrate the grades on a history exam below.

A: 12 students, B: 19 students, C: 14 students, D: 4 students, F: 5 students

Answer





Quantitative, or numerical, data can also be summarized into frequency tables.

Example 7

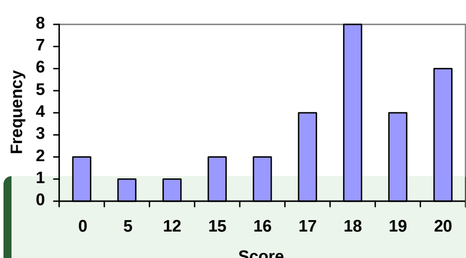
A teacher records scores on a 20-point quiz for the 30 students in his class. The scores are:

19 20 18 18 17 18 19 17 20 18 20 16 20 15 17 12 18 19 18 19 17 20 18 16 15 18 20 5 0 0

These scores could be summarized into a frequency table by grouping like values:

Score	Frequency
0	2
5	1
12	1
15	2
16	2
17	4
18	8
19	4
20	6

Using this table, it would be possible to create a standard bar chart from this summary, like we did for categorical data:



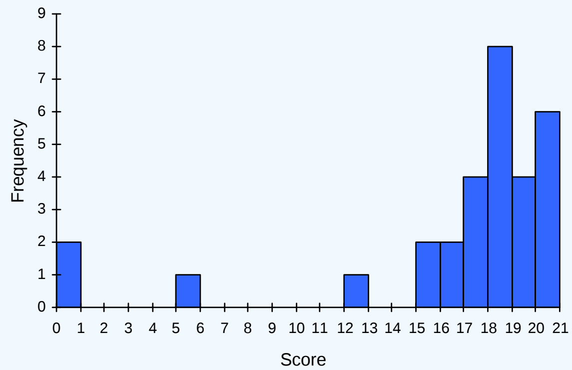
However, since the scores are numerical values, this chart doesn't really make sense; the first and second bars are five values apart, while the later bars are only one value apart. It would be more correct to treat the horizontal axis as a number line. This type of graph is called a **histogram**.

Definition: Histogram

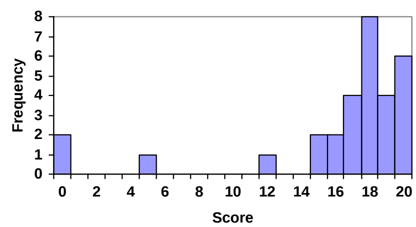
A histogram is like a bar graph, but where the horizontal axis is a number line.

Example 8

For the values above, a histogram would look like:



Notice that in the histogram, a bar represents values on the horizontal axis from that on the left hand-side of the bar up to, but not including, the value on the right hand side of the bar. Some people choose to have bars start at ½ values to avoid this ambiguity.



Unfortunately, not a lot of common software packages can correctly graph a histogram. About the best you can do in Excel or Word is a bar graph with no gap between the bars and spacing added to simulate a numerical horizontal axis.

If we have a large number of widely varying data values, creating a frequency table that lists every possible value as a category would lead to an exceptionally long frequency table, and probably would not reveal any patterns. For this reason, it is common with quantitative data to group data into **class intervals** or **bins**.

Definition: Class Intervals

Class intervals are groupings of the data. In general, we define class intervals so that:

- Each interval is equal in size. For example, if the first class contains values from 120-129, the second class should include values from 130-139.
- We have somewhere between 5 and 20 classes, typically, depending upon the number of data we're working with.
- The class interval width is the lowest value in one class minus the lowest value in the previous class.

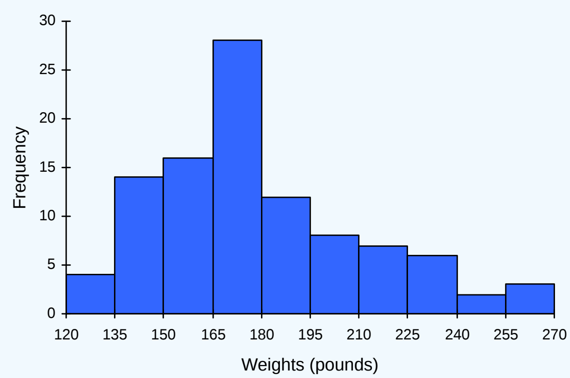
Example 9

Suppose that we have collected weights from 100 male subjects as part of a nutrition study. For our weight data, we have values ranging from a low of 121 pounds to a high of 263 pounds, giving a range of 263-121 = 142. We could create 7 intervals with a width of around 20, 14 intervals with a width of around 10, or somewhere in between. Often time we have to experiment with a few possibilities to find something that represents the data well.

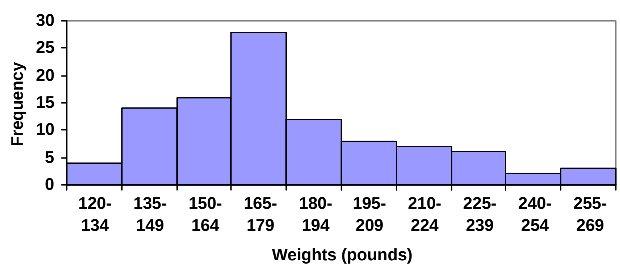
Let us try using an interval width of 15. We could start at 121, or at 120 since it is a nice round number. With a class width of 15, the next class begins at 120 + 15 or 135, the following one at 135 + 15 or 150, and so on until there are classes for all the data.

Interval	Frequency
120 — 134	4
135 — 149	14
150 — 164	16
165 — 179	28
180 — 194	12
195 — 209	8
210 — 224	7
225 — 239	6
240 — 254	2
255 — 269	3

A histogram of this data would look like:

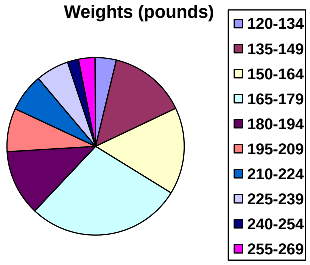


In many software packages, you can create a graph similar to a histogram by putting the class intervals as the labels on a bar chart.



Histograms also display something very important called **shape**. In this particular example above, you can see a majority of the data is the the left, and it gets smaller and smaller the more you go to the right. We consider this the tail-end to the right and therefore call this **skewed to the right**. If a majority of the data was on the right and its tail-end was to the left, we would call this **skewed to the left**. If there was approximately equally distributed data to both the right and left, we would call this **symmetric**. If all rectangles in a histogram had the same height, we would call this **uniform**.

Other graph types such as pie charts are possible for quantitative data. The usefulness of different graph types will vary depending upon the number of intervals and the type of data being represented. For example, a pie chart of our weight data is difficult to read because of the quantity of intervals we used.



Try it Now 3

The total cost of textbooks for the term was collected from 36 students. Create a histogram for this data.

\$140 \$160 \$160 \$165 \$180 \$220 \$235 \$240 \$250 \$260 \$280 \$285

\$285 \$285 \$290 \$300 \$300 \$305 \$310 \$310 \$315 \$315 \$320 \$320

\$330 \$340 \$345 \$350 \$355 \$360 \$360 \$380 \$395 \$420 \$460 \$460

Answer

Using a class intervals of size 55, we can group our data into six intervals:

cost interval	Frequency
$\$140-194$	5
$\$195-249$	3
$\$250-304$	9
$\$305-359$	12
$\$360-414$	4
$\$415-469$	3

We can use the frequency distribution to generate the histogram.

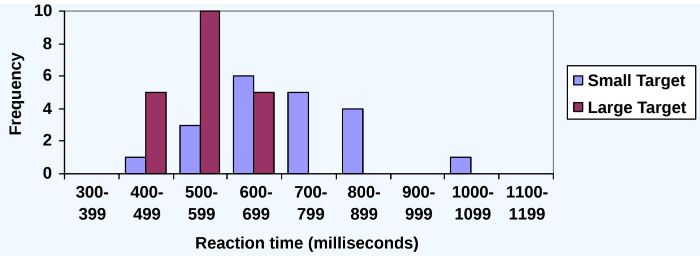
When collecting data to compare two groups, it is desirable to create a graph that compares quantities.

Example 10

The data below came from a task in which the goal is to move a computer mouse to a target on the screen as fast as possible. On 20 of the trials, the target was a small rectangle; on the other 20, the target was a large rectangle. Time to reach the target was recorded on each trial.

Interval (milliseconds)	Frequency small target	Frequency large target
$300-399$	0	0
$400-499$	1	5
$500-599$	3	10
$600-699$	6	5
$700-799$	5	0
$800-899$	4	0
$900-999$	0	0
$1000-1099$	1	0
$1100-1199$	0	0

One option to represent this data would be a comparative histogram or bar chart, in which bars for the small target group and large target group are placed next to each other.

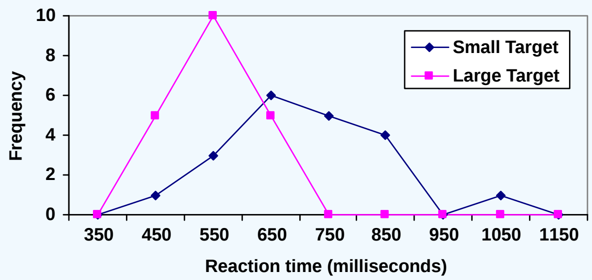


Definition: Fequency Polygon

An alternative representation is a **frequency polygon**. A frequency polygon starts out like a histogram, but instead of drawing a bar, a point is placed in the midpoint of each interval at height equal to the frequency. Typically the points are connected with straight lines to emphasize the distribution of the data.

Example 11

This graph makes it easier to see that reaction times were generally shorter for the larger target, and that the reaction times for the smaller target were more spread out.



[1] Gallup Poll. March 5-8, 2009. http://www.pollingreport.com/enviro.htm

[2] For example: http://nces.ed.gov/nceskids/createAgraph/ or http://docs.google.com

[3]CNN/Opinion Research Corporation Poll. Dec 19-21, 2008, from http://www.pollingreport.com/civil.htm

3.11.1: Exercises

1. The table below shows scores on a Math test.

80	50	50	90	70	70	100	60	70	80	70	50
90	100	80	70	30	80	80	70	100	60	60	50

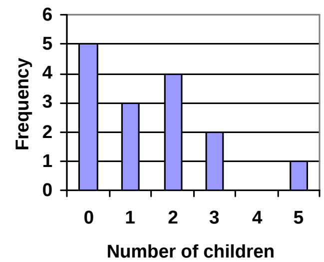
   a. Complete a frequency table for the Math test scores.
   b. Construct a bar graph of the data. Label both axes.
   c. Construct a pie chart of the data. Label the sectors.

2. A group of adults where asked how many cars they had in their household. Their responses are listed below.

1	4	2	2	1	2	3	3	1	4	2	2
1	2	1	3	2	2	1	2	1	1	1	2

   a. Complete a frequency table for the number of cars data.
   b. Construct a bar graph of the data. Label the axes.
   c. Construct a Pareto chart of the data. Label the axes.
   d. Construct a pie chart of the data. Label the sectors.

3. A group of adults were asked how many children they have in their families. The bar graph to the right shows the number of adults who indicated each number of children.

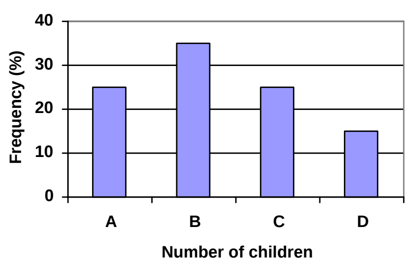


   a. How many adults where questioned?
   b. What percentage of the adults questioned had 0 children?
   c. Create a pie graph for the number of children in these families. Label the sectors.

4. Jasmine was interested in how many days it would take an order from Amazon to arrive at her door. The graph below shows the data she collected.

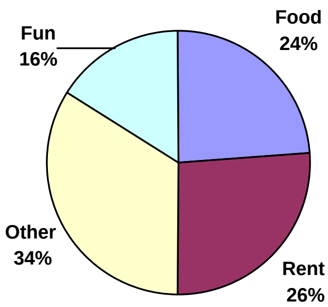
   a. How many items did she order?
   b. What percentage of the items arrived in one day?
   c. Create a pie graph for her data. Label the sectors.

5. The bar graph below shows the *percentage* of students who received each letter grade on their last English paper. The class contains 20 students. What number of students earned an A on their paper?

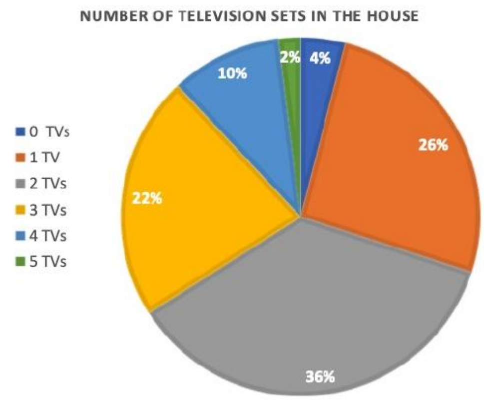


   a. How many students earned an A on their papers?
   b. Create a frequency bar graph for these data. How do the two bar graphs compare?

6. Kori categorized her spending for this month into four categories: Rent, Food, Fun, and Other. The percents she spent in each category are pictured here. If she spent a total of $2600 this month, how much did she spend on rent?



7. 50 people were asked about the number of television sets in their houses. The pie chart below shows their responses.



a. How many television sets did the largest number of people have?
b. How many television sets did the fewest number of people have?
c. How many people had no television sets?
d. How many people had at least 3 television sets?
e. What percent of the people surveyed had at most 2 television sets?
f. How many people had not more than 2 television sets?

8. In 2018, 487,890 people in California graduated from college with various degrees. The numbers are shown in the chart below.

Associate's Degree	170,890
Bachelor's Degree	216,810
Master's Degree	80,480
Doctorate or Professional Degree	19,710

a. Is the data categorical or numerical?
b. Which type(s) of graphs would be appropriate way(s) to display this information? Bar Graph, Pareto Chart, Pie Chart, Frequency Histogram, or Frequency Polygon

9. The ages of the U.S.Presidents who were inaugurated in the 20th century are given.

42; 51; 56; 55; 51; 54; 51; 60; 62; 43; 55; 56; 61; 52; 69; 64; 46

a. Create a grouped frequency table for these scores starting at 40 with a class width of 10.
b. Create a frequency histogram for these scores.
c. Describe the shape of the frequency histogram.

10. Results for 40 students on a recent math exam are given.

83; 76; 98; 75; 56; 47; 67; 92; 69; 74; 72; 84; 96; 66; 51; 64; 58; 80; 85; 78; 58; 88; 90; 76; 74; 82; 76; 66; 50; 84; 66; 79; 83; 43; 54; 93; 77; 67; 54; 76

a. Create a grouped frequency table for these scores starting at 40 with a class width of 10.

b. Create a frequency histogram for these scores.
c. Describe the shape of the frequency histogram.
d. Make a frequency polygon for this data.

11. Suppose that a new AIDS antibody drug is currently under study. It is given to patients once the AIDS symptoms have revealed themselves. Of interest is the average length of time in months patients live once starting the treatment. A researcher follows a set of 40 AIDS patients from the start of treatment until their deaths. The following data (in months) are collected.

    3; 4; 11; 15; 16; 17; 22; 44; 37; 16; 14; 24; 25; 15; 26; 27; 33; 29; 35; 44; 13; 21; 22; 10; 12; 8; 40; 32; 26; 27; 31; 34; 29; 17; 8; 24; 18; 47; 33; 34

    a. Create a frequency table starting at 0 with a class width of 6 months.
    b. What percent of the AIDS patients in the study survived at least 3 years after beginning treatment?
    c. What percent of the AIDS patients in the study survived less than 18 months?
    d. Create a frequency histogram using the above data. Label the axes.

12. As part of a study, the weights of 42 medium-sized dogs were measured in pounds. The results are shown below.

26.1	16.2	21.1	34.7	45.3	18.5	41.6
38.8	22.9	48.1	27.1	22.4	30.6	39.1
62.8	25.1	25.0	38.6	29.9	31.7	28.9
20.3	56.1	60.5	24.0	61.7	28.6	32.9
33.5	18.0	23.5	27.9	46.8	30.0	34.3
62.2	49.3	59.7	19.6	20.8	23.2	24.4

    a. Create a frequency table using classes. Start at 16 and use a class width of 8.
    b. What percent of the dogs weigh no more than 24 pounds?
    c. What percent of the dogs studied weigh at least 48 pounds?
    d. What percent of the dogs studied weigh at least 24 pounds, but do not exceed 40 pounds?

13. People were asked how long they waited in line to ride a popular attraction at a local theme park. The frequency table below shows the results.

x = Time (minutes)	Frequency
$0 \le x < 5$	32
$5 \le x < 10$	47
$10 \le x < 15$	36
$15 \le x < 20$	22
$20 \le x < 25$	13
$25 \le x < 30$	10

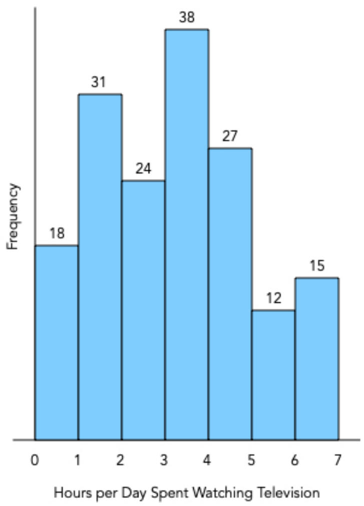
    a. Construct a histogram to represent the data. Label the axes of your histogram.
    b. Describe the shape of the histogram.

14. Hourly wages of workers at a local company are shown in the frequency table below.

x = Hourly Wage (dollars)	Frequency
$13.00 \le x < 14.50$	21
$14.50 \le x < 16.00$	35
$16.00 \le x < 17.50$	42
$17.50 \le x < 19.00$	27
$19.00 \le x < 20.50$	18
$20.50 \le x < 22.00$	9

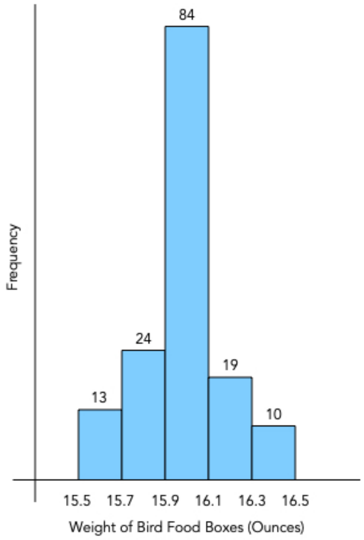
a. Construct a histogram to represent the data. Label the axes of your histogram.
b. Describe the shape of the histogram.

15. The frequency histogram below shows the number of hours per day a random sample of teenagers spent watching television.



a. How many teenagers were included in the sample?
b. What percent of the teenagers in the sample spent less than 4 hours watching television?
c. What percent of the teenagers in the sample spent at least 5 hours watching television?
d. What percent of the teenagers in the sample spent at least 1 hour watching television?
e. What percent of the teenagers in the sample spent less than 2 hours watching television?
f. What percent of the teenagers in the sample spent at least 2 hours but less than 4 hours watching television?
g. What percent of the teenagers in the sample spent more than 3.5 hours watching television?

16. A pet food company is testing the accuracy of the machine that fills its boxes with bird food. The weights of the sample boxes are shown in the frequency histogram.



a. How many boxes did the company test?
b. How many of the boxes tested weighed at least 16.1 ounces?
c. What percent of the boxes tested weighed less than 15.9 ounces?
d. What percent of the boxes tested weighed within 0.1 ounces of the 16 ounce weight stated on the box? That is, $15.9 \le w < 16.1$ where w is the weight of the box.
e. If boxes weighing within 0.1 ounces of the 16-ounce stated weight are acceptable, are there more underweight boxes or overweight boxes among the boxes that do not meet the weight standard?

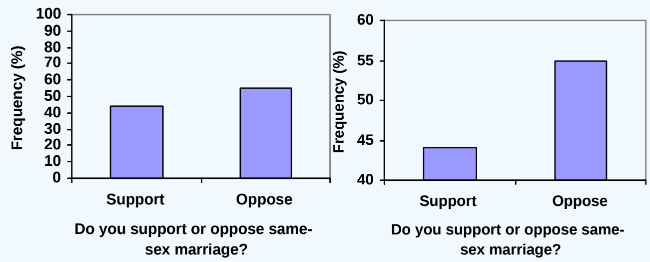
3.12: Reading Graphs in Media

Don’t get fancy with graphs! People sometimes add features to graphs that don’t help to convey their information. For example, 3-dimensional bar charts are usually not as effective as their two-dimensional counterparts.

Another distortion in bar charts results from setting the baseline to a value other than zero. The baseline is the bottom of the vertical axis, representing the least number of cases that could have occurred in a category. Normally, this number should be zero.

Example 1

Compare the two graphs below showing support for same-sex marriage rights from a poll taken in December 2008[3]. The difference in the vertical scale on the first graph suggests a different story than the true differences in percentages; the second graph makes it look like twice as many people oppose marriage rights as support it.



Here is another way that fanciness can lead to trouble. Instead of plain bars, it is tempting to substitute meaningful images. This type of graph is called a **pictogram**.

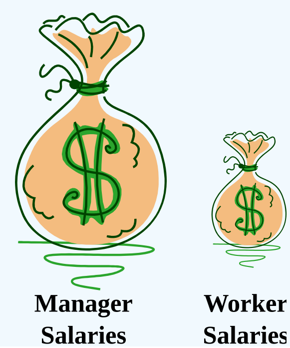
Pictogram

A **pictogram** is a statistical graphic in which the size of the picture is intended to represent the frequencies or size of the values being represented.

Example

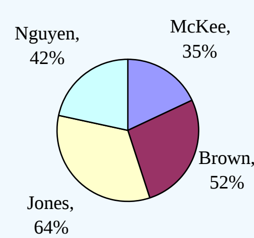
A labor union might produce the graph to the right to show the difference between the average manager salary and the average worker salary.

Looking at the picture, it would be reasonable to guess that the manager salaries is 4 times as large as the worker salaries – the area of the bag looks about 4 times as large. However, the manager salaries are in fact only twice as large as worker salaries, which were reflected in the picture by making the manager bag twice as tall.



Try it Now 2

A poll was taken asking people if they agreed with the positions of the 4 candidates for a county office. Does the pie chart present a good representation of this data? Explain.



Answer

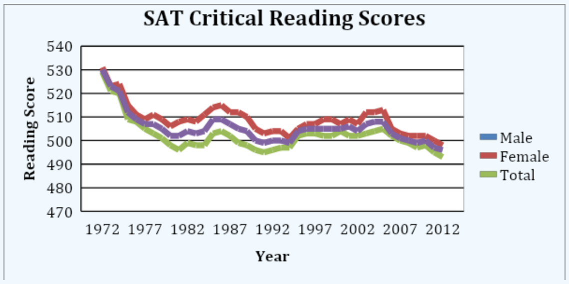
While the pie chart depicts the relative size of the people agreeing with each candidate as compared to the other candidates, the chart is confusing. Percents on a pie chart usually represent the percentage out of 100% that the slice represents.

There are many other types of graphs you will encounter in the media. The following example is a **multiple line graph**.

Example 3

The average score for SAT critical reading has been going down over the years. You can also see that the average score for male students is higher than the average score for female students for all of the years. Also, the average scores for male and female used to be closer to each other, then they separated fairly far, and look to be getting closer to each other again. What do you think could be a reason for this?

Multiple Line Graph for SAT Critical Reading Scores in AZ



(College Board: Arizona, 2012)

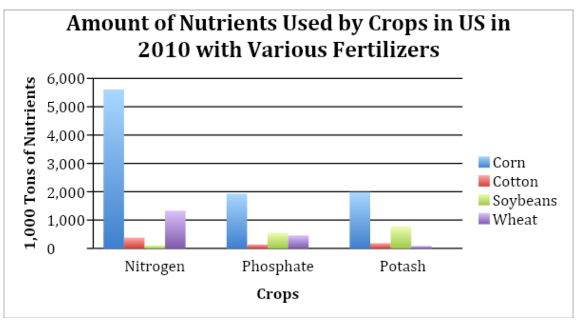
Solution

In this case the vertical axis did not start at zero, because if it did start at zero, the different lines would be very close together and difficult to see. When at all possible, the scale on the vertical axis should start at zero. Always carefully consider that if a vertical scale of a graph does not start at zero, the researchers could be attempting to exaggerate insignificant differences in the data.

Multiple Bar Graphs:

Sometimes you have information for multiple variables and instead of putting the information on different bar graphs, you can put them all on one so that you can compare the variables. The following is an example of where you might use this. The data is the amount of nutrients used by crops with various fertilizers. By creating a graph that has all of the fertilizers and all of the crops, you can see that corn with nitrogen uses the most nutrients, and soybeans with nitrogen uses the least amount of nutrients. You can also see that phosphate seems to use low amounts of nutrients for all of its crops. So, a multiple bar graph is useful to make all of these observations.

Multiple Bar Graph for the Amount of Nutrients Used by Crops: 2010

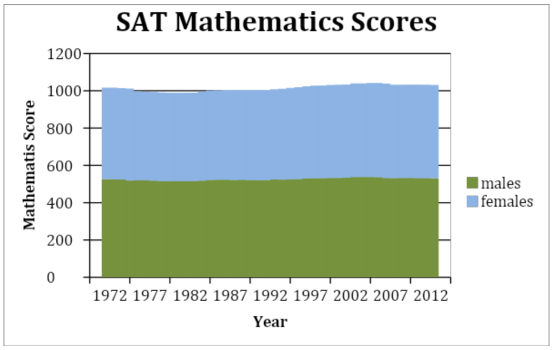


(United States Department of Agriculture [USDA], 2010)

Stack Plots:

A stack plot is basically a multiple bar graph, but with the bars separated (or stacked) on top of each other instead of next to each other. This can be useful when it is difficult to interpret a multiple bar graph since the bar heights are so close to one another. To read a given bar on a stack plot, you must subtract the height of the bar from the height of the bar below it. In the example of a stack plot below, you can see that the SAT Mathematics score for males in 1972 is about 530 while the SAT Mathematics score for females in 1972 is about 1010 – 530 = 480.

Stack Plot of SAT Mathematics Scores

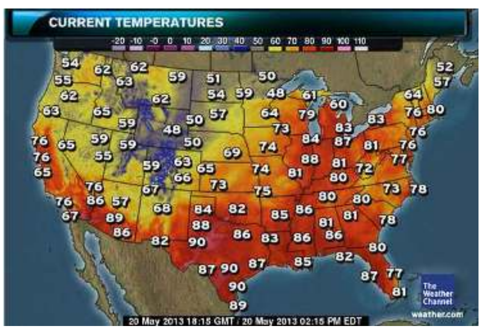


(College Board: Arizona, 2012)

Geographical Graphs:

Weather maps, topographic maps, population distribution maps, gravity maps, and vegetation maps are examples of geographical graphs. They allow you to see a trend of information over a geographic area. The following is an example of a weather map showing temperatures. As you can see, the different colors represent certain temperature ranges. From this graph, you can see that on this date, the red in the south means that the temperature was in the 80s and 90s there, and the blue in the Rockies area means that the temperature was in the 40s there.

Geographical Graph

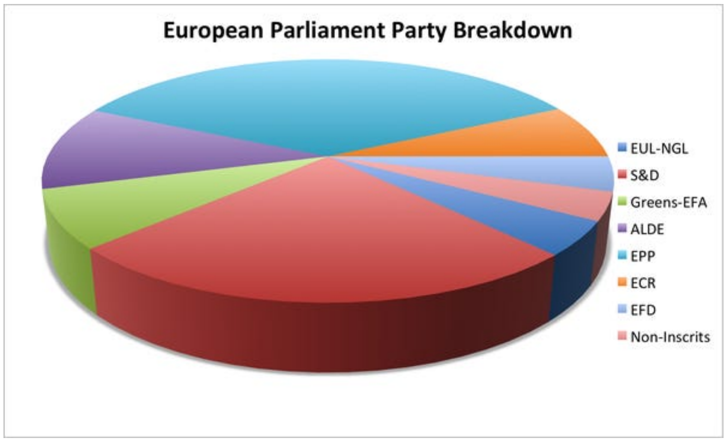


(Weather Channel, 2013)

Three-Dimensional Graphics

Some people like to show graphs in three-dimensions. This type of graph can be misleading when the media presents it this way. The following is misleading because the red party looks almost as equivalent to the teal party, which is not true.

Three-Dimensional Graph

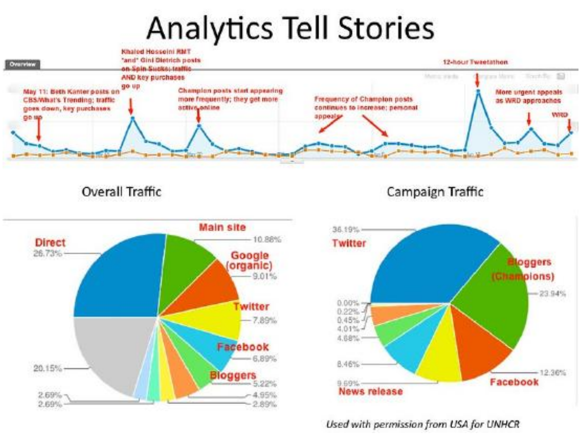


(Business Insider, 2013)

Combination Graphics

Some graphs are created so that they combine the data table and a graph or combine two types of graphs in one. The advantage is that you can see a graphical representation of the data, and still have the data to find exact values. The disadvantage is that they are busier, and usually people show graphical representation of data because people do not like looking at the data.

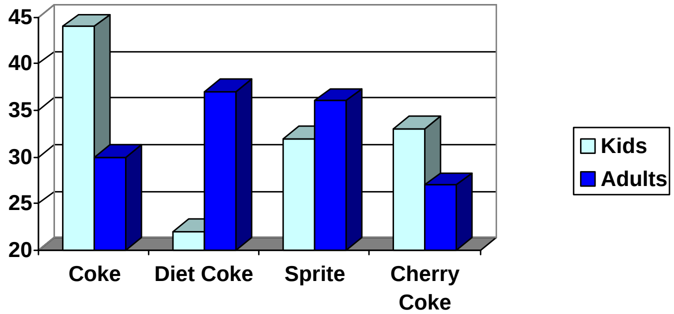
Combination Graph



These are just a few of the different types of graphs that exist in the world. There are many other ones. A quick Google search on statistical graphs will show you many more. Just open up a newspaper, magazine, or website and you are likely to see others. The most important thing to remember is that you need to look at the graph objectively, and interpret for yourself what it says. Also, do not read any cause and effect into what you see which will be discussed in more depth in the next section.

3.12.1: Exercises

1. A graph appears below showing the number of adults and children who prefer each type of soda. There were 130 adults and kids surveyed. Discuss some ways in which the graph below could be improved



3.13: Correlation and Causation, Scatter Plots

5.9 Learning Objectives

- Create a scatter plot from bi-variate data
- Describe the type of linear correlation in a scatter plot
- Explain why correlation is not the same as causation

The label on a can of Planters Cocktail Peanuts says, "Scientific evidence suggest but does not prove that eating 1.5 ounces per day of most nuts, such as peanuts, as part of a diet low in saturated fat and cholesterol & not resulting in increased caloric intake may reduce the risk of heart disease. See nutritional information for fat content (1.5 oz. is about 53 pieces)." Why is it written this way and what does this statement mean? There are many studies that exist that show that two variables are related to one another. The strength of a relationship between two variables is called **correlation**. Variables that are strongly related to each other have strong correlation. However, if two variables are correlated it does not mean that one variable caused the other variable to occur. The above example from the Planters Cocktail Peanuts label is an example of this. There is a strong correlation between eating a diet that is low in saturated fat and cholesterol and heart disease. But that correlation does not mean that eating a diet that is low in saturated fat and cholesterol will cause your risk of heart disease to go down. There could be many different variables that could cause both variables in question to go down or up. One example is that a person's genetic makeup could make them not want to eat fatty food and also not develop heart disease. No matter how strong a correlation is between two variables, you can never know for sure if one variable causes the other variable to occur without conducting experimentation. The only way to find out if eating a diet low in saturated fat and cholesterol actually lowers the risk of heart disease is to do an experiment. This is where you tell one group of people that they have to eat a diet low in saturated fat and cholesterol and another group of people that they have to eat a diet high in saturated fat and cholesterol, and then observe what happens to both groups over the years. You cannot morally do this experiment, so there is no way to prove the statement. That is why the word "may" is in the statement. We see many correlations like this one. Always be sure not to make a correlation statement into a causation statement.

Example 3.13.1: Correlation vs Causation

For each of the following scenarios answer the question and give an example of another variable that could explain the correlation.

1. There is a negative correlation between number of children a woman has and her life expectancy. Does that mean that having children causes a woman to die earlier?

A correlation between two variables does not mean that one causes the other. A possible cause for both variables could be better health care. If there is better health care, then life expectancy goes up, and also with better health care birth control is more readily available.

2. There is a positive correlation between ice cream sales and the number of drownings at the beach. Does that mean that eating ice cream can cause a person to drown?

A correlation between two variables does not mean that one causes the other. The cause for both could be that the temperature is going up. The higher the temperature, the more likely someone will buy ice cream and the more people at the beach.

3. There is a correlation between waist measures and wrist measures. Does this mean that your waist measurement causes your wrist measurement to change?

A correlation between two variables does not mean that one causes the other. The cause of both could be a person's genetics, eating habits, exercise habits, etc.

How do we tell if there is a correlation between two variables? The easiest way is to graph the two variables together as ordered pairs on a graph called a **scatter plot**. To create a scatter plot, consider that one variable is the independent variable and the other is the dependent variable. This means that the dependent variable depends on the independent variable. We usually set up these two variables as ordered pairs where the independent variable is first and the dependent variable is second. Thus, when graphed, the independent variable is graphed along the horizontal axis and the dependent variable is graphed along the vertical axis. You do not connect the dots after plotting these ordered pairs. Instead look to see if there is a pattern, such as a line, that fits the data well. Here are some examples of scatter plots and how strong the linear correlation is between the two variables.

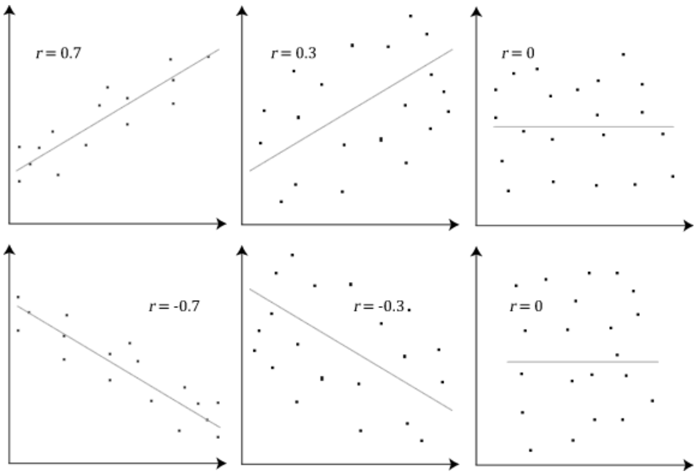


Figure 3.13.1: Scatter Plots Showing Types of Linear Correlation

Creating a scatter plot is not difficult. Just make sure that you set up your axes with scaling before you start to plot the ordered pairs.

Example 3.13.2: Creating a Scatter Plot

Data has been collected on the life expectancy and the fertility rate in different countries ("World health rankings," 2013). A random sample of 10 countries was taken, and the data is:

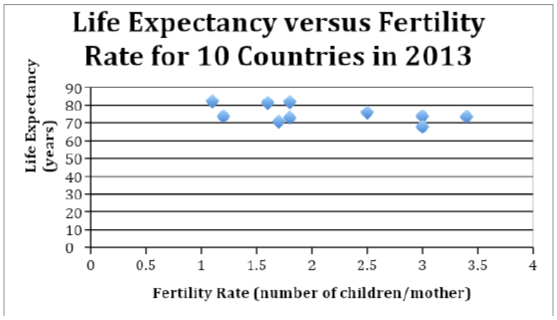
Table 3.13.2: Life Expectancy and Fertility Rate in 2013

Country	Life Expectancy (years)	Fertility Rate (number of children per mother)
SINGAPORE	82.3	1.1
MONACO	81.9	1.8
CANADA	81.5	1.6
ECUADOR	76	2.5
MALAYSIA	73.9	3
LITHUANIA	73.8	1.2
BELIZE	73.6	3.4
ALGERIA	73	1.8
TRINIDAD/TOB.	70.8	1.7
TAJIKISTAN	67.9	3

To make the scatter plot, you have to decide which variable is the independent variable and which one is the dependent variable. Sometimes it is obvious which variable is which, and in some case it does not seem to be obvious. In this case, it seems to make more sense to predict what the life expectancy is doing based on fertility rate, so choose life expectancy to be the dependent variable and fertility rate to be the independent variable. The horizontal axis needs to encompass 1.1 to 3.4, so have it range from zero to four, with tick marks every one unit. The vertical axis needs to encompass the numbers 70.8 to 81.9, so have it range from zero to 90, and have tick marks every 10 units.

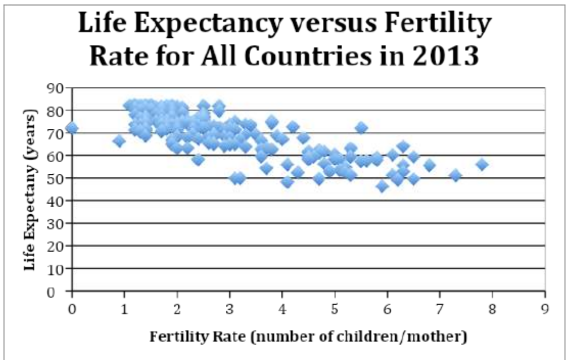
Note: Always start the vertical axis at zero to avoid exaggeration of the data.

Graph 2.5.3: Scatter Plot of Life Expectancy versus Fertility Rate



From the graph, you can see that there is somewhat of a downward trend, but it is not prominent. What this says is that as fertility rate increases, life expectancy decreases. The trend is not strong which could be due to not having enough data or this could represent the actual relationship between these two variables. Let's see what the scatter plot looks like with data from all countries in 2013 ("World health rankings," 2013).

Graph 2.5.4: Scatter Plot of Life Expectancy versus Fertility Rate for All Countries in 2013



Again, there is a downward trend. It looks a little stronger than the previous scatter plot and the trend looks more obvious. This correlation would probably be considered moderate negative correlation. It appears that there is a trend that the higher the fertility rate, the lower the life expectancy. Caution: just because there is a correlation between higher fertility rate and lower life expectancy, do not assume that having fewer children will mean that a person lives longer. The fertility rate does not necessarily cause the life expectancy to change. There are many other factors that could influence both, such as medical care and education. Remember a correlation does not imply causation.

3.13.1: Exercises

The table shows the GPA of 15 randomly selected students and the number of hours spent on studying their lessons before an exam. Use this information for problems 1 - 4.

Student	Number of Hours	GPA
1	3	2.5
2	3	2.4
3	6	3.5
4	4	3
5	5	3.2
6	6	3.8
7	3	2.7
8	4	2.6
9	2	2.8
10	5	3.4
11	5	3
12	4	3.3
13	5	3
14	4	2.7
15	8	4

1. Draw a scatter plot of the data. Label the axes.
2. Based on the information provided, which best describes the relationship between students' GPA and study habits?
    a. Negative
    b. Positive
    c. No correlation
3. Based on the scatter plot, which statement accurately describes the information provided?
    a. The correlation is positive. As the number of hours spent on studying increases, the students' GPA tends to increase.
    b. The correlation is negative. As the number of hours spent on studying increases, the students' GPA tends to increase.
    c. The correlation is negative. As the number of hours spent on studying decreases, the students' GPA tends to increase.
    d. The data show no relationship between students' GPA and study habits.
4. Do the data imply that the length of time spent on studying before an exam causes the change in students' GPA? Why or why not?
    a. Yes. Since hours spent on studying and GPA are correlated, it means the time spent on studying causes the GPA to increase.
    b. No. The increase in GPA could be attributed to other factors such as gender, class size, socioeconomic status, faculty qualification, and many others.
    c. No. The two variables are not correlated. Hence, the increase in time spent on studying does not affect the GPA.
    d. Yes. There is sufficient evidence to support the effect of study habits on students' GPA.

5. If there is a strong correlation between marijuana use and use of other drugs, can we conclude that using marijuana leads to using other drugs?
6. Describe whether the scatter plot of the following variables will show a positive correlation, a negative correlation, or no correlation.
    a. Height and weight
    b. Height and grade in a course
    c. Grade in a course and shoe size
    d. Hours of television watched and grade in a course
    e. Height and shoe size
    f. Hours of television watched and weight
7. The table below shows the number of students in a class and the absences in a week.

Class Size	15	18	20	20	25	25	30	35	35	40
Number of Absences	5	2	10	4	8	12	2	5	7	10

    a. Draw a scatter plot and determine the relationship that best describes the data.
    b. Is the correlation between class size and the number of absences positive, negative, or zero? Fill in the blank: As the class size increases, the number of absences ___________________.
8. The table below shows the recorded maximum average monthly temperature readings of a certain state in the US and the shipment of residential storage water heaters.

Month	Avg. Monthly Temperature °C	Shipment of water heaters (in units)
Jan	-1 °C	384, 213
Feb	0 °C	745, 570
Mar	9 °C	1, 162, 074
Apr	15 °C	1, 473, 757
May	22 °C	1, 806, 054
June	27 °C	2, 217, 780
July	30 °C	2, 610, 644
Aug	26 °C	3, 011, 506
Sep	24 °C	3, 377, 667
Oct	16 °C	3, 803, 921
Nov	12 °C	4, 162, 667
Dec	5 °C	4, 584, 367

    a. Draw a scatterplot.
    b. Describe the correlation that best describes the data provided.
9. In order to determine whether or not there is a relationship between the temperature and the chirp rate of a certain variety of cricket, the following data were collected. Let x = ˚C and y = # chirps per second.

x	31.4	22.0	34.1	29.1	27.0	24.0	20.9	27.8	20.8	28.5	26.4	28.1	27.9	28.6	24.5
y	20.0	16.0	19.8	18.4	17.1	15.5	14.7	17.1	15.4	16.2	15.0	17.2	16.0	17.0	14.4

    a. Draw a scatterplot of the data given.
    b. Is the correlation between temperature and chirp rate positive, negative, or zero?
    c. If the correlation is strong, can we conclude that temperature affects the chirp rate?
10. During the spring and summer, an ice cream shop kept track of the average daily temperature (in ˚F) and the amount of ice cream sales (in dollars).

Temperature (˚F)	52	55	61	66	72	75	77	84	90	94	97
Ice Cream Sales ($)	3500	4200	5000	5300	6600	6800	7200	8000	8400	9100	9500

    a. Create a scatterplot with this data.

b. Use your scatterplot to make a prediction for the ice cream sales when the temperature is 70˚F.

c. If ice cream sales were $7500, what do you think the average temperature was that day?

3.14: Measures of Central Tendency and the Normal Distribution

This topic requires a leap of faith. It is one of the rare times when this textbook will say "don't worry about *why* it's true; just accept it."

A normal distribution, often referred to as a bell curve, is symmetrical on the left and right, with the mean, median, and mode being the value in the center. There are lots of data values near the center, then fewer and fewer as the values get further from the center. A normal distribution describes the data in many real-world situations: heights of people, weights of people, errors in measurement, scores on standardized tests (IQ, SAT, ACT)…

One of the best ways to demonstrate the normal distribution is to drop balls through a board of evenly spaced pegs, as shown here. [1] Each time a ball hits a peg, it has a fifty-fifty chance of going left or right. For most balls, the number of lefts and rights are roughly equal, and the ball lands near the center. Only a few balls have an extremely lopsided number of lefts and rights, so there are not many balls at either end. As you can see, the distribution is not perfect, but it is approximated by the normal curve drawn on the glass.

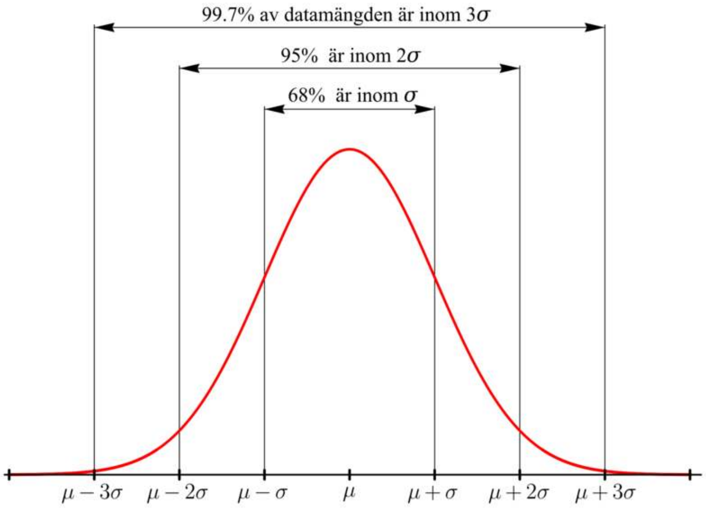
The standard deviation is a measure of the spread of the data: data with lots of numbers close to the mean has a smaller standard deviation, and data with numbers spaced further from the mean has a larger standard deviation. (In this textbook, you will be given the value of the standard deviation of the data and will never need to calculate it.) The standard deviation is a measuring stick for a particular set of data.

In a normal distribution…

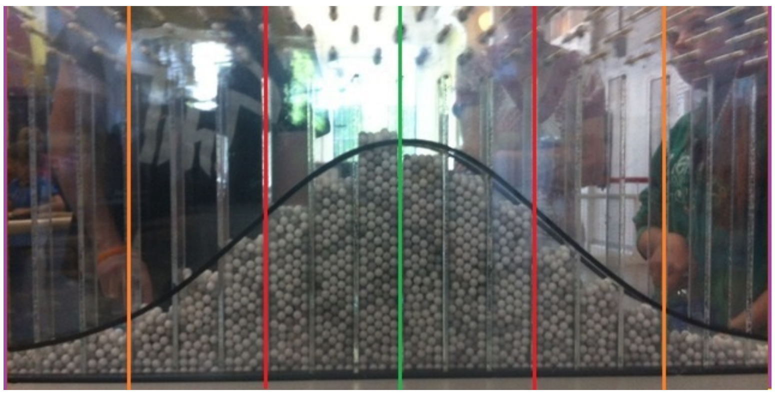
- roughly 68% of the numbers are within 1 standard deviation above or below the mean
- roughly 95% of the numbers are within 2 standard deviations above or below the mean
- roughly 99.7% of the numbers are within 3 standard deviations above or below the mean

This **68-95-99.7 rule** is called an *empirical rule* because it is based on observation rather than some formula. Nobody discovered a calculation to figure out the numbers 68%, 95%, and 99.7% until after the fact. Instead, statisticians looked at lots of different examples of normally distributed data and said "*Mon Dieu*, it appears that if you count up the data values that are within one standard deviation above or below the mean, you have about 68% of the data!" and so on.[2]

The following image is in Swedish, but you can probably decipher it because math is an international language.



Let's go back to the ball-dropping experiment, and let's assume that the standard deviation is three columns wide.[3] In the picture below, the **green** line marks the center of the distribution.



First, the two **red** lines are each three columns away from the center, which is **one standard deviation** above and below the **center**, so about **68%** of the balls will land between the red lines.

Next, the two **orange** lines are another three columns farther away from the center, which is six columns or **two standard deviations** above and below the **center**, so about **95%** of the balls will land between the orange lines.

And finally, the two **purple** lines are another three columns farther away from the center, which is nine columns or **three standard deviations** above and below the **center**, so about **99.7%** of the balls will land between the purple lines. We can expect that 997 out of $1,000$ balls will land between the purple lines, leaving only 3 out of $1,000$ landing beyond the purple lines on either end.

Here are Damian Lillard's game results for points scored, in increasing order, for the 80 games he played in the 2018-19 NBA season.[4] This is broken up into eight rows of ten numbers each, and this is a total of $2,069$ points.

$$\begin{gathered}11,13,13,13,14,14,15,15,15,16,\\16,16,17,17,17,18,18,19,19,20,\\20,20,20,20,21,21,22,22,23,23,\\23,23,24,24,24,24,24,24,24,24,\\25,25,25,26,26,26,28,28,28,29,\\29,29,29,30,30,30,30,30,31,31,\\33,33,33,33,33,33,34,34,34,35,\\36,36,37,39,40,40,41,41,42,51\end{gathered}$$

Exercises 3.14.1

This is a review of mean, median, and mode; you'll need to know the mean in order to complete the standard deviation exercises that follow.

1. What is the mean of the data? (Round to the nearest tenth.)

2. What is the median of the data?

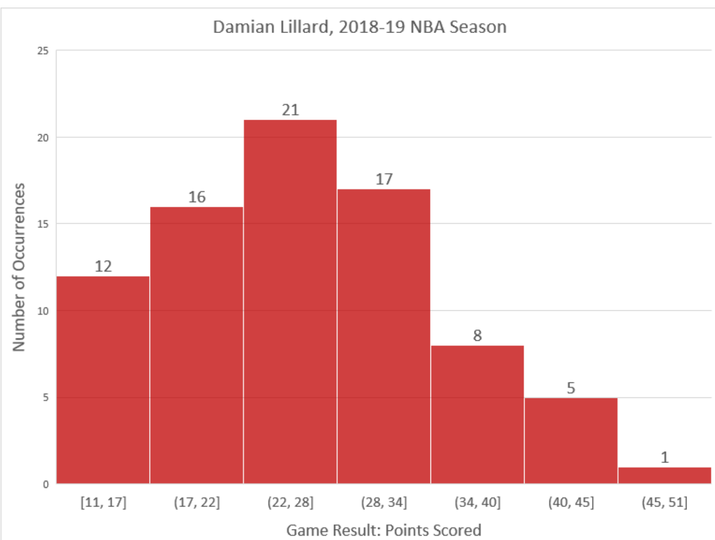
3. What is the mode of the data?

4. Do any of the mean, median, or mode seems misleading, or do all three seem to represent the data fairly well?

Answer

1. 25.9 points
2. 24.5 points
3. 24 points (which occurred eight times)
4. all three seem to represent the typical number of points scored; the mean is a bit high because there are no extremely low values but there are a few high values that pull the mean upwards.

Here is a histogram of the data, arbitrarily grouped in seven equally-spaced intervals. It shows that the data roughly follows a bell-shaped curve, somewhat truncated on the left and with an outlier on the right.



If we enter the data into a spreadsheet program such as Microsoft Excel or Google Sheets, we can quickly find that the standard deviation is 8.2 points.

Based on the empirical rule, we should expect approximately 68% of the results to be within 8.2 points above and below the mean.

Exercises 3.14.1

5. Determine the range of points scored that are within one standard deviation of the mean.

6. How many of the 80 game results are within one standard deviation of the mean?

7. Is the previous answer close to 68% of the total number of game results?

Answer

5. 17.7 to 34.1 points
6. 54 of the 80 game results

7. yes; $54 \div 80 = 67.5\%$

And we should expect approximately 95% of the results to be within $2 \cdot 8.2 = 16.4$ points above and below the mean.

Exercises 3.14.1

8. Determine the range of points scored that are within two standard deviations of the mean.

9. How many of the 80 game results are within two standard deviations of the mean?

10. Is the previous answer close to 95% of the total number of game results?

Answer

8. 9.5 to 42.3 points

9. 79 of the 80 game results

10. sort of close but not really; $79 \div 80 = 98.75\%$

And we should expect approximately 99.7% of the results to be within $3 \cdot 8.2 = 24.6$ points above and below the mean.

Exercises 3.14.1

11. Determine the range of points scored that are within three standard deviations of the mean.

12. How many of the 80 game results are within three standard deviations of the mean?

13. Is the previous answer close to 99.7% of the total number of game results?

Answer

11. 1.3 to 50.5 points

12. 79 of the 80 game results, again

13. yes, this is pretty close; $79 \div 80 = 98.75\%$

Notice that we could think about the standard deviations like a measurement error or tolerance: the mean ± 8.2, the mean ± 16.4, the mean $\pm 24.6 \ldots$

Exercises 3.14.1

For U.S. females, the average height is around 63.5 inches (5 ft 3.5 in) and the standard deviation is 3 inches. Use the empirical rule to fill in the blanks.

14. About 68% of the women should be between _______ and _______ inches tall.

15. About 95% of the women should be between _______ and _______ inches tall.

16. About 99.7% of the women should be between _______ and _______ inches tall.

For U.S. males, the average height is around 69.5 inches (5 ft 9.5 in) and the standard deviation is 3 inches. Use the empirical rule to fill in the blanks.

17. About 68% of the men should be between _______ and _______ inches tall.

18. About 95% of the men should be between _______ and _______ inches tall.

19. About 99.7% of the men should be between _______ and _______ inches tall.

Answer

14. 60.5; 66.5

15. $57.5; 69.5$

16. $54.5; 72.5$

17. $66.5; 72.5$

18. $63.5; 75.5$

19. $60.5; 78.5$

This graph at https://tall.life/height-percentile-calculator-age-country/ shows that, because the standard deviations are equal, the two bell curves have essentially the same shape but the women's graph is centered six inches below the men's.

Exercises 3.14.1

Around 16% of U.S. males in their forties weigh less than 160 lb and 16% weigh more than 230 lb.[5] Assume a normal distribution.

20. What percent of U.S. males weigh between 160 lb and 230 lb?

21. What is the average weight? (Hint: think about symmetry.)

22. What is the standard deviation? (Hint: You have to work backwards to figure this out, but the math isn't complicated.)

23. Based on the empirical rule, about 95% of the men should weigh between _______ and _______ pounds.

Answer

20. 68% because $100\% - (16\% + 16\%) = 68\%$

21. 195 lb because this is halfway between 160 and 230 lb

22. 35 lb because $195 - 35$ lb and $195 + 35$ lb encompasses 68% of the data

23. $125; 265$

If you are asked only one question about the empirical rule instead of three in a row $(68\%, 95\%, 99.7\%)$, you will most likely be asked about the 95%. This is related to the "95% confidence interval" that is often mentioned in relation to statistics. For example, the margin of error for a poll is usually close to two standard deviations.[6]

Let's finish up by comparing the performance of three NFL teams since the turn of the century.

The numbers of regular-season games won by the New England Patriots each NFL season from 2001-19:[7]

year	wins
2001	11
2002	9
2003	14
2004	14
2005	10
2006	12
2007	16
2008	11
2009	10
2010	14
2011	13
2012	12
2013	12
2014	12
2015	12
2016	14
2017	13
2018	11
2019	12

Exercises 3.14.1

For the Patriots, the mean number of wins is 12.2, and a spreadsheet tells us that the standard deviation is 1.7 wins.

24. There is a 95% chance of the Patriots winning between _______ and _______ games in a season.

25. In 2020, the Patriots won 7 games. Could you have predicted that based on the data? How many standard deviations from the mean is this number of wins?

Answer

24. 8.8; 15.6

25. You would not have predicted this from the data because it is more than two standard deviations below the mean, so there would be a roughly 2.5% chance of this happening randomly. In fact, $(12.2-7)\div 1.7$ is slightly larger than 3, so this is more than three standard deviations below the mean, making it even more unlikely. (You might have predicted that the Patriots would get worse when Tom Brady left them for Tampa Bay, but you wouldn't have predicted only 7 wins based on the previous nineteen years of data.)

The numbers of regular-season games won by the Buffalo Bills each NFL season from 2001-19:[8]

year	wins
2001	3
2002	8
2003	6
2004	9
2005	5
2006	7
2007	7
2008	7
2009	6
2010	4
2011	6
2012	6
2013	6
2014	9
2015	8
2016	7
2017	9
2018	6
2019	10

Exercises 3.14.1

For the Bills, the mean number of wins is 6.8, and a spreadsheet tells us that the standard deviation is 1.7 wins.

26. There is a 95% chance of the Bills winning between _______ and _______ games in a season.

27. In 2020, the Bills won 13 games. Could you have predicted that based on the data? How many standard deviations from the mean is this number of wins?

Answer

26. 3.4; 10.2

27. You would not predict this from the data because it is more than two standard deviations above the mean, so there would be a roughly 2.5% chance of this happening randomly. In fact, $(13-6.8) \div 1.7 \approx 3.6$, so this is more than three standard deviations above the mean, making it even more unlikely. This increased win total is partly due to external forces (i.e., the Patriots becoming weaker and losing two games to the Bills) but even 11 wins would have been a bold prediction, let alone 13.

The numbers of regular-season games won by the Denver Broncos each NFL season from 2001-19:[9]

year	wins
2001	8
2002	9
2003	10
2004	10
2005	13
2006	9
2007	7
2008	8
2009	8
2010	4
2011	8
2012	13
2013	13
2014	12
2015	12
2016	9
2017	5
2018	6
2019	7

? Exercises 3.14.1

For the Broncos, the mean number of wins is 9.1, and a spreadsheet tells us that the standard deviation is 2.6 wins.

28. There is a 95% chance of the Broncos winning between _______ and _______ games in a season.

29. In 2020, the Broncos won 5 games. Could you have predicted that based on the data? How many standard deviations from the mean is this number of wins?

Answer

28. 3.9; 14.3

29. The trouble with making predictions about the Broncos is that their standard deviation is so large. You could choose any number between 4 and 14 wins and be within the 95% interval. $(9.1-5) \div 2.6 \approx 1.6$, so this is around 1.6 standard deviations below the mean, which makes it not very unusual. Whereas the Patriots and Bills are more consistent, the Broncos' win totals fluctuate quite a bit and are therefore more unpredictable.

---

1. The Plinko game on *The Price Is Right* is the best-known example of this; here's a clip of Snoop Dogg helping a contestant win some money. ↵
2. Confession: This paragraph gives you the general idea of how these ideas developed but may not be perfectly historically accurate. ↵
3. I eyeballed it and it seemed like a reasonable assumption. ↵

4. Source: https://www.basketball-reference.com/players/l/lillada01/gamelog/2019↵
5. Source: https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=17&ved=2ahUKEwjm-d-whavhAhWCFXwKHQxMDz4QFjAQegQIARAC&url=https%3A%2F%2Fwww2.census.gov%2Flibrary%2Fpublications%2F2010%2Fcompendia%2Fstatab%2F130ed%2Ftables%2F11s0205.pdf&usg=AOvVaw1DFDbil78g-qXbIgK6JirW↵
6. Source: https://en.Wikipedia.org/wiki/Standard_deviation↵
7. Source: https://www.pro-football-reference.com/teams/nwe/index.htm ↵
8. Source: https://www.pro-football-reference.com/teams/buf/index.htm ↵
9. Source: https://www.pro-football-reference.com/teams/den/index.htm ↵

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